The Hidden Costs of Urban Renewal[*]

Matéo Moglia[†]

November 11, 2026

JOB MARKET PAPER

Frequently updated, last version here

Abstract

Three wedges hinder urban redevelopment: (i) market structure (concentrated developers); (ii) market frictions (uncertainty regarding future rents, high fixed costs, and construction time); and (iii) market regulation (taxes and land-use regulations). I formalise these constraints within an option value model of parcel redevelopment, in which land is traded between landowners and oligopsonistic developers. Developers mark down the price of land, a market failure that delays only parcels that carry a building. I calibrate the model on data from 600,000 parcels in the Paris metropolitan area, recover a parcel-specific fixed cost net of the other wedges, and compute medium-run (20-year) supply responses under alternative policies. Lifting density caps changes the quality-adjusted housing stock by -4.29% after twenty years and by 6.72% after fifty, because taller allowances also raise the value of waiting. Within legal bounds and under a balanced budget, the best rent-minimising mix of instruments cuts the taxes on new construction, taxes land instead, and lowers market rents by 14.02% on average over twenty years, mostly through a transfer from incumbent landlords to renters. Neither restores efficiency. A tax on the standing value of under-used parcels, at a rate set by local developer concentration, does so and raises revenue. It brings the redevelopment of the next twenty years forward by 1.9 years.

Keywords: Redevelopment, monopsony, concentration, fixed costs, real options, housing supply, land taxation. **JEL**: R14, R31, R33, D43, L13, H21, H23

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[*]I am extremely grateful to Pierre Boyer and Benoît Schmutz-Bloch for their invaluable guidance. This paper greatly benefited from discussions with Suzanne Bellue, Gilles Duranton, Bertrand Garbinti, Jessie Handbury, Yurou He, Jonathan Goupille-Lebret, Alain Trannoy, Etienne Wasmer, and seminar and workshop participants at AFSE (Nantes), Aix-Marseille School of Economics, Erasmus University Rotterdam, Humboldt University, CREST, HEC Paris, New York University Abu Dhabi, UEA (Barcelona), Università della Svizzera italiana, and the Wharton School. This research was supported by subsidies from the Agence Nationale de la Recherche (Labex ECODEC No. ANR-11-LABEX-047), the Chaire Professorale Jean Marjoulet, and the Chaire DDX. I also thank the Berlin Quantitative Spatial Economics Group and the Wharton School for their hospitality during the writing of the paper. Data are accessed through the Centre d'accès sécurisé aux données.

[†]Centre de Recherche en Économie et Statistique (CREST), École polytechnique, Institut Polytechnique de Paris, Palaiseau, France. *Contact*: mateo.moglia@gmail.com.

1. Introduction

In large cities, new housing is supplied by replacing the existing structure. In Paris and its inner suburbs, 71% of the housing units authorised since 2014 replace a building demolished on the same parcel. This margin will soon be the only one, since French law now requires cities to stop the net consumption of open land by 2050.[1] Yet redevelopment responds little and late to prices. Even where rents are high, the supply response is low (Saiz, 2010; Baum-Snow and Han, 2024) and slow (Rollet, 2025), and in the Paris area many parcels zoned for housing have stayed empty, vacant, or under-built since 2014.

Whether this rate of redevelopment is inefficient is a first-order question for housing policy. Where supply cannot follow demand, housing costs outgrow incomes and workers are kept away from productive places (Glaeser and Gyourko, 2018; Hsieh and Moretti, 2019). Governments have answered with subsidies to clear brownfields, such as the French *Fonds Friches*, reduced VAT on new housing, upzoning (Chapelle et al., 2025), and taxes on vacant housing (Segú, 2020). These policies are often justified as if slow redevelopment were a market failure. Much of the delay, however, reflects costs that any planner would bear, such as demolition, irreversibility, uncertainty about rents, and the time it takes to build. A policy that offsets such a cost moves resources from taxpayers to builders, and it raises welfare only if the fiscal revenue it brings forward pays for it. A policy that corrects a market failure raises welfare on its own. Telling the two apart requires knowing which part of the delay is a failure and which part is a cost, parcel by parcel. This paper provides that measurement for the redevelopment of Paris and its inner suburbs.

Three main results follow. First, market failures delay redevelopment. Buyer power on the land market raises the rent at which built parcels are sold, delays their redevelopment by years, and blocks some of them for ever. Second, a model that measures each wedge separately is a tool to evaluate housing policies. Every supply policy changes one or several wedges, so the same framework ranks policies by their effect on the stock, on rents, on the budget, and on each group. Third, corrective policies exist and can be implemented. A tax on the standing value of under-used parcels, at a rate set by local developer concentration, restores the constrained-efficient allocation, raises welfare, and raises fiscal revenue.

The model starts from the real-options theory of land development (Titman, 1985; Capozza and Helsley, 1990; Dixit and Pindyck, 1994). Each parcel carries a standing building and can be redeveloped once.

[1] Law 2021-1104 of 22 August 2021 (*Climat et Résilience*) sets the objective of zero net land take (*zéro artificialisation nette*) by 2050, and halves land take over 2021–2031 relative to the previous decade. Containment policies of this kind trade open space against housing (Turner et al., 2014). Where they bind, as the greenbelts of English cities do, they raise house prices and lower welfare (Hilber and Vermeulen, 2016; Koster, 2024).

Rents follow a geometric Brownian motion, so the landowner holds an option to sell the parcel for redevelopment and exercises it the first time the local rent reaches a threshold. The developer who buys the parcel demolishes the standing building, pays a fixed cost of clearing the site, and builds up to the density cap after a construction lag. The novelty is the land market between them. In each municipality, a few developers buy most of the land. As employers do in labour markets (Card et al., 2018; Berger et al., 2022), each developer faces an upward-sloping supply of parcels, because landowners have private preferences over whether to sell and to whom. Each developer therefore pays landowners only a share of what a parcel is worth to it. This land share falls with the local Herfindahl index of land purchases. The markdown is the part of that value the landowner does not receive, so it rises with the developer's share of local land purchases.

The model has three groups of wedges, and each enters the threshold in a different place. Market structure, the markdown, scales down what the landowner receives. Frictions raise the cost of a project. They are the fixed cost of clearing a site, the option value of waiting that irreversibility and uncertainty create, and the construction lag. Regulation lowers the return on a project, through taxes on land, floorspace, and new housing, and through density caps. Where the density cap set by land-use regulation binds, the threshold has a closed form. The fixed cost and the development tax raise the cost of the project. The lag, VAT, and the property tax lower the return on each square metre built. The cap sets the floorspace gained. The option value of waiting scales the whole threshold up. The markdown enters only through the standing value, which the landowner gives up in full while receiving a share of the developer value.

From the model, I derive theoretical results. First, buyer power distorts redevelopment only when the landowner gives something up. On empty land, the landowner gives up nothing, so the markdown transfers rents to developers without changing when the parcel is sold. On a built parcel, the markdown inflates the standing value relative to what the landowner receives, so the parcel is sold later than in a competitive land market. Where the standing building is large relative to what could replace it, the parcel is never sold. Competitive firms exercise investment options at the socially optimal time (Leahy, 1993). Buyer power breaks this result, but only on a set of parcels that the data identify, those that carry a building. Because the distortion binds on some parcels and not on others, it both delays redevelopment and misallocates it across parcels.

Second, the model separates failures from costs. I define two benchmarks that face the same technology, the same uncertainty, and the same zoning plan as private parties. The first-best allocation counts tax revenue as social value. The constrained-efficient allocation takes the tax code as given. A wedge is a

cost relative to a benchmark if it enters the benchmark threshold as well, and a distortion if it enters only the private one. The fixed cost, the option value, and the lag are costs relative to both benchmarks. Taxes are distortions relative to the first best only. The markdown is a distortion relative to both, and a competitive land market implements the constrained-efficient allocation. A subsidy for a friction is then a transfer that changes total surplus only to second order. It raises welfare if and only if the fiscal revenue it brings forward covers its outlay, because taxes on redevelopment hold private thresholds above the first best.

The cost of redevelopment is not observed, and the usual approach recovers it from the observed rate of construction (Murphy, 2018; Rollet, 2025). That approach attributes every source of delay to the cost, the markdown included. The structure of the model allows a different strategy, in which each wedge is measured from its own source and the fixed cost is the only residual. The option premium follows from the volatility of the transaction price index, the yield of new units, and the perceived growth of rents. The construction lag is the observed time from the sale of a parcel to completion, by initial use. Taxes come from the tax code and the municipal rate files, and the density cap from the buildings of each cadastral section. The slope of the land share in concentration comes from sales of land followed by a permit on the same parcel, and its level from the operating margins of listed developers. Given these wedges, the price of an unsold parcel equals its standing value plus the landowner's share of the option to redevelop it. The option value falls strictly with the fixed cost, so a unique fixed cost rationalises each price within two bounds. The absence of a sale adds a lower bound, since a parcel still unsold once rents reached a given level has a threshold above that level. The land market matters for this measurement. Ignoring the markdown would overstate the aggregate fixed cost by 35%, because a landowner who kept the whole option would need a higher cost to rationalise the same price.

The data make this strategy possible. I assemble a parcel-level panel of Paris and its inner suburbs from 2014 to 2025 that links the land tax records, building footprints and heights, and the universes of building permits and property sales. Four steps make these sources usable. I aggregate deeds into redevelopment operations, measure demolitions by comparing successive vintages of buildings, consolidate the project companies of developers into their parent groups, and compare local concentration with a random allocation of developers across projects. In the median municipality, the effective number of developer groups is 3.0. The model rationalises 63.1% of observed prices, mostly on lightly built parcels in the inner suburbs, and every counterfactual is computed on these parcels. The slope of the land share is imprecise, so I report the main results at its point estimate, at the lower end of its confidence interval,

and with no slope at all.

The first main result is that market failures delay redevelopment. The markdown delays the parcels it reaches. At the calibrated land shares, landowners receive 81% of the developer value on average, and the median parcel with a standing building starts 2.4 years later in expectation than in the constrained-efficient allocation. Across all unsold parcels, 0.3% are blocked by concentration for ever. Holding demand fixed, concentration lowers the number of parcels redeveloped at every horizon, and the gap converges to the floorspace of the blocked parcels. On the estimable stock, the markdown accounts for 0.9% of the twenty-year supply shortfall, a small share of a large gap. Frictions and regulation account for the rest. The fixed cost alone accounts for 30.9%, and taxes and density caps for 45.3%. The markdown also reorders redevelopment, but the parcels it swaps are close substitutes, so its welfare cost is mostly a cost of delay. The estimable stock is lightly built by construction, so these shares understate the failure where it bites, on heavily built parcels.

The second main result is that the model is a tool to evaluate housing policies. Because each wedge is a separate object, every supply policy maps into the model as a change in one of them. A reduced VAT rate raises the return on each square metre built, a subsidy in the spirit of the *Fonds Friches* lowers the fixed cost, upzoning raises the cap, and a subsidy to sellers raises the land share. I solve every policy in general equilibrium, on common demand paths, and report its effect on the stock and on rents, its net cost to the budget, its incidence by group, and a marginal value of public funds (Hendren and Sprung-Keyser, 2020). Three results show what this evaluation adds. Lifting density caps changes the stock by -4.29% after twenty years and by 6.72% after fifty, because a taller allowance also raises the value of waiting (Capozza and Li, 1994). Upzoning is a long-run policy. Lower taxes on redevelopment and a transfer for the fixed cost bring projects forward, and some of them pay for themselves within twenty years, because the projects they induce pay taxes. Finally, the balanced-budget mix of instruments that lowers rents most cuts the taxes on new construction and taxes land instead. It lowers market rents by 14.02% on average over twenty years, but mostly through a transfer from incumbent landlords to renters, 120.5 billion euros against a surplus of 28.8 billion on the additional housing. None of these policies restores efficiency, because they lower thresholds where frictions and taxes bite rather than where the markdown bites.

The third main result is that corrective policies restore efficiency. The theory also gives the corrective instrument. A policy restores the constrained-efficient allocation if and only if it raises what landowners receive relative to what they give up by the inverse of the land share. Two instruments do so: a subsidy

to sellers, and a tax on the standing value of a parcel, paid until it is redeveloped. Because the markdown is a proportional wedge within a municipality, the corrective rate needs neither the fixed cost nor the preferences of landowners. It needs a public statistic, the local Herfindahl index of land purchases, and two elasticities. The tax is also implementable. It can be levied on the cadastral rental value that already serves as the base of the property tax, with new buildings exempt, and French law already taxes undeveloped constructible land, unused commercial premises, and vacant dwellings. At its corrective rate, 17% of the standing flow on average, the tax brings the redevelopment of the next twenty years forward by 1.9 years and raises 23.3 billion euros of fiscal revenue by year twenty. A subsidy to sellers reaches the same thresholds at 158.5 thousand euros per additional unit, since most of it goes to landowners who would have sold anyway. Its revenue can also replace the transfer tax, which falls on transactions rather than value, as public finance recommends (Mirrlees, 2011).

These conclusions rest on measurement that can be improved. Concentration is under-measured, because the consolidation of project companies is incomplete, and the slope of the land share is imprecise. The tax is large relative to the current property tax, and the model holds maintenance, vacancy, and the pass-through to tenants fixed. The fixed cost is a residual that may contain hold-out rents and other failures. I sign the bias that each of these choices induces on the measured effect of the markdown. Taken together, the paper offers a theory of when buyer power distorts land use, a framework that separates failures from costs and prices every supply policy on the same footing, and a corrective tax that requires only public statistics.

Related literature The paper contributes to four literatures. First, it measures the cost of redevelopment from land prices. Structural models recover large fixed costs from the observed rate of construction (Murphy, 2018; Peng, 2023; Rollet, 2025), and a long tradition measures regulation as the gap between prices and construction costs (Glaeser and Gyourko, 2003; Gyourko and Molloy, 2015). Closer to this paper, Cunningham (2006) shows that uncertainty raises vacant land prices and delays development, and Clapp and Salavei (2010) and McMillen and O'Sullivan (2013) show that property prices capitalise the option to redevelop. I invert this capitalisation parcel by parcel, and recover costs net of the markdown, taxes, and the construction lag, which a quantity-based approach folds into a single number.[2]

Second, the paper brings buyer power into the theory of real options in land development (Titman,

[2]The theory of redevelopment goes back to Brueckner (1980) and Rosenthal and Helsley (1994), and Combes et al. (2021) and Chapelle et al. (2023) measure housing production and land-use regulation in France. Housing constraints also misallocate workers across cities (Herkenhoff et al., 2018; Ganong and Shoag, 2017).

1985; Capozza and Helsley, 1990; Dixit and Pindyck, 1994). In this literature, market structure works through strategic exercise among builders, who build earlier and in clusters when rivals hold similar options (Grenadier, 1996, 2002; Bulan et al., 2009). My mechanism works on the other side of the transaction, and less competition delays exercise. The closest paper is Lebret et al. (2026). In their model of redevelopment in New York, tax exemptions accelerate projects near the threshold, density bonuses expand projects but can delay them, and affordability mandates weaken redevelopment unless paired with strong incentives. The last two results hold here as well. The two papers differ in who holds the option. In their model, developers hold it. Here, the landowner holds it and sells it on a concentrated land market. The timing of redevelopment then carries a distortion that no incentive on the project corrects, and their policy trade-offs become a distinction between efficiency and redistribution.[3]

Third, the paper measures oligopsony on an input market other than labour, and the misallocation it causes. Firms face upward-sloping input supply when sellers have idiosyncratic preferences (Card et al., 2018; Berger et al., 2022; Yeh et al., 2022), and buyer concentration shapes input prices and productivity outside labour markets as well (Rubens, 2023). I apply this structure to landowners choosing among developers. Quintero (2023) shows that seller concentration among US homebuilders lowers housing production, a restriction on quantity whose remedy is competition among builders. Buyer power on the land market instead distorts the timing of redevelopment, and its remedy is a tax on standing value, which deconcentration alone need not achieve. Land assembly is the mirror case, in which many sellers face one buyer and hold-out rents delay projects (Eckart, 1985; Strange, 1995; Miceli and Sirmans, 2007; Brooks and Lutz, 2016). Following the misallocation literature (Restuccia and Rogerson, 2008; Hsieh and Klenow, 2009; Baqaee and Farhi, 2020), I separate the level of the distortion, which delays redevelopment, from its dispersion, which changes which parcels are redeveloped first.[4]

Finally, the paper brings the taxation of land into the evaluation of supply policies. That a tax on current-use value brings development forward is an old result (Skouras, 1978; Bentick, 1979; Mills, 1981; Anderson, 1986). In these models, private timing is efficient, so the tax is a distortion. I extend the result to

[3]Other extensions of the theory consider time to build (Majd and Pindyck, 1987; Bar-Ilan and Strange, 1996; Oh et al., 2024), predictable demand growth (Lange and Teulings, 2024), and the vacancy of commercial real estate (Moszkowski and Stackman, 2023).

[4]In real estate, market power has been studied mostly on the seller side. Concentrated landlords raise rents in New York (Watson and Ziv, 2026), and developers charge large markups in São Paulo (Alvarez and Biderman, 2026). Kroft et al. (2025) show that construction firms hold power in both labour and product markets. Markusen and Scheffman (1977) ask whether developers held buyer power in the Toronto land market. Duranton et al. (2015) document the misallocation of land across firms in India. Search models of the housing market offer another channel (Wheaton, 1990; Genesove and Han, 2012; Piazzesi et al., 2020). Under efficient bargaining, however, concentration lengthens the search for a buyer but leaves the decision to redevelop unchanged.

a land market with buyer power, in which private timing is too late and a tax at a rate set by buyer concentration restores the efficient date. Unlike a floor on land prices, the analogue of a minimum wage under monopsony (Robinson, 1933; Manning, 2003), the tax leaves developers' rents in place. The other instruments are evaluated on their net fiscal cost and their incidence, as in the analysis of place-based and urban housing policies (Busso et al., 2013; Kline and Moretti, 2014; Favilukis et al., 2023; Rollet and Weiwu, 2025).[5]

2. Setting, data and facts

2.1. Setting

The analysis covers Paris and the three départements that surround it, Hauts-de-Seine, Seine-Saint-Denis, and Val-de-Marne. The area has 142 municipalities, counting the 20 districts of Paris separately. The area hosts about 10% of the French population and produces about 20% of French GDP. Demand for floorspace is strong, in both the residential and the commercial markets.

Each municipality, or group of municipalities, writes a zoning plan that sets where housing can be built and how dense it can be, and the mayor grants building permits (Chapelle et al., 2023, 2025). Since a permit that complies with the plan can still be refused, and the applicant bears both the delay and the risk of refusal, regulation adds risk and delay to every project. In the model, these costs are part of the fixed cost of redeveloping a parcel and of the lag before the new building is delivered.

Land and the building on it are sold together, so a developer who wants a parcel must buy the standing building, and its landowner gives up the income it yields. The price of a built parcel therefore includes the value of its standing building, which is why this value enters the landowner's decision to sell in the model (Section 3.2).

A developer is a firm that buys land and builds new housing on it. Developers prospect for land locally, approach landowners directly, and sign a conditional sale agreement that gives them exclusive rights to the parcel while they apply for a permit, so a parcel is rarely auctioned among many buyers. Many developers create a new company for each project. I therefore measure market shares at the level of the parent group (Appendix A.6). The paper studies private land redeveloped by private developers, which buy land on the market. Social landlords, mixed-economy companies, and public bodies build

[5] Housing subsidies are capitalised and crowd out unsubsidised construction (Fack, 2006; Eriksen and Rosenthal, 2010), inclusionary zoning acts as a tax on development (Schuetz et al., 2011; Soltas, 2024), and transfer taxes distort transactions and mobility (Dachis et al., 2012; Kopczuk and Munroe, 2015; Best and Kleven, 2018; Bérard and Trannoy, 2018).

under other rules and with other funding. Land held by the State, local authorities, social landlords, and other public or non-profit bodies is sold under different rules. I leave both out of the sample of parcels and of the flows of redevelopment. Non-private builders stay in the market shares, because a private landowner can also sell to them (Appendix A.2).

Four taxes fall on redevelopment, and they are set by different levels of government. The State levies value-added tax on the sale of new housing. Départements levy most of the transfer tax on the sale of land. Municipalities set the rate of the annual property tax, and municipalities, départements, and the region levy a development tax on new floorspace at the permit date. Section 3.3 shows how each tax enters the model, and Section 4.3 reports their rates.

2.2. Data

I assemble a parcel-level panel of the area from 2014 to 2025. Four sources are measured at the parcel and linked through the cadastral identifier of the parcel. The land tax records cover every taxed unit, with its floorspace, its use, and its construction date. I aggregate units to the parcel and call a parcel mixed when no single use accounts for 90% of its units. A national building database gives the footprint and height of every building in successive years. Comparing years separates actual demolitions and extensions from measurement noise. The permit register covers every authorised building permit, with its floorspace, its number of units, its date, and its applicant. Dates of the start and end of works are reported for about 60% of permits. The transaction register records every property sale, with its price and its date.

Three links between these sources identify the land market. When the applicant of a permit is a firm, its business identifier matches the permit to a developer group. 73% of collective permits are matched this way (Appendix Table A6). The legal category of the applicant separates private developers from social landlords, mixed-economy companies, and public bodies. Private developers file 88.8% of the floorspace of matched permits. Some land sales are followed by a permit on the same parcel. For these sales, I observe both the price and the identity of the buyer, which identifies the markdown (Section 4.2).

Other sources are measured at a coarser level. Rents come from a municipal index of market rents, available from 2015. Zoning, population and income, ground conditions, and utility networks come from administrative sources. Construction and demolition costs come from a survey of realised project costs. Appendix A describes every source and the restrictions on the sample.

2.3. Facts

Five facts motivate the model. The first is the puzzle the paper explains. Each of the others motivates one ingredient of the model.

Fact 1. Many parcels stay idle where rents are high I call a parcel idle if it is zoned for housing and is empty, vacant, or under-built. An empty parcel carries no building. A vacant parcel carries a building with no taxed unit and a documented past economic activity. An under-built parcel carries a low building, less than a quarter of the mean height of the buildings on immediate neighbouring parcels.[6] There are 4,328 privately owned idle parcels in the area (Table 1). Building them up to the density cap of their section[7] would add 10.6 million square metres of floorspace net of what already stands, or about 176 thousand dwellings. Empty and vacant parcels cluster in the inner suburbs closest to Paris (Figure J3), in municipalities whose rents rank among the highest 2% in France. 65% of idle parcels have not changed since 2014, and 1,947 of them enter the estimation sample of Section 5.1. That they are not redeveloped at those rents is the puzzle the rest of the paper explains.

Table 1: Potential housing supply on idle parcels

Type	Parcels	Gross potential (m^2)	Standing floorspace (m^2)	Net potential (m^2)	Net potential units
Empty	759	449,874	0	449,874	7,498
Under-built	1,862	8,208,597	172,486	8,038,808	133,980
Vacant	1,707	4,250,423	2,475,445	2,100,557	35,009
Total	4,328	12,908,895	2,647,931	10,589,240	176,487

Data: land tax records, building database (2024). Parcels in areas zoned for housing. Gross potential: the inferred density cap of the cadastral section (Section 4.1), as a floor-area ratio, times the parcel area. Standing floorspace: footprint times height of the building on the parcel, divided by a 2.5-metre floor height. Net potential: gross potential minus standing floorspace, truncated at zero parcel by parcel, so it exceeds the difference of the two column totals wherever standing floorspace exceeds the cap. Units of 60 square metres. Private land only: parcels held in 2025 by the State, local authorities, social landlords, or other public or non-profit bodies are excluded, such as railway land and public parks. Parcels outside the size range of private redevelopment, from 54 to 9,829 square metres, are also excluded. Both restrictions are those of the estimation sample (Appendix A.2). The largest 5% of the remaining parcels by potential are excluded.

Fact 2. Redevelopment is the main supply margin I match every permit that creates housing between 2014 and 2024 with the building on its parcel before and after the permit. This separates three types of project: new buildings on empty land, conversions of an existing building without changing its size,

[6] Parcels in natural, green, equipment, or business zones are excluded. I also drop buildings built below the 10th and above the 90th percentiles of density in the cadastral section to avoid false positives, such as parking structures or unusually shaped buildings.

[7] Cadastral sections are an aggregation of parcels, the equivalent of three blocks in New York. The median section hosts 95 parcels and its median area is 66,000 square metres.

and demolition and rebuilding. Over the period, 15,694 permits authorised 285,343 units and 16.5 million square metres of residential floorspace. Demolition and rebuilding delivers 71% of the new units, conversions 26%, and new buildings on empty land 3% (Figure 1). Most new housing therefore replaces a building that already stands: in a dense city, housing supply is replacement. This motivates a model in which the decision to build is a decision to replace a standing building.

Land is scarce enough that population grows faster than built-up land. Between 1999 and 2021, the population of the area grew by 11%, while its built-up land grew by 0.8% (Appendix Figure J2). Building more densely absorbed the difference. Between 2015 and 2023, housing floorspace grew by 5.7% and population by 1.7%, so floorspace per resident rose by 3.9%. New housing came from building up on land already built, which is what redevelopment does.

Redevelopment also reallocates land across uses. Mixed-use parcels deliver 42% of new units, and former office and service parcels 16%. Replacing an office building yields the densest projects. Their median floor-area ratio is 1.76, against 0.53 when a single-family house is replaced (Appendix Table I1). The model therefore indexes parcels by their initial use. It has no conversion margin. Where a building can be converted, as many offices can, the price of the parcel also capitalises the option to convert. The inversion of Section 3.7 attributes this value to redevelopment, so it understates the fixed cost of these parcels and its share of the supply shortfall.

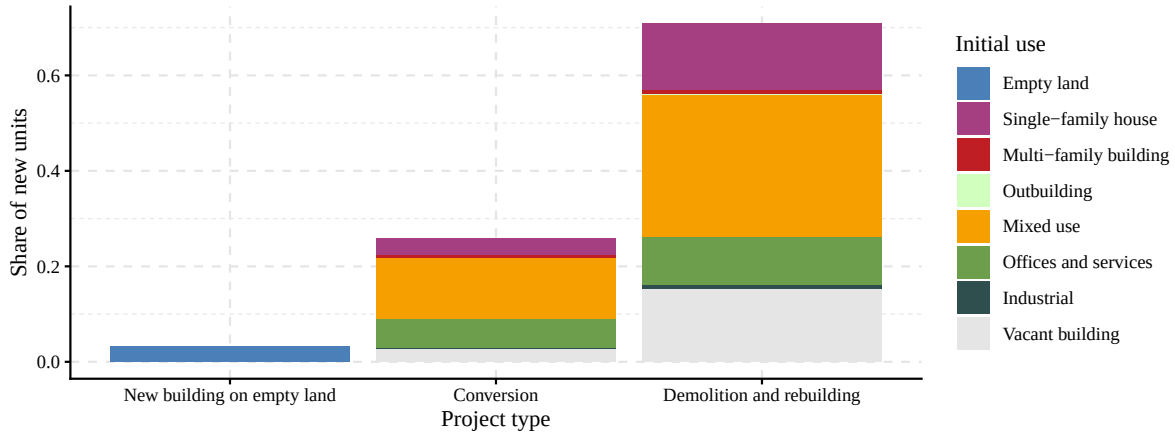


Figure 1: Housing supply by type of project and initial use

Data: permit register, land tax records, building database (2014–2025). Share of new housing units by type of project and initial use of the parcel. New building on empty land: units built on a parcel without a building. Conversion: units created in a building whose size does not change. Demolition and rebuilding: units created where the building is demolished and replaced. Initial use: the main use of the parcel before the permit.

Fact 3. Land is bought by a handful of developers in each municipality Developers take rents as given when they sell housing (Fact 5), but the land market is different. It is fragmented across the region and concentrated within municipalities. Across the region, the top seven of the 3,857 developer groups active over 2013–2024 supply 16% of redeveloped floorspace. The region-wide Herfindahl index is 0.007, which corresponds to 153 groups of equal size (Figure 2). Within a municipality, the picture changes. The effective number of developer groups, the inverse of the Herfindahl index, has a median of 3.0 across municipalities. The largest group holds a median share of 50% of land purchases (Appendix Table I3). Throughout, a purchase is valued at the floorspace of its permit times the local price of new housing, a proxy for the land payment, which I do not observe for every permit.

Two checks bear on the municipality as the relevant market. First, developers buy locally, but not only in one municipality. A developer group with several projects makes on average 56% of its purchases in a single municipality, and 66% in a single group of municipalities. The rest of its purchases are made elsewhere, so the assumption that each group buys in one municipality only (Assumption 4) is an approximation. A group that also buys in neighbouring municipalities faces more elastic landowners than the model assumes, which makes the measured markdown too large. Second, local concentration is more than the result of small numbers. A municipality has a median of 12 projects over three years, and a few projects mechanically produce a high Herfindahl index. I therefore compare the observed index with the index obtained when developer groups are drawn at random across the projects of the region, bin by bin of project count (Table 2). Over all municipalities, the median index is 0.333 in the data and 0.20 under random allocation, and 80% of municipalities are above the 95th percentile of their random index. Among the municipalities with at least twenty projects, the median index is 0.20, against 0.10 under random allocation. Where projects are numerous, the observed index is still twice its random benchmark, and 91% of these municipalities are above the 95th percentile of theirs.

These figures are a lower bound on concentration. Many developers create a new company for each project, and I cannot yet trace every such company to its parent group (Appendix A.6). Local concentration motivates a land market in which a few developers in each municipality make offers to landowners (Section 3.4).

Concentration is not higher where rents are higher. The correlation between the municipal Herfindahl index and municipal rents is -0.06 ($p = 0.47$). Whether concentration distorts supply in the most or in the least valuable locations is therefore an empirical question (Section 5.4).

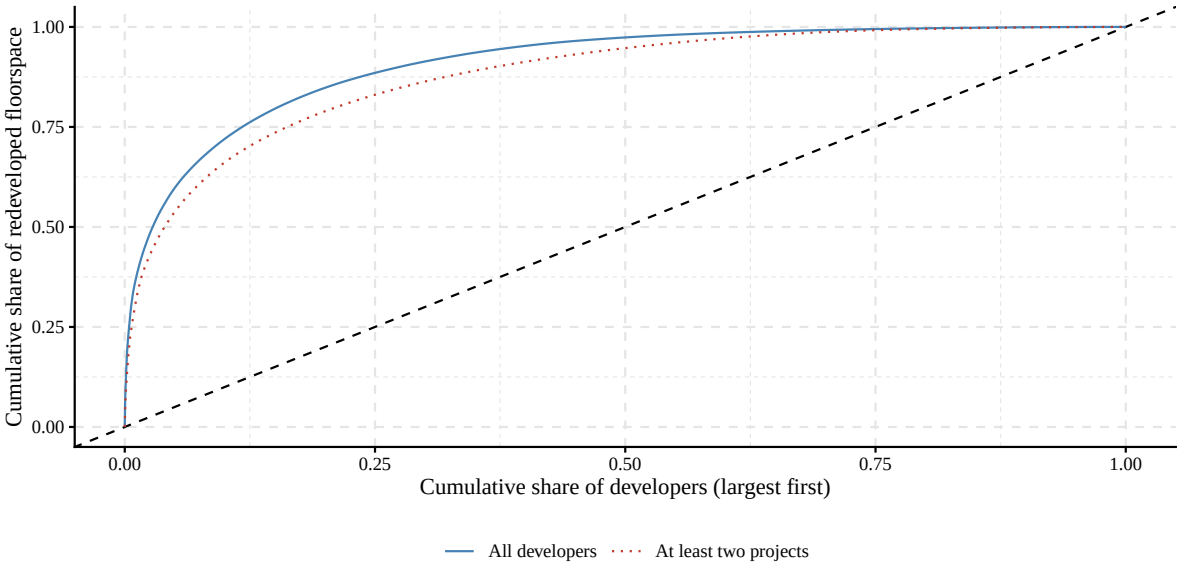


Figure 2: Region-wide developer concentration

Data: permit register, business register, 2013–2024. Cumulative share of redeveloped floorspace against the cumulative share of developer groups, ranked largest first. The dashed 45° line denotes perfect equality. Plain line: all projects with a recorded developer. Dotted line: developer groups with at least two projects.

Table 2: Local concentration of land purchases

Municipalities	N	Median HHI, observed	Median HHI, random allocation	Above the 95th percentile of random allocation (%)
All with a permit in the last window	137	0.33	0.20	80
Fewer than 10 projects	55	0.59	0.37	64
10 to 19 projects	39	0.31	0.17	90
20 projects or more	43	0.20	0.10	91

Data: permit register, business register, 2013–2024. Shares of developer groups in the purchases of each municipality over the last three-year window, each permit valued at its floorspace times the local price of new housing. Rows: municipalities with a permit in the window, by number of projects. Columns: number of municipalities, median observed Herfindahl index, median index under random allocation, and share of municipalities above the 95th percentile of their own random index. Random allocation: developer groups permuted across the projects of the region in the same window, 500 draws. Each group keeps its number of projects and each municipality its projects.

Fact 4. Redevelopment takes years, and longer on some initial uses For the permits of private developers, I measure the time a project takes from the sale of the parcel to the completion of the new building, in three steps: from the sale to the permit, from the permit to the start of works, and the works themselves. From the sale to completion, the median project takes 4.6 to 6.7 years, depending on the initial use of the parcel (Table 3). The works alone take 2.2 to 2.5 years. Projects that replace mixed-use buildings take the longest, followed by offices, services, and industrial buildings, and new buildings on empty land are the shortest (Appendix Figure J1). The differences in construction time across initial uses are significant (Appendix Table I2). This motivates a construction lag indexed by the initial use of the parcel (Assumption 3).

Table 3: Time from the sale of a parcel to completion, by initial use

Initial use	Sale to permit	Permit to start	Works	Sale to completion	Permits
Empty	1.7	0.8	2.2	4.6	284
Housing	1.9	1.0	2.5	5.4	1,617
Mixed	3.2	1.0	2.5	6.7	1,896
Non-housing	2.7	1.0	2.5	6.2	1,030

Data: transaction register, permit register. Median years by step for permits creating housing, mapped to the four initial uses of the model. Empty: new buildings on empty land. Housing: demolition of single-family houses, multi-family buildings, and outbuildings. Mixed: demolition of mixed-use buildings. Non-housing: demolition of office, service, industrial, and vacant buildings. Within each initial use, the mean of the medians of its project types. Sale to completion is the sum of the three steps and is the construction lag ℓ_u of the model. The steps may not sum to the total because of rounding.

Fact 5. Individual projects are small relative to the housing stock No single project is large relative to the housing stock. The largest project in a municipality represents a median of 1.35% of its dwellings. This motivates the assumption that no developer internalises its effect on rents (Assumption 2).

A donut difference-in-differences around new projects tests this assumption directly. It compares resale prices within 100 metres of a project with those between 150 and 500 metres, before and after the permit. For projects of at least 50 dwellings, the effect on nearby prices is -0.056 (s.e. 0.169), and no project size yields a significant estimate (Appendix A.7). The design has little power, so it cannot rule out the effects of a few percent found in dense US cities (Asquith et al., 2023; Li, 2022; Pennington, 2021).

3. A model of redevelopment

Each parcel can be redeveloped once, rents are uncertain, and its landowner chooses when to sell it for redevelopment. This is a real-options problem in the tradition of Titman (1985), Capozza and Helsley (1990), and Dixit and Pindyck (1994). I add a land market in which a few developers make offers to landowners. The section describes demand, the standing building, the project, the land market, the timing of redevelopment, efficiency and misallocation, what the price of a parcel reveals about its fixed cost, and how parcels add up to a path of supply.

3.1. Environment and demand

Each parcel i is defined by its area T_i, its initial use $u(i)$ (housing, mixed, non-housing, or empty), a standing building, and a density cap $\bar{h}_i$. It is located in municipality $m(i)$, written m when there is no ambiguity. Time is continuous, and every agent discounts at rate ρ.

Households rent housing floorspace across the region, and they value a square metre more in some municipalities than in others. I summarise these differences by a location premium ϕ_m. Households pay

ϕ_m times as much for a square metre in municipality m as in a municipality with the median premium. The premium is a local demand shifter. It captures amenities and access to jobs, and I measure it by the municipal rent relative to the regional median (Section 4.1).

Assumption 1 (Demand)**.** The regional rent is $R_t = A_t H_t^{-\alpha}$, where $H_t = \sum_i \phi_m h_{it}$ is the housing stock weighted by location premia, h_{it} is the housing floorspace on parcel i at date t, and $\alpha > 0$ is the inverse elasticity of demand. The rent per square metre on parcel i is $R_{it} = \phi_m R_t$. The demand shifter A_t follows a geometric Brownian motion, $dA_t = \mu A_t\, dt + \sigma A_t\, dW_t$, with drift μ and volatility $\sigma > 0$.

I call H_t the quality-adjusted housing stock. A square metre built where ϕ_m is high adds more to H_t than a square metre built where it is low, because households are willing to pay more for it. Every parcel faces the local rent $\phi_m R_t$.

Assumption 2 (Perceived rent process)**.** Each parcel has measure zero. Landowners and developers treat the rent as a geometric Brownian motion with drift $\delta < \rho$ and volatility σ: $dR_t = \delta R_t\, dt + \sigma R_t\, dW_t$.

Each parcel is negligible, so no landowner or developer internalises the effect of its own project on rents, which Fact 5 supports. Agents do anticipate the effect of all future projects together. The drift δ is the growth of rents they expect, net of the supply that future redevelopment will add. Along the equilibrium path, rents grow at $\mu - \alpha g_H$, where g_H is the growth rate of the stock. Section 5.2 compares δ with this realised growth.

Every value is computed under this perceived process and discounted at ρ, the required return of the landowners who hold the option. The cash flows of a parcel are not spanned by traded assets. Landowners hold a single undiversified parcel, so the option to redevelop cannot be replicated, and I discount at a risk-adjusted rate rather than under a risk-neutral measure. The yield $y = \rho - \delta$ capitalises rents into prices. From here on, R denotes the local rent of the parcel under consideration.

3.2. The standing building

Until it is redeveloped, a parcel yields two flows. The first is housing. The standing housing floorspace h_i^{old} earns the local rent, so it yields $R\, h_i^{\text{old}}$ per year. Its value follows the local rent. A standing building is sold whole, with its tenants, and trades at a higher yield than new units sold one by one. Sitting tenants, rent regulation, and the cost of renovation all lower its price. I capitalise its housing flow at the yield $y^w \geq y$, where $y \equiv \rho - \delta$ is the yield of new units.

The second flow is commercial. Shops and offices are let under commercial leases, whose rents are revised only with a lease index and may not follow the local housing rent. The commercial rental flow is $v_i^C = R_i^C h_i^C$, the commercial rent times the commercial floorspace. I observe the product v_i^C directly, as the rental value of the business premises recorded in the tax files. The tax files also assign a rental value to dwellings. I leave it out of v_i^C, since dwellings enter through h_i^{old}. I capitalise it at the commercial yield y^C and hold its value $V_i^C \equiv v_i^C / y^C$ constant over time as I do not observe commercial rental values over time.

The value of keeping the building in place, which I call the standing value, is

$$V_i^w(R) = \frac{R\,\tilde{h}_i^{\text{old}}}{\rho - \delta} + \tilde{V}_i^C. \tag{1}$$

Both flows are net of the property tax that the standing building pays, at the municipal rate t_m^F on a base equal to a share ψ^{new} of the rental flow (Section 3.3): $\tilde{h}_i^{\text{old}} \equiv (1 - t_m^F \psi^{\text{new}})\,(y/y^w)\,h_i^{\text{old}}$ and $\tilde{V}_i^C \equiv (1 - t_m^F \psi^{\text{new}})\,V_i^C$. The factor y/y^w capitalises the standing housing flow at y^w in (1), so every result below holds with two yields. A tilde always denotes these after-tax quantities, and only them. Without a tilde, h_i^{old} and V_i^C denote physical floorspace and the gross commercial value. A housing parcel has $V_i^C = 0$, a non-housing parcel $h_i^{\text{old}} = 0$, and a mixed parcel both flows. An empty parcel, which carries no building, has neither, so $V_i^w = 0$.

3.3. The project

Assumption 3 (Technology and time to build)**.** A project started on parcel i at date t delivers h square metres of housing floorspace at $t + \ell_i$, where the lag $\ell_i = \ell_{u(i)}$ depends on the initial use (Fact 4). The density h is chosen at the start and cannot exceed the cap $\bar{h}_i$. The project cannot be abandoned once started, and the standing building is emptied at the start. The fixed cost F_i is paid at the start. The construction cost is $C_i = c_0 T_i^{-1/\zeta} h^{1+1/\zeta}$, a Cobb-Douglas technology in land and capital (Combes et al., 2021), paid at a constant rate between the start and the completion of works.

In a dense city with binding land-use regulations, most projects build up to the density cap. Among parcels with a finite threshold, 99% build at the cap at that threshold (Appendix Table F7). The standard decision is then whether and when to redevelop a parcel, not necessarily how much to build on it. Appendix B.1 derives the density chosen below the cap, the threshold in that interior regime, and the condition under which it is finite. The numerical solution covers both regimes. The split matters for

policy. Below the cap, waiting also raises the density built, so the option premium is larger, 2.32 against 1.63 at the cap. The threshold also rises with the fixed cost less than proportionally, at the power $1/(\zeta + 1)$. Interior parcels therefore respond less than constrained parcels to a transfer for the fixed cost, and not at all to a higher cap. Comparisons of policies across parcels partly reflect which regime each parcel is in.

Two discount factors summarise the lag. Rents start at completion, so their present value is scaled by $\lambda_i^R = e^{-(\rho-\delta)\ell_i}$. Construction costs are spread over the works, so their present value is scaled by $\lambda_i^C = (1 - e^{-\rho\ell_i})/(\rho\ell_i)$.

Two taxes fall on the new building. Value-added tax, at rate τ^V, is levied on the sale price of new units. The developer deducts the VAT paid on construction, so VAT lowers the value of the building by the factor $1/(1+\tau^V)$. The property tax is paid every year after an exemption of e years, at the municipal rate t_m^F on a base equal to a share ψ^{new} of the rent. Its present value is a share $\tau_m^F \equiv t_m^F \psi^{\text{new}} e^{-(\rho-\delta)e}$ of the value of the building. I drop the subscript m when there is no ambiguity.[8] I absorb the lag and the two taxes into one effective discount on the value of a square metre:

$$\kappa_i \equiv \lambda_i^R \frac{1-\tau^F}{1+\tau^V}. \tag{2}$$

At the start date, a square metre of new floorspace is worth $\kappa_i R/(\rho-\delta)$ to the developer.

In the constrained regime, the payoff of the project at rent R, before the fixed cost, is

$$\pi_i(R) = \kappa_i \frac{R\,\bar{h}_i}{\rho-\delta} - C_i, \qquad C_i \equiv \lambda_i^C\, c_0 T_i^{-1/\zeta} \bar{h}_i^{1+1/\zeta}. \tag{3}$$

The first term is the value of the new building, net of the two taxes on it and discounted for the lag. The second, C_i, is the construction cost at the cap, discounted over the works.

3.4. The land market

The most a developer is willing to pay for a parcel is the value of the project net of every cost the developer bears. Two more taxes enter here. The development tax, τ^A per square metre of new floorspace, is paid at the permit date, which I treat as the start date. The transfer tax, at rate τ^M, is paid by the buyer on the price of the land. I call developer value the most the developer can pay the landowner for parcel

[8] The standing building pays the property tax too, at the same rate and on the same base share, with no exemption, which the after-tax quantities $\tilde{h}_i^{\text{old}}$ and $\tilde{V}_i^C$ of the standing value (1) account for. Appendix C.5 details every tax.

i at rent R:

$$V_i^D(R) \equiv \frac{\pi_i(R) - F_i - \tau^A \bar{h}_i}{1 + \tau^M}. \tag{4}$$

The numerator is the payoff of the project, π_i, minus the fixed cost F_i and the development tax on the new floorspace, $\tau^A \bar{h}_i$. Dividing by $1 + \tau^M$ leaves room for the transfer tax on the payment to the landowner. The gain from redeveloping, which I call the surplus, is the developer value minus the standing value the landowner gives up:

$$S_i(R) \equiv V_i^D(R) - V_i^w(R). \tag{5}$$

The land market determines how the developer value is split between the landowner and the developer. Section 3.5 shows that the split also changes when parcels are sold. Three assumptions describe the land market.

Assumption 4 (Market structure)**.** Municipality m has N_m developer groups, indexed by j. Each group buys land in its municipality only, and market shares are stable over time.

Assumption 4 is an approximation. Developer groups make about half of their purchases in their main municipality (Fact 3). A group that also buys elsewhere can substitute across municipalities, so landowners in each municipality face more competition than the municipal Herfindahl index measures, and the markdown the model assigns to a given index is too large.

Landowners of ripe parcels in municipality m choose how much land to sell to each developer, L_{jm}, measured in euros of developer value. Developer j pays a share θ_{jm} of the developer value. Landowners have private preferences over whether to sell and to whom, as workers do over employers (Card et al., 2018; Berger et al., 2022). Their choices solve

$$\max_{\{L_{jm}\}_j} \sum_j \theta_{jm} L_{jm} - \bar{\Theta}\, \bar{L}_m^{-1/\eta^S} \frac{\mathcal{L}_m^{1+1/\eta^S}}{1 + 1/\eta^S}, \qquad \mathcal{L}_m \equiv \Big(\sum_j L_{jm}^{\frac{1+\eta^D}{\eta^D}} \Big)^{\frac{\eta^D}{1+\eta^D}}, \tag{6}$$

with $\eta^D > \eta^S > 0$. The second term is the cost of selling. The elasticity η^S measures how readily landowners sell when offers rise. The elasticity η^D measures how readily they switch from one developer to another. The constant $\bar{L}_m$ is the stock of ripe parcels, in euros of developer value, and $\bar{\Theta}$ is the offer at which all of them would sell. Each developer chooses how much land to buy, taking the purchases of its rivals as given. It pays every landowner in its municipality the same share θ_{jm} of the developer value, the share at which landowners supply that quantity.

As in Berger et al. (2022), the supply of parcels to developer j, in euros of developer value, is (Lemma B.2)

$$L_{jm} = \left(\frac{\theta_{jm}}{\Theta_m}\right)^{\eta^D} \left(\frac{\Theta_m}{\bar{\Theta}}\right)^{\eta^S} \bar{L}_m, \qquad \Theta_m \equiv \left(\sum_{k=1}^{N_m} \theta_{km}^{1+\eta^D}\right)^{\frac{1}{1+\eta^D}}. \tag{7}$$

The offer index Θ_m summarises the offers of all developers in the municipality. The first factor of (7) splits sales across developers, and the second sets total sales. In the dynamic problem below, the outside option $\bar{\Theta}$ is replaced by each parcel's standing value V_i^w.

Let w_{jm} denote developer j's share of land payments in municipality m, and $HHI_m = \sum_j w_{jm}^2$ the Herfindahl index of these shares. In the data, I proxy land payments by the floorspace of each permit times the local price of new housing (Appendix A.6).

Adapting Berger et al. (2022) to the land market, developer j pays the share (Proposition B.3)

$$\theta_{jm} = \left[1 + \frac{w_{jm}}{\eta^S} + \frac{1 - w_{jm}}{\eta^D}\right]^{-1}, \tag{8}$$

and the share of the developer value that landowners receive in municipality m, $\bar{\theta}_m \equiv \sum_j \theta_{jm} L_{jm} / \sum_j L_{jm}$, is

$$\bar{\theta}_m = \left[1 + \frac{HHI_m}{\eta^S} + \frac{1 - HHI_m}{\eta^D}\right]^{-1}. \tag{9}$$

I call $\bar{\theta}_m$ the land share and $1 - \bar{\theta}_m$ the markdown. The share a developer pays falls with its market share. A developer with a negligible share faces the elasticity η^D of landowners switching between developers. A developer with the whole market faces only the elasticity η^S of landowners deciding whether to sell. Aggregation is exact: the land share of a municipality depends only on its Herfindahl index and on the two elasticities.

The relation between market share and offer comes from the form of competition. Developers here compete for quantities and post one offer to all landowners. Under price competition for each parcel, or under an auction among several bidders, the price would not fall with market share in this way.

A single developer pays the land share $\eta^S / (1 + \eta^S)$, and with many small developers each pays $\theta^{\text{atom}} \equiv \eta^D / (1 + \eta^D) < 1$. Deconcentration therefore raises the land share only up to θ^{atom}: landowners receive the full developer value only if developers are perfect substitutes for them, $\eta^D \to \infty$. In the calibration, θ^{atom} is 90.3%, against a mean land share of 81% at observed concentration. The rest of the markdown reflects imperfect substitution between developers, which deconcentration alone does not remove (Proposition 3(iii)).

The elasticities η^S and η^D and the market shares are exogenous. I do not model why few developers operate. Concentration could reflect efficient scale or barriers to entry. Either way, given the number of developers, the markdown distorts the landowners' decision, which is what the welfare analysis requires.

Developers pay landowners less than the developer value because they post one offer. To buy one more parcel, a developer must raise its offer, and the higher offer applies to every parcel it would have bought anyway. A posted offer fits two features of the land market. Developers do not observe what a landowner would accept, and with private values no trading mechanism is both efficient and voluntary (Myerson and Satterthwaite, 1983). A buyer facing sellers with private reservation values then optimally posts an offer below its own value (Riley and Zeckhauser, 1983). Parcels are also not auctioned, since a developer prospects landowners directly and secures a parcel with an exclusive conditional sale agreement. Under an auction, or under efficient bilateral bargaining, the markdown would be smaller or would transfer value without delaying the sale.[9] In the timing problem below, I follow a landowner who values the standing building at its market level. The preferences of the other landowners set the elasticities, and hence the share this landowner is offered, but not this landowner's exercise rule.

3.5. Timing: the redevelopment threshold

The landowner holds the option to sell. When the landowner sells at rent R, the landowner receives $\bar{\theta}_m V_i^D(R)$ and gives up the standing value $V_i^w(R)$.

The value of waiting depends on how rents evolve. An option on an asset whose value follows a geometric Brownian motion with drift δ and volatility σ, discounted at ρ, has a value proportional to R^β, where β is the positive root of

$$\tfrac{1}{2}\sigma^2\beta(\beta-1)+\delta\beta-\rho=0 \tag{10}$$

(Dixit and Pindyck, 1994, ch. 5). Since $\delta<\rho$, $\beta>1$. A larger β means that waiting is worth less: rents are less volatile, grow more slowly, or are discounted more heavily.

The value of a parcel to its landowner is (Lemma B.4)

$$W_i(R)=V_i^w(R)+\bar{\theta}_m\,\mathcal{O}_i^\theta(R),\qquad \mathcal{O}_i^\theta(R)\equiv\sup_\tau \mathbb{E}\left[e^{-\rho\tau}S_i^\theta(R_\tau)\right],\qquad S_i^\theta(R)\equiv V_i^D(R)-\frac{V_i^w(R)}{\bar{\theta}_m}. \tag{11}$$

[9] The posted offer predicts that the ratio of the price to the developer value is the same within a developer and a municipality, up to the fixed cost. In the sales followed by a permit, developer-by-municipality effects explain 74% of the variance of this ratio, across 283 cells with at least two sales. The fixed cost also enters the ratio, so this is weak evidence in either direction.

The landowner's value is the standing value plus the landowner's share of the option to sell. The landowner exercises this option on the private surplus S_i^θ.[10] The private surplus differs from the surplus S_i in one place: the standing value is divided by the land share $\bar{\theta}_m$. The landowner gives up the full standing value but receives only a share of the developer value. Selling is therefore worth it only if the developer value exceeds the standing value by a margin, which the division by $\bar{\theta}_m$ captures.

The exercise rule is the standard one of Dixit and Pindyck (1994), applied to the private surplus (Proposition B.5). The landowner sells, and the parcel is redeveloped, the first time the rent reaches $R_i^\star$, the smallest root of $\beta\, S_i^\theta(R) = R\, S_i^{\theta\prime}(R)$ with $S_i^\theta(R) > 0$. In the constrained regime, the private surplus is linear in the rent, $S_i^\theta(R) = a_i^\theta R - K_i^\theta$, with

$$a_i^\theta = \frac{1}{\rho - \delta}\left[q_i - \frac{\tilde{h}_i^{\text{old}}}{\bar{\theta}_m}\right], \qquad K_i^\theta = \frac{F_i + C_i + \tau^A \bar{h}_i}{1+\tau^M} + \frac{\tilde{V}_i^C}{\bar{\theta}_m}, \qquad q_i \equiv \frac{\kappa_i \bar{h}_i}{1+\tau^M}.$$

Provided $a_i^\theta > 0$,

$$R_i^\star = \frac{\beta}{\beta - 1}\,\frac{K_i^\theta}{a_i^\theta}. \tag{12}$$

The two terms of the private surplus have a direct interpretation. The slope a_i^θ is the gain in value per euro of rent. It counts the new floorspace q_i, after taxes and the lag, minus the standing housing floorspace marked up by $1/\bar{\theta}_m$. The intercept K_i^θ is the cost of redeveloping. It counts the fixed cost, the construction cost, and the development tax, deflated by the transfer tax, plus the commercial value of the standing building marked up by $1/\bar{\theta}_m$.

The ratio K_i^θ / a_i^θ is the static break-even rent, at which the private surplus is zero. A landowner who ignored the option to wait would sell at that rent. The landowner waits longer. The factor $\beta/(\beta-1) > 1$ is the premium the landowner requires for giving up the option to wait. At the calibrated $\beta = 2.59$, it equals 1.63: the landowner sells only once the rent is 1.63 times the break-even rent. The constrained-efficient allocation faces the same irreversibility and uncertainty, and requires the same premium. In the estimation, I compute $R_i^\star$ numerically for every parcel, by maximising $S_i^\theta(R) R^{-\beta}$ over R, in both regimes. The closed form serves as a check.

Every parcel faces the same rent process, up to its location premium, so timing is driven by the common rent rather than by shocks specific to the profits or costs of each parcel. A single state variable, the running maximum of the regional rent, determines which parcels have been redeveloped, and the

[10] A landowner who sells to developer j receives $\theta_{jm} V_i^D$. Lemma B.4 replaces the buyer-specific share by the municipal land share $\bar{\theta}_m$, the payment-weighted harmonic mean of the θ_{jm} in (9). The approximation error is of second order in the dispersion of market shares within the municipality.

information on each parcel lies in its threshold, which its price reveals, rather than in the date at which it happens to be sold.

Let R_i^{CE} denote the threshold that the landowner would choose if the land market were competitive, that is the threshold of Proposition B.5 with $\bar{\theta}_m = 1$. I call it the constrained-efficient threshold: it maximises the surplus of redeveloping the parcel, given the tax code and the zoning plan (Section 3.6).

Proposition 1 (Concentration raises the threshold)**.** *(i) The threshold $R_i^\star$ decreases with the land share $\bar{\theta}_m$, so $R_i^\star \geq R_i^{CE}$, with equality if and only if the parcel has no standing value, $V_i^w = 0$. (ii) In the constrained regime, if $\bar{\theta}_m \leq \tilde{h}_i^{old}/q_i < 1$, the parcel is redeveloped in the constrained-efficient allocation but never privately. Proof in Appendix B.*

The markdown matters only for parcels that carry a building. On empty land, the landowner gives up nothing, so receiving a share of the developer value scales the landowner's payoff without changing when the landowner sells. Concentration then only transfers value. On a built parcel, the markdown inflates the standing value and delays redevelopment. Where the standing housing is large relative to what could be built, it can block redevelopment for ever.[11] Concentration is therefore a failure specific to redevelopment, not to construction on open land.

3.6. Efficiency and misallocation

I compare the private economy with two benchmarks. Both face the same frictions as private parties, and both take the zoning plan as given. They differ in how they treat taxes.

Let $S_i^0(R)$ denote the pre-tax surplus: the surplus (5) with every tax rate set to zero, in the developer value and in the standing value alike. Taxes are transfers between private parties and the government, so S_i^0 is the social value of redeveloping parcel i at rent R.

Definition 1 (First-best allocation)**.** The first-best allocation redevelops each parcel the first time the rent reaches its first-best threshold R_i^{FB}, the threshold that maximises the expected pre-tax surplus $\mathbb{E}[e^{-\rho\tau} S_i^0(R_\tau)]$ over stopping times τ. It faces the same technology as private parties: the fixed cost F_i, the construction lag ℓ_i, irreversibility, the rent process, and the zoning plan.

[11] Standing buildings do not depreciate, and standing floorspace earns the rent of new floorspace. Both assumptions overstate the standing value of old buildings and keep the problem one-dimensional. With depreciation, every parcel with a positive constrained-efficient slope would eventually become ripe, and the parcels blocked for ever would be parcels that concentration delays beyond any horizon of interest.

Definition 2 (Constrained-efficient allocation)**.** Given the tax code and the zoning plan, the constrained-efficient allocation redevelops each parcel the first time the rent reaches its constrained-efficient threshold R_i^{CE}, the threshold that maximises the expected after-tax surplus $\mathbb{E}[e^{-\rho\tau}S_i(R_\tau)]$ over stopping times τ. It is the threshold that the landowner would choose if the land market were competitive, $\bar{\theta}_m = 1$.

Both allocations count the developer value and the standing value at par. The price of land is a transfer between landowners and developers, so it enters neither. Both pay the fixed cost, wait out the construction lag, and value the option to wait. The zoning plan is held fixed in both, because the model does not contain the externalities that density caps may correct. The first-best allocation counts tax revenue as social value. The constrained-efficient allocation takes the tax code as given, as private parties do. With zero taxes, the two coincide. Since the constrained-efficient threshold is the private threshold at $\bar{\theta}_m = 1$, a competitive land market implements the constrained-efficient allocation. This is the land-market version of the result that competitive firms exercise investment options at the socially optimal time (Leahy, 1993).

Following Restuccia and Rogerson (2008) and Hsieh and Klenow (2009), I separate costs from distortions. Relative to a benchmark, a wedge is a cost if it enters the benchmark threshold as well as the private one, and a distortion if it enters only the private one. The frictions are costs relative to both benchmarks. Taxes are costs relative to the constrained-efficient allocation and distortions relative to the first best. The markdown is a distortion relative to both. As the rent rises, each benchmark converts parcels in increasing order of its regional threshold, $r_i^{\mathrm{FB}} \equiv R_i^{\mathrm{FB}}/\phi_m$ or $r_i^{\mathrm{CE}} \equiv R_i^{\mathrm{CE}}/\phi_m$. Write the private regional threshold as $r_i = (1+\omega_i^b)\, r_i^b$ for $b \in \{\mathrm{FB}, \mathrm{CE}\}$. The distortion $\omega_i^b > -1$ measures how far parcel i is from its benchmark threshold.

Definition 3 (Efficiency, delay, and misallocation)**.** Relative to benchmark b, an allocation is *efficient* if $\omega_i^b = 0$ for every parcel. It is *delayed* if the distortion is positive on average: with a common distortion $\omega > 0$, every parcel starts when the rent reaches $1+\omega$ times its benchmark threshold. It is *misallocated* if, for some quantity of new floorspace, the parcels it converts to deliver that quantity differ from those that benchmark b converts first.

Delay is about the mass of parcels redeveloped by a given date. Misallocation is about the order in which they are redeveloped. Efficiency implies no misallocation. The converse is false. A common positive distortion delays every parcel without changing the order, so an allocation can be free of misallocation and still inefficient. I write $\omega_i \equiv \omega_i^{\mathrm{CE}}$ for the markdown distortion relative to the constrained-efficient

allocation. Relative to the first best, taxes add a distortion, $1 + \omega_i^\tau \equiv r_i^{\text{CE}} / r_i^{\text{FB}}$, which can take either sign. Taxes on the new building and on the sale of land raise the threshold, whereas the property tax on the standing building lowers the standing value and can bring the threshold forward.

Let $\lambda_g \geq 0$ be the shadow cost of public funds, and let welfare be the sum of the values of landowners and developers plus $1 + \lambda_g$ times the public balance. The next proposition separates the market failure from the frictions, and gives the condition under which a policy that offsets a friction is still worth paying for.

Proposition 2 (Efficiency)**.** *(i)* The markdown is a market failure. *Without the markdown, $\bar{\theta}_m = 1$, the private economy implements the constrained-efficient allocation. With it, the private threshold exceeds R_i^{CE} on every parcel with a standing value and equals it on every other parcel. Raising the land share raises total surplus to first order.*

(ii) Frictions are not failures. *The fixed cost, the construction lag, and the option premium enter the first-best, constrained-efficient, and private thresholds alike. At the constrained-efficient allocation with zero taxes, a marginal subsidy to the fixed cost changes total surplus only to second order, and its net welfare effect is $-\lambda_g$ times the expected outlay to first order. The same holds for transfers that compensate the construction lag or the risk of waiting.*

(iii) Self-financing policies raise welfare. *Consider a marginal policy that pays an outlay X to private parties, net of the taxes levied on the outlay itself, lowers private thresholds, and changes the fiscal revenue from the projects whose start date it moves by T, all in expected present value. Holding rents fixed, its welfare effect is, to first order,*

$$dW = (1 + \lambda_g)\, T - \lambda_g\, X + M, \tag{13}$$

where $M \geq 0$ is the gain on parcels whose threshold exceeds R_i^{CE} because of the markdown, and $M = 0$ if $\bar{\theta}_m = 1$. A self-financing policy, $T \geq X$, raises welfare for every $\lambda_g \geq 0$.

Proof in Appendix B.

Part (i) is the land-market counterpart of a monopsony distortion. Landowners give up the whole standing value but receive only a share of the developer value, so they sell too late, and the loss of surplus is first order. Part (ii) explains why a subsidy for a friction is a transfer that has to be paid for. Part (iii) gives the condition under which such a transfer is still worth paying. Taxes on redevelopment raise private thresholds above the first-best ones, so a policy that brings projects forward collects fiscal revenue

that private parties ignore, a fiscal externality in favour of the budget. If that revenue covers the outlay, the policy raises welfare whatever the cost of public funds, which is why every supply policy below is evaluated by its net cost to the budget. Rents are held fixed in (13). When supply rises, lower rents transfer value from landlords to renters and add a surplus on the new housing, which the counterfactuals account for.

A self-financing supply policy raises welfare, but it restores efficiency only if it removes the markdown distortion. Two instruments act on the markdown. Let τ_m^{sell} be an ad valorem subsidy on the land price paid to landowners who sell for redevelopment in municipality m. Let t_m^w be a tax on the standing flows of a parcel, its after-tax rent $R\tilde{h}_i^{\text{old}}$ and commercial flow $y^C \tilde{V}_i^C$, paid every year until the parcel is redeveloped, with new buildings exempt. Its present value is a share t_m^w of the standing value. I call it the tax on standing value.

Proposition 3 (Restoring efficiency)**.** *Hold the elasticities η^S and η^D fixed.*

(i) The efficiency-restoring policies. *The pair (τ_m^{sell}, t_m^w) implements the constrained-efficient threshold R_i^{CE} on every parcel of municipality m if and only if*

$$(1 + \tau_m^{sell})\, \bar{\theta}_m = 1 - t_m^w. \tag{14}$$

The rates depend only on the local Herfindahl index, η^S, and η^D. They leave offers and developer rents per parcel unchanged, and they unblock every parcel blocked by concentration. Along (14), thresholds and supply are the same. The pure subsidy, $\tau_m^{sell} = 1/\bar{\theta}_m - 1$, has a positive outlay. The pure tax, $t_m^w = 1 - \bar{\theta}_m$, raises revenue. If the pure subsidy is not self-financing and the pure tax is, exactly one pair along (14) balances the budget at a given horizon.

(ii) Under-used land. *Restricting the base of the tax on standing value to parcels whose standing housing floorspace is below the cap, $h_i^{old} < \bar{h}_i$, which includes every parcel without housing, leaves part (i) unchanged on every parcel that the constrained-efficient allocation redevelops, provided $\kappa_i \leq (1 + \tau^M)(1 - t_m^F \psi^{new})$. Restricting it to parcels without housing, the secondary and tertiary uses, restores R_i^{CE} on those parcels only.*

(iii) Instruments that do not restore efficiency. *A lump-sum subsidy to the fixed cost implements R_i^{CE} only if it is set parcel by parcel, as a function of the fixed cost and the standing value. A uniform subsidy raises total surplus to first order on parcels with a standing value and lowers it to second order on parcels without one. No subsidy bounded by the fixed cost unblocks a parcel blocked by concentration. A tax at a uniform*

rate τ^U on the forgone rent of the gap $(\chi\bar{h}_i - h_i^{old})^+$ implements R_i^{CE} on a parcel only if $\tilde{V}_i^C = 0$ and $\tau^U = (1-\bar{\theta}_m)\tilde{h}_i^{old}/(\chi\bar{h}_i - h_i^{old})$. On a parcel without standing value, it lowers the threshold below R_i^{CE}. Finally, as $HHI_m \to 0$, the land share tends to θ^{atom}, which is below one unless $\eta^D \to \infty$, so deconcentration need not restore efficiency.

Proof in Appendix B.

The intuition is simple. The landowner sells when what the landowner receives, a share of the developer value, exceeds what the landowner gives up, the standing value. The markdown scales the first term down. Efficiency requires the ratio of the two terms to be the same as in a competitive land market. Raising what landowners receive, with a subsidy, or lowering what they give up, with a tax on standing value, both restore the ratio. Because the markdown is a common proportional distortion within a municipality, neither instrument needs information on landowners' preferences, fixed costs, or standing values. They need only the local Herfindahl index and two elasticities. The two instruments differ in who pays. The subsidy is paid by taxpayers and goes to every seller, including those who would have sold anyway. The tax is paid by landowners of standing buildings and raises revenue.

A parcel can be under-used in quantity, when its standing floorspace falls short of what the zoning plan allows, or in use, when it carries no housing. The quantity definition is the right base for efficiency. A parcel whose standing floorspace reaches the cap has no redevelopment option in the constrained-efficient allocation, so leaving it out of the base changes no threshold. The use definition is narrower and closer to existing French taxes on vacant and commercial premises. It restores efficiency on non-housing parcels and leaves the markdown on housing parcels, so it removes part of the delay and creates misallocation between housing and non-housing parcels.

The other instruments act elsewhere. A subsidy to the fixed cost acts where fixed costs are high, which is not where the markdown is. It offsets a friction and, on built parcels, part of the markdown, but it brings empty parcels forward of their efficient date. A tax on the gap between the cap and the standing building has its largest base where the standing building is smallest, which is where the markdown is zero. It brings empty land forward of its constrained-efficient date and corrects too little on heavily built parcels. Both instruments can still raise welfare by part (iii) of Proposition 2, because taxes on redevelopment hold private thresholds above the first best. Their case then rests on the budget, or on an externality of idle land that the model does not contain, not on the markdown. The same budget criterion applies to relaxing density caps, whose welfare effect otherwise depends on externalities the model does not contain.

The tax on standing value can be levied on the cadastral rental value that already serves as the base of the property tax, as a surcharge on standing buildings with new buildings exempt. French law already contains taxes in this spirit. Municipalities in tight housing markets may raise the base of undeveloped constructible land (article 1396 of the tax code), tax commercial premises left unused (article 1530), and tax vacant dwellings (article 232), a tax that raises the share of dwellings in use (Segú, 2020). The tax considered here differs in its base, the standing value of under-used parcels, and in its rate, set by local developer concentration.

Proposition 4 (Uniform and dispersed distortions)**.** *(i) If the distortion is the same for every parcel, $\omega_i^b = \omega$, the private economy converts the same parcels as benchmark b, in the same order. For any quantity of new floorspace, the set of converted parcels coincides. Every parcel starts when the regional rent reaches $1+\omega$ times its benchmark threshold. A dispersed distortion is therefore necessary for misallocation. (ii) In the constrained regime, the markdown distortion relative to the constrained-efficient allocation is*

$$1+\omega_i = \frac{1+\left(\frac{1}{\bar{\theta}_m}-1\right)\frac{\tilde{V}_i^C}{K_i}}{1-\left(\frac{1}{\bar{\theta}_m}-1\right)\frac{\tilde{h}_i^{old}}{q_i-\tilde{h}_i^{old}}}, \tag{15}$$

where K_i is the intercept of the surplus S_i and q_i the after-tax, time-discounted new floorspace of Proposition B.5. It is zero for parcels without standing value. It increases with the markdown $1/\bar{\theta}_m - 1$, with the commercial standing value relative to the cost of redeveloping, and with the standing housing relative to the new floorspace. Proof in Appendix B.

Part (i) is the land-market counterpart of the result that a tax levied at the same rate on every firm distorts the scale of production but does not misallocate inputs across firms. Part (ii) shows that the markdown distortion is not uniform. It is zero on empty land and positive on built parcels, larger where concentration is higher and where the standing building is worth more relative to what replaces it. The markdown therefore delays redevelopment and, in principle, misallocates it: relative to the constrained-efficient order, built parcels are converted too late compared with empty ones. Because demand is over quality-adjusted floorspace, the markdown also changes where housing is built whenever its distortion is correlated with the location premium ϕ_m.

3.7. Identification of the fixed cost

The price of a parcel that has not been redeveloped is its value to its landowner, W_i in (11). It equals the standing value plus the landowner's share of the option to redevelop. Given the land share, the price therefore reveals the option, and the option reveals the fixed cost.

Proposition 5 (Identification of the fixed cost)**.** *Fix the structural parameters, the current rent R_{i0}, and the land share $\bar{\theta}_m$. The option value $\mathcal{O}_i^{\theta}(R_{i0}; F)$ is continuous and strictly decreasing in the fixed cost F, from $\mathcal{O}_i^{\theta}(R_{i0}; 0)$ to zero. A unique $F_i \geq 0$ rationalises the observed price $\hat{p}_i^L$ if and only if*

$$V_i^w(R_{i0}) \;<\; \hat{p}_i^L \;\leq\; V_i^w(R_{i0}) + \bar{\theta}_m \, \mathcal{O}_i^{\theta}(R_{i0}; 0). \tag{16}$$

Proof in Appendix B.

The land share comes from sales of ripe parcels and from developer margins, and it is solved jointly with the fixed costs (Section 4.2). Given the land share, the fixed cost is a residual determined parcel by parcel. No parameter is chosen to reproduce when parcels are redeveloped. Location premia and the location component of the land price are both measured at the municipal level. Within a municipality, the recovered fixed cost therefore varies with parcel and building characteristics, not with location.

Ignoring the markdown would overstate fixed costs. With a land share of one, the landowner keeps the whole option, so a given price implies a smaller option, which only a higher fixed cost rationalises. Holding prices and parameters fixed, setting $\bar{\theta}_m = 1$ in the inversion raises the aggregate fixed cost of the estimation sample by 35%. The fixed cost of the median parcel rises by 36%. The effect is large for two reasons. Landowners receive 81% of the developer value on average (Section 4.2). And the fixed cost is a residual on top of the construction cost, so a small change in the option that the price implies is a large change in the fixed cost that rationalises it. A model fitted to the rate of construction without a markdown would make the same attribution.

F_i is a property of parcel i: what redeveloping it would cost, whether or not it is ever redeveloped. A parcel priced at or below its standing value violates the lower bound of (16). Its building is worth more than any building the market expects on it, and it would not be redeveloped at any fixed cost. The price bound, not the absence of a sale, distinguishes it from a parcel with a high but finite fixed cost.

The absence of a sale adds a second bound. Let M_Y be the running maximum of the regional rent up to year Y. A parcel still unsold in Y, although its local rent reached $\phi_m M_Y$, has a threshold above

$\phi_m M_Y$. The threshold increases with the fixed cost (Proposition B.5), so the fixed cost is bounded below, $F_i \geq F_i^{\min}$ with $R_i^\star(F_i^{\min}) = \phi_m M_Y$. A price that implies $F_i < F_i^{\min}$ is inconsistent with the absence of a sale, and I leave the parcel out of the estimation sample. The bound uses rents up to $Y = 2024$, the year of the rents. By construction, no unsold parcel of the estimation sample is then ripe at the current rent.

3.8. Aggregation and margins

Redevelopment is irreversible, so the state variable for the extensive margin is the running maximum of the regional rent, $\mathcal{M}_t \equiv \max_{s \leq t} R_s$. Let $\Delta_u(\mathcal{M})$ be the net quality-adjusted floorspace, $\sum_i \phi_m (h_i - h_i^{\text{old}})$, delivered by parcels of initial use u whose threshold satisfies $R_i^\star \leq \phi_m \mathcal{M}$, each building at the density chosen at its start date. A parcel starts when the running maximum crosses its threshold and delivers ℓ_u years later. The quality-adjusted stock and the rent are

$$H_t = \bar{H} + \sum_u \Delta_u(\mathcal{M}_{t-\ell_u}), \qquad R_t = A_t H_t^{-\alpha}, \tag{17}$$

where $\bar{H}$ is the initial stock. Supply at t depends on running maxima dated $t - \ell_u$ or earlier, so it is predetermined, and the equilibrium rent follows without solving a fixed point.

Holding the demand path fixed, the number of parcels redeveloped by any date is weakly lower under the markdown than in the constrained-efficient allocation. If the running maximum of rents grows without bound, the gap converges to the floorspace of parcels blocked by concentration, and to zero if no parcel is blocked (Proposition B.6). Concentration therefore delays supply everywhere and caps it where standing housing is valuable. Its effect at a finite horizon mixes both, which is why Section 6 reports the time saved alongside the gain in supply.

Let $r_i \equiv R_i^\star / \phi_m$ be the regional rent at which parcel i is redeveloped, $G(r)$ the mass of parcels with $r_i \leq r$, and g its density. Parcel i adds $\Delta_i \equiv \phi_m [h_i(R_i^\star) - h_i^{\text{old}}]$ to the quality-adjusted stock. Abstracting from lags, the stock at running maximum r is $H(r) = \bar{H} + \int \mathbb{1}\{r_i \leq r\} \Delta_i \, di$, and the supply elasticity $\varepsilon = rH'(r)/H(r)$ satisfies

$$H'(r) = \underbrace{g(r)}_{\text{extensive margin}} \times \underbrace{\mathbb{E}[\Delta_i \mid r_i = r]}_{\text{intensive margin}}. \tag{18}$$

The first factor counts the parcels that cross their threshold at r. The second is the net floorspace each delivers.

Table 4 summarises where each wedge acts (Proposition B.7). The threshold rises with the fixed cost, the

standing value, the option premium, every tax, and the lag when $\delta < \rho/2$, as in the calibration. It falls with the land share on every parcel with a standing value. The density built at a given rent depends only on the lag, VAT, the property tax, and the cap. The time between start and delivery is the lag alone.

Table 4: Where each wedge acts

				Enters the threshold of the	
Wedge	Extensive margin	Intensive margin	Delay after start	Constrained-efficient allocation	First best
Frictions					
Fixed cost F_i	✓			✓	✓
Option value $\beta/(\beta-1)$	✓			✓	✓
Time to build ℓ_i	✓	✓	✓	✓	✓
Regulation					
Transfer tax, development tax	✓			✓	
VAT, property tax	✓	✓		✓	
Land-use regulation (cap $\bar{h}_i$)	✓	✓		✓	✓
Market structure					
Markdown $1-\bar{\theta}_m$	✓				

Source: Proposition B.7. A tick means the wedge enters the object. Every wedge that acts on the extensive margin also delays the date at which a parcel starts. The markdown acts only on parcels with a standing value (Proposition 1). A tick in the last two columns means the wedge enters the benchmark threshold of Definitions 1 and 2, so it is a cost relative to that benchmark. A wedge without a tick separates the private economy from the benchmark. Caps are held fixed in both benchmarks.

Where the cap binds for most redeveloping parcels, the intensive margin is fixed by the plan, and the supply elasticity is mostly a statement about how many parcels cross their threshold. Housing supply in a built-out city is then a question of whether to build at all, rather than how much. The markdown acts on the same margin as the fixed cost. Only the benchmark columns separate them: the fixed cost would constrain any planner, the markdown would not.

A higher cap never lowers the payoff π_i. It does not follow that it brings redevelopment forward. A higher cap raises the payoff only at rents above the saturation rent at which the cap starts to bind, and by more the higher the rent (Appendix B.1). It raises the value of waiting as well as the value of building. For a parcel with $a_i^\theta > 0$ that stays in the constrained regime, the threshold satisfies (Proposition B.8)

$$\frac{d\ln R_i^\star}{d\ln \bar{h}_i} = \varepsilon_i^K - \varepsilon_i^a, \qquad \varepsilon_i^K \equiv \frac{(1+1/\zeta)\,C_i + \tau^A \bar{h}_i}{F_i + C_i + \tau^A \bar{h}_i + (1+\tau^M)\tilde{V}_i^C/\bar{\theta}_m}, \qquad \varepsilon_i^a \equiv \frac{q_i}{q_i - \tilde{h}_i^{\text{old}}/\bar{\theta}_m}. \tag{19}$$

Since $\varepsilon_i^a \geq 1$, a higher cap brings redevelopment forward whenever $C_i/\zeta \leq F_i + (1+\tau^M)\tilde{V}_i^C/\bar{\theta}_m$. On a parcel without standing value, it delays redevelopment if and only if $C_i/\zeta > F_i$.

Construction cost rises more than proportionally with height. Where it is large relative to the fixed cost,

a taller allowance makes the landowner wait for a rent that justifies the taller building. The standing building works the other way. A higher cap raises the floorspace gained relative to the floorspace given up, which brings redevelopment forward. Which force dominates is an empirical matter, parcel by parcel (Section 6.2).

4. Calibration

The calibration has three steps. I fix the structural parameters from the literature or from aggregate moments. I estimate how the land share falls with local concentration and set its level to the operating margins of listed developers. I then recover the fixed cost parcel by parcel from the price of land. No parameter is chosen to match the timing of redevelopment, and the absence of a sale only bounds the fixed cost from below. Table 5 reports every parameter, and Appendix C gives the details.

4.1. Structural parameters

Market clearing gives $\log A_t = \log R_t + \alpha \log H_t$, which I build from the regional rent index and the quality-adjusted housing stock in the land tax records. I measure the volatility of the demand shifter on transaction prices, $\sigma = 3.3\%$, since the option to redevelop is written on the value of the building. The rent index, which moves with indexation and rent control, gives a lower bound of 0.64%. The drift is the long difference of the series plus $\sigma^2/2$, which gives $\mu = 1.5\%$.

Metropolitan rents respond to the metropolitan stock through two margins. When rents fall, households consume more floorspace, and households also move into the region. With idiosyncratic location preferences, the inverse demand elasticity satisfies (Lebret et al., 2026)

$$\frac{1}{\alpha} = s_H \, \varepsilon^L + \varepsilon^h, \tag{20}$$

where $s_H = 30\%$ is the median housing expenditure share, $\varepsilon^L = 3.4$ the elasticity of the population of the area to its real income (Ahlfeldt et al., 2020), and $\varepsilon^h = 0.90$ the household price elasticity of floorspace demand (Combes et al., 2019). This gives $\alpha = 0.52$, against 1.11 in a closed city. Every counterfactual is solved under the same simulated paths of the demand shifter, so a difference between two scenarios is a difference in policy, not in demand.

The yield of new units is $y = 2.6\%$, as reported by developers, applied to market rents net of charges and vacancy. The standing building is capitalised at $y^w = 5.5\%$. I choose it on the parcels bought by

a developer since 2023 and still in their pre-sale state. These parcels sold, so their price exceeded their standing value. With a single yield, 20.5% of them are priced below it. y^w is the smallest yield at which at most 10% are (9.8%, Appendix C.2). Shops and offices are capitalised at $y^C = 6.5\%$, the yield of secondary retail and offices. I set the perceived drift close to the drift of the simulated rent, $\delta = 1.5\%$, so the discount rate of the landowner, who holds the option, is $\rho = y + \delta = 4.1\%$. These rates give $\beta = 2.59$ and an option premium $\beta/(\beta - 1) = 1.63$. Since σ is small relative to δ, the premium reflects expected rent growth more than uncertainty (Lange and Teulings, 2024). The curvature of construction cost is $\zeta = 1.17$ (Combes et al., 2021), and the construction cost is $c_0 = 1,186$€ per square metre, from the survey of realised project costs. The construction lag ℓ_u is the time from the sale of the parcel to completion, by initial use (Table 3). It ranges from 4.6 to 6.7 years.

The density cap $\bar{h}_i$ is the 90th percentile of building height and coverage within each cadastral section, since no digitised cap exists for every parcel. Most new buildings stay below it. Standing housing floorspace is the habitable surface of dwellings in the land tax records, and the commercial rental flow is the rental value of business premises. The location premium ϕ_m is the ratio of the municipal rent to the regional median, with Paris divided into its 20 districts. Within a municipality, the model therefore treats parcels as identical in location and different in everything else. The two yields are regional. Yields are lower in central Paris, so the inversion attributes part of the location value of Paris parcels to their fixed cost. This bears on where a transfer for the fixed cost would go.

4.2. The land share

By (9), the land share of a municipality depends on concentration only through its Herfindahl index, $\bar{\theta}_m = \theta^{\text{atom}}/(1 - \varsigma\, HHI_m)$, where $\theta^{\text{atom}} = \eta^D/(1 + \eta^D)$ is the land share with atomistic developers and $\varsigma \equiv -(1/\eta^S - 1/\eta^D)\theta^{\text{atom}}$ the slope. I compute each group's share of land purchases from redevelopment permits over a rolling three-year window, after consolidating project companies into their parent group. The consolidation is still incomplete, so measured indices are too low.

I estimate the slope on 1,631 sales of housing parcels followed by a permit on the same parcel. At such a sale the parcel is ripe, so its price is $\bar{\theta}_m V_i^D$, and the ratio of the price to the developer value is the land share, up to the fixed cost. The estimating equation is

$$\ln \frac{x_i}{g_i} = \gamma_{d(i)} + \gamma_{t(i)} + b_T \ln T_i + b\, HHI_{m(i)t(i)} + u_i, \tag{21}$$

where x_i is the price per square metre of land, g_i the developer value per square metre at a zero fixed cost, T_i the parcel area, and γ_d and γ_t département and year effects. Concentration may respond to local land values, so I instrument the municipal index with the average index of the other municipalities of the same group of municipalities, across which developer groups operate. The slope is imprecise: $\varsigma = -0.29$, with a 95% confidence interval of $[-1.38, 0.50]$. I calibrate the model at the point estimate and report the main results at the lower end of the interval, $\varsigma = -1.38$, the largest markdown the data allow, and at $\varsigma = 0$.

I set θ^{atom} so that the median model-implied operating margin matches the median margin of listed French developers, jointly with the fixed costs. Both elasticities then follow in closed form:

$$\frac{1}{\eta^D} = \frac{1}{\theta^{\text{atom}}} - 1, \qquad \frac{1}{\eta^S} = \frac{1}{\eta^D} - \frac{\varsigma}{\theta^{\text{atom}}}. \tag{22}$$

The margin target gives $\theta^{\text{atom}} = 90.3\%$, and the median model margin is 8.0%, against 8.0% for listed developers. The calibration gives $\eta^D = 9.30$ and $\eta^S = 2.31$. At observed concentration, landowners receive 81% of the developer value on average, between 72% and 86% at the 10th and 90th percentiles across municipalities.

4.3. Taxes

VAT is $\tau^V = 20\%$, with input VAT deductible. The transfer tax τ^M combines the departmental duty, the municipal tax, and a state levy, 6.32% in total. The development tax τ^A is the statutory value per square metre in the Paris region times the sum of the municipal, departmental, and regional rates. The property tax rate t_m^F is the municipal rate, levied on a base share $\psi^{\text{new}} = 0.25$ of the rental value, and new buildings are exempt for $e = 2$ years.

5. Baseline results

5.1. The recovered fixed costs

I recover the fixed cost of every parcel whose price lies strictly within the bounds (16), by bisection between zero and 100,000 euros per square metre of parcel area. The model rationalises 63.1% of observed prices, which is the main fit statistic of the inversion, and these 314,271 parcels form the estimation sample (Appendix Table A1). Of these, 3,782 were bought by a developer by 2025. The 310,489 others form the unsold stock, on which every counterfactual is computed. The other parcels fall into four groups.

Table 5: Calibration parameters

Par.	Name	Source	Value
μ	Drift of the demand shifter (%)	Rounded long difference (1.54)	1.5
σ	Volatility of the demand shifter (%)	Transaction price index	3.3
s_H	Housing expenditure share (%)	Median share of income spent on housing	30
ε^L	Elasticity of population to real income	Ahlfeldt et al. (2020)	3.4
ε^h	Household price elasticity of floorspace demand	Combes et al. (2019)	0.90
α	Inverse demand elasticity, open city	Equation (20)	0.52
y	Net yield of new units (%)	Developer reports	2.6
y^w	Net yield of the standing building (%)	Recent developer purchases	5.5
	Rent net of charges and vacancy (% of rent, charges included)	Rent index, household survey	90
y^C	Commercial yield (%)	Secondary retail and offices	6.5
δ	Perceived drift of rents (%)	Drift of the simulated rent	1.5
ρ	Discount rate (%)	$y + \delta$	4.1
β	Root of the value of waiting	Equation (10)	2.59
ζ	Curvature of construction cost	Combes et al. (2021)	1.17
c_0	Construction cost (€ per m^2)	Cost survey	1,186
ℓ_u	Construction lag, by initial use (years)	Table 3	4.6–6.7
ϕ_m	Location premium (P10–P90)	Rent index	0.90–1.54
ς	Slope of the land share (point estimate)	Sales followed by a permit	-0.29
η^D	Elasticity across developers	Slope ς, developer margins	9.30
η^S	Elasticity of selling	Slope ς, developer margins	2.31
HHI_m	Municipal Herfindahl index (median)	Permit register	0.333
$\bar{\theta}_m$	Land share, mean (%)	Equation (9)	81
θ^{atom}	Land share with many small developers (%)	Developer margins	90.3
τ^V	VAT on new housing	Tax code	20%
τ^M	Transfer tax rate (%)	Tax code	6.32
τ^A	Development tax (€/m^2)	Planning code	88
t_m^F	Municipal property tax rate, median (P10–P90, %)	Local tax rates	34.1 (20.5–43.5)
ψ^{new}, e	Property tax base share, exemption (years)	Tax code	0.25, 2

13.2% of parcels are priced at or below their standing value, so their building is worth more than any building the market expects on them. Some carry a cap that makes the option worthless. 4.3% are priced as if their fixed cost were so low that they would have sold by 2024, which they did not (Section 3.7). And 21.7% of the parcels priced above their standing value are priced above the value of redevelopment at a zero fixed cost, which measurement error in the hedonic price or a value the model omits, such as the option to assemble neighbouring parcels, can explain. I exclude the latter rather than give them a negative cost, and Appendix A.3 describes every group. The estimation sample is less built than the average parcel. Its median standing floor-area ratio is 0.30, against 0.39 over all parcels. It contains almost no parcel of Paris itself, whose prices lie above the zero-cost bound or below the standing value (Appendix Table A2), so every counterfactual concerns the inner suburbs. Since the markdown bites where the standing value is high, its measured effect on this sample is small partly by construction. Empty parcels, conversely, are well represented, with 15,123 parcels.

The fixed cost is a residual. It contains the physical cost of clearing the site: demolition, decontamination, and ground works. It also contains every cost of redevelopment that the model does not represent and that lowers the price a parcel commands: co-ownership, which requires every co-owner to agree to the

sale, tenant protection, heritage rules, the premium needed to assemble contiguous parcels, the risk of a refused permit, and errors in the computed developer value. The hedonic price uses municipal effects, so within a municipality the fixed cost varies with parcel and building characteristics, not with location. The median fixed cost is 3,157 euros per square metre of land, or 2,143 euros per square metre of floorspace at the cap, 1.81 times the construction cost of a square metre of floorspace. Predicted demolition costs account for a small part of it (Appendix H).

5.2. Validation

The calibration inverts land prices and targets developer margins, and no parameter is fit to a timing moment or an aggregate quantity, so every check below is out of sample (Table I4).

I first check the timing the thresholds predict. For this test, I invert the fixed costs again with the non-exercise bound at 2019, using rents up to 2019 only, so the thresholds use no information on the sales of 2019–2024. A parcel is predicted to start in 2019–2024 if the running maximum of its local rent crosses its threshold in that window. The model predicts a start for 2.6% of parcels. Among them, 0.6% are bought by a developer in the window, against a base rate of 0.7% over all parcels. The thresholds do not predict purchases within the window, with characteristics measured in 2025 and location measured by municipality.

The model predicts the order in which parcels are redeveloped better than the rate. No unsold parcel is ripe at the current rent, by the non-exercise bound, so supply does not jump at the start. The private equilibrium starts 1.13 million square metres of new floorspace a year on the unsold stock over its first ten years. The permit register records 0.32 million a year over 2014–2024 on the parcels of the estimation sample, and 0.80 million on all parcels of the base data, counting only permits with a recorded developer. The gap lies on the extensive margin. The model starts 1.0% of the unsold stock each year, against a purchase rate of 0.11% a year on the estimation sample. I read it as an unmodelled hazard of permit approval and land assembly, which the model does not represent. A hazard common to all parcels scales every scenario in about the same proportion, so the decomposition and the ranking of policies, which are ratios, are little affected. Levels are reported relative to a baseline that shares the same gap.

Chapelle et al. (2023) estimate supply elasticities from long differences over 1999–2019 with a shift-share instrument for local demand. Their preferred value for France is 0.5. For the Paris area, the relevant comparison here, it is 0.35. In the model analogue, two economies face the same demand draws, one with the demand shifter permanently 10% higher. After 20 years, the ratio of the difference in the log

housing stock to the difference in the log rent is 0.39. Only unsold parcels of the estimation sample can add floorspace in the model, so the two elasticities are not measured on the same base.

The perceived drift is consistent if it is close to the growth of rents the model delivers. Along the simulated private equilibrium, the regional rent grows at 1.5% a year over twenty years, and at $\mu - \alpha g_H = 1.2\%$ on the observed growth of the quality-adjusted stock, against a perceived drift of $\delta = 1.5\%$. The perceived drift is therefore consistent with the growth of rents the model delivers, and above the observed growth. Appendix Table F5 reports the results with δ half a point higher and lower.

5.3. Decomposing the supply shortfall

Let H^{private} be the quality-adjusted stock at the 20-year horizon in the private equilibrium, and H^0 the stock when every wedge is switched off, under the same demand paths. The wedges are $\mathcal{K} = \{\theta, \beta, \ell, F, \tau, \bar{h}\}$: the markdown (land share $\bar{\theta}_m \to 1$), the option value (exercise at the static break-even rent, $\beta/(\beta-1) \to 1$), time to build ($\ell_u \to 0$), the fixed cost ($F_i \to 0$), taxes (all rates set to zero), and the density cap (lifted to 1.5 times the 99th percentile of the inferred caps, a floor-area ratio of 12.47). For a subset $\mathcal{J} \subseteq \mathcal{K}$, let $H(\mathcal{J})$ be the stock once every wedge in $\mathcal{J}$ is switched off. With $K = 6$ wedges, the Shapley value of wedge k is

$$\varphi_k = \sum_{\mathcal{J} \subseteq \mathcal{K} \setminus \{k\}} \frac{|\mathcal{J}|!\,(K - |\mathcal{J}| - 1)!}{K!} \left[H(\mathcal{J} \cup \{k\}) - H(\mathcal{J}) \right], \tag{23}$$

the average marginal contribution of k over every order in which the wedges could be removed. The values sum exactly to $H^0 - H^{\text{private}}$. The decomposition is an accounting of thresholds, not a set of counterfactuals. No planner can remove the option value or the construction lag, and both are costs relative to every benchmark (Table 4). The groups of Table 6 therefore separate the wedges that a policy can move, the markdown, taxes, and caps, from the frictions that it can only offset with transfers. 0.9% of the shortfall comes from the markdown, 53.8% from frictions, and 45.3% from taxes and land-use regulations (Table 6). The fixed cost alone accounts for 30.9%. Without any wedge, quality-adjusted new floorspace at twenty years is 22.5 times that of the private equilibrium, so these shares divide a very large gap. Removed alone from the private equilibrium, the fixed cost is the largest wedge (31.2% of the gap), ahead of the option value (Table 6) and taxes (9.3%). Removing the fixed cost alone adds 6.7 times the new floorspace of the private equilibrium. Removed alone, caps lower the stock at twenty years (-3.6% of the gap), because a taller allowance also raises the value of waiting (Section 6.2). Their

Shapley share comes from their interactions with the other wedges.

Table 6: Shapley decomposition of the supply shortfall

Wedge	Group	Shapley share of the gap (%)	Removed alone (% of the gap)
Markdown ($\bar{\theta}_m \to 1$)	Market failure	0.9	0.8
Option value	Friction	18.2	15.3
Time to build	Friction	4.6	4.7
Fixed cost	Friction	30.9	31.2
Taxes	Policy	21.2	9.3
Density cap	Policy	24.1	-3.6
Market failure	Total	0.9	0.8
Frictions	Total	53.8	41.5
Policy wedges	Total	45.3	10.4

Data: model estimation. Each cell is the Shapley value φ_k of (23), as a share of $H^0 - H^{\text{private}}$ at the 20-year horizon, computed over the $2^6 = 64$ subsets of wedges. Group rows: the sum of the Shapley values of the wedges of the group, and, in the last column, the gain from removing the wedges of the group jointly, which is not the sum of the gains from removing each.

Parcel by parcel, frictions alone hold back 95% of the unsold parcels that would be ripe at the current rent without any wedge, and most of the others are held back by several wedges at once. Almost none is held back by the markdown alone or by taxes and caps alone (Table 7). On a parcel with a standing building, the markdown delays the expected start date by 2.4 years at the median.[12]

The markdown also acts on parcels outside the estimation sample. Across all unsold parcels, 1,392 (0.3%) are blocked by concentration for ever. Their private option is worthless, so their price reveals no fixed cost, and they are neither in the decomposition nor in the unsold stock of the counterfactuals. The two numbers belong together. The markdown accounts for 0.9% of the shortfall on the estimable stock, and it blocks 0.3% of all unsold parcels indefinitely.

Several choices move the measured effect of the markdown, in opposite directions (Appendix Table F4). The attribution of the whole developer margin to the markdown makes it larger. Incomplete consolidation of project companies, the exclusion of blocked parcels, and an estimation sample with little standing value make it smaller. The slope is imprecise. At the lower bound of its confidence interval, the markdown accounts for 1.8% of the shortfall.

[12] Under the perceived process, if $\delta > \sigma^2/2$, the expected time for the rent to rise from R to a threshold $R^\star$ is $\ln(R^\star/R)/(\delta - \sigma^2/2)$. The expected delay due to the markdown on parcel i is therefore $d_i = \ln(R_i^\star/R_i^{\text{CE}})/(\delta - \sigma^2/2)$, which does not depend on the current rent. It is a partial-equilibrium measure. The counterfactuals report its general-equilibrium counterpart, the time saved by correcting the markdown.

Table 7: Which unsold parcels are held back by what

Row	Parcels	Share of the unsold stock (%)	Share of potential floorspace (%)	Median delay due to the markdown (y)
Not ripe without wedges	150	0.0	0.1	10.8
Held back by markdown only	0	0.0	0.0	–
Held back by frictions only	295,537	95.2	95.4	2.3
Held back by taxes+caps only	0	0.0	0.0	–
Held back by several wedges	14,802	4.8	4.6	5.4
Ripe at the current rent	0	0.0	0.0	–
Memo: blocked by concentration, all unsold parcels	1,392	0.3	–	–

Data: model estimation. Unsold parcels of the estimation sample. Rows: not ripe without wedges, held back by the markdown only, by frictions only, by taxes and caps only, by several wedges, ripe at the current rent. Columns: parcel count, share of the unsold stock, share of the floorspace these parcels would add without any wedge, median delay due to the markdown on parcels with a standing building. Memo row: parcels blocked by concentration for ever, among all unsold parcels.

5.4. Misallocation of redevelopment

The markdown delays redevelopment and changes the order in which parcels are redeveloped, but the misallocation it causes is small in value. Its distortion relative to the constrained-efficient threshold is zero on empty parcels. On built parcels, its median is 0.03 and its interquartile range is [0.02, 0.06], so the median built parcel starts when the regional rent reaches $1 + 0.03$ times its constrained-efficient threshold. This dispersion reorders parcels at the margin. The rank correlation between private and constrained-efficient thresholds is 1.00, yet over twenty years 24% of new floorspace is built on parcels that the constrained-efficient allocation would not have converted to deliver the same physical quantity (Table 8). The two statements are consistent. A high rank correlation leaves room for many swaps between parcels whose thresholds are close, and such swaps move a large quantity of floorspace without moving much value.

Two measures of value confirm that these swaps cost little. Following Hsieh and Klenow (2009), I hold the physical floorspace delivered privately fixed and reallocate it to the parcels that the constrained-efficient allocation converts first, and the allocation component of the supply gap is negligible (Table 8). Holding output and timing fixed, converting parcels in the constrained-efficient order would also barely change the resource cost of the parcels converted, $\sum_i (F_i + C_i + V_i^w(R_{i0}))$. Both measures are close to zero because the markdown distortion is, if anything, larger where the location premium is low (their correlation is -0.11), so the private economy builds slightly more in high-premium municipalities. The average location premium of new floorspace is 1.07 in the private equilibrium and 1.07 in the constrained-efficient allocation. The location premium is measured by municipality, so these measures miss misallocation within municipalities. Within the estimable stock, the welfare cost of the markdown is therefore mostly a cost of delay, not of misallocation.

Table 8: Misallocation of redevelopment

Wedge	Med. wedge	IQR	Corr. with ϕ_m	Rank corr. of thresholds	Floorspace built out of order (%)	Allocation share of the gap (%)	ϕ of new floorspace, with wedge	ϕ of new floorspace, without
Markdown	0.03	[0.02, 0.06]	-0.11	1.00	23.51	-1.27	1.07	1.07
Taxes	0.40	[0.36, 0.45]	-0.27	0.99	25.13	-0.68	1.07	1.06
Density cap	-0.25	[-0.31, -0.14]	-0.05	0.94	76.74	–	1.07	1.13

Data: model estimation. Rows: the markdown (relative to the constrained-efficient allocation), taxes (relative to the first best), the density cap (relative to the uncapped allocation). Columns: median and interquartile range of the distortion (built parcels for the markdown, all parcels for taxes and the cap; a negative distortion means that the tax or the cap brings the threshold forward), correlation of the distortion with the location premium ϕ_m, rank correlation between private and benchmark thresholds, share of new floorspace built by year 20 on parcels outside the benchmark order, allocation share of the supply gap, and mean location premium of new floorspace with and without the distortion.

5.5. Threshold distribution

Figure 3 plots, by initial use, the distribution of the threshold $R_i^\star$ relative to the current local rent R_{i0}. For parcels with a standing building, Appendix Figure J5 plots the ratio of the private to the constrained-efficient threshold, which rises with the standing value of the parcel.

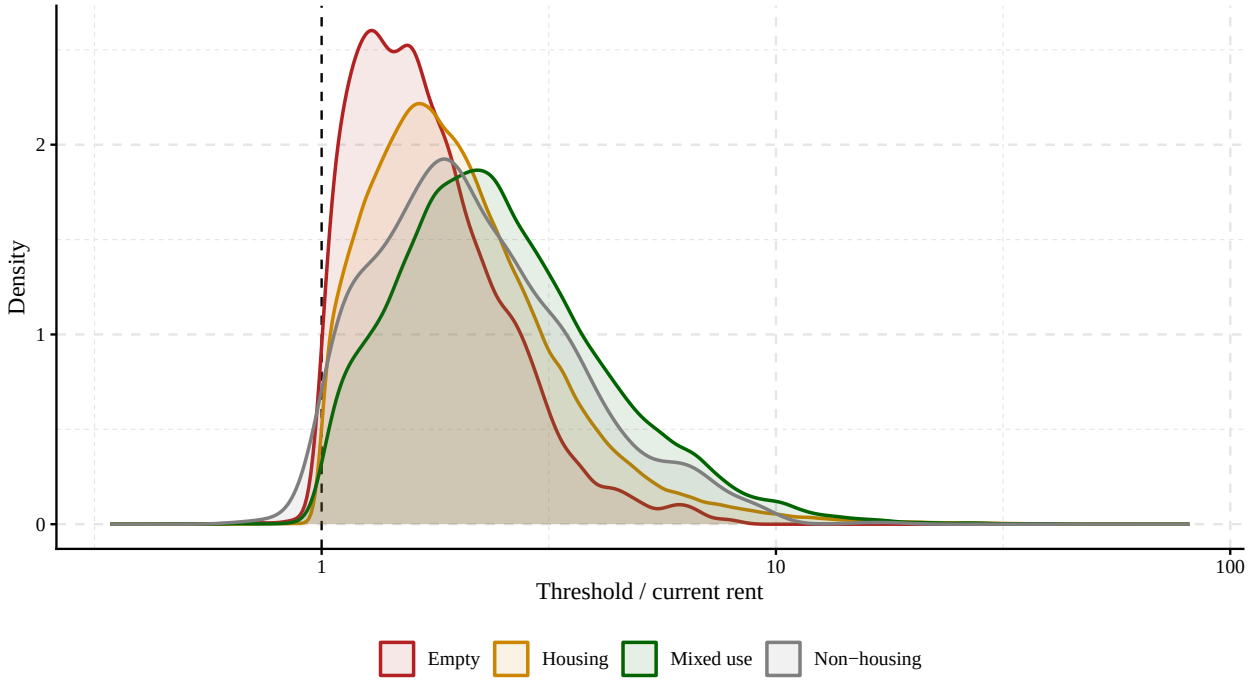


Figure 3: Redevelopment threshold relative to the current rent

Data: model estimation. Distribution, by initial use, of $R_i^\star / R_{i0}$ over the estimation sample. Horizontal axis in logs.

6. Counterfactual policy analysis

The counterfactuals separate two objectives that supply policy often conflates, lowering rents and restoring efficiency. I ask three questions. Does relaxing land-use regulations raise supply? How far can supply policy lower rents without costing the budget? And which policy removes the inefficiency that

the markdown creates? Section 6.2 lifts the density caps. Section 6.3 searches for the combination of instruments that lowers rents most. Section 6.4 searches for the combination that maximises welfare, and isolates the instrument that restores efficiency. Appendix E evaluates every instrument alone: lower taxes on redevelopment, a transfer to developers for the fixed cost, a subsidy to sellers, and a tax on standing value. Table I5 reports all of them side by side.

6.1. Outcomes and welfare accounting

Every counterfactual is solved under the same demand paths, on the 310,489 unsold parcels of the estimation sample (Section 5.1). I report it at a 20-year horizon $\bar{t}$, and Appendix Table F1 reports the 50-year horizon. Every policy is reported with four outcomes.

Stock and rents The change in the quality-adjusted stock dH and in the market-clearing rent dR after $\bar{t}$ years, relative to the private equilibrium, in general equilibrium. Every scenario faces the same demand path, so $R_t = A_t H_t^{-\alpha}$ gives $dR \approx -\alpha\, dH$. I also report the additional physical floorspace delivered by $\bar{t}$.

Net cost Let $B^\pi(\bar{t})$ be the expected present value, at the government's discount rate $r_g = 3.0\%$, of the fiscal revenue of the projects started by $\bar{t}$ under policy π, counted until $\bar{t}$, net of the outlays of the policy:

$$B^\pi(\bar{t}) = \sum_i \mathbb{E}\left[e^{-r_g t_i^\pi}\, \mathbb{1}\{t_i^\pi \le \bar{t}\} \left(\Phi_i(\bar{t} - t_i^\pi) - s_i^\pi \right) \right], \tag{24}$$

where t_i^π is the start date of parcel i, s_i^π the outlay it receives, and $\Phi_i(x)$ the revenue of the taxes the project pays in its first x years, net of the property tax the standing building stops paying (Appendix E.1). The net cost is $NC^\pi(\bar{t}) = B^0(\bar{t}) - B^\pi(\bar{t})$. Throughout the paper, a net cost is positive when the policy costs the budget, and the balance of a group, the government included, is positive when the group gains. A policy is self-financing if $NC^\pi(\bar{t}) \le 0$, which by Proposition 2(iii) is sufficient for it to raise welfare. I divide the net cost by the additional floorspace, in units of 60 square metres of living area (69 square metres of floorspace). Counting revenue until $\bar{t}$ favours policies that bring projects forward into the window. A project moved from year 25 to year 15 enters with its full revenue. The 50-year accounts of Appendix Table F1 correct this.

Time saved Let $\hat{H}^0(t)$ and $\hat{H}^\pi(t)$ be the mean quality-adjusted stock at date t without and with policy π. The time saved is

$$\Delta t^\pi \;=\; \bar{t} - \min\{t:\; \hat{H}^\pi(t) \;\geq\; \hat{H}^0(\bar{t})\}. \tag{25}$$

A policy that reaches in twelve years the stock the baseline reaches in twenty saves eight years. The measure separates a policy that builds more from a policy that builds the same earlier.

Who gains A policy moves value between five groups: renters, incumbent landlords, selling landowners, developers, and the government. In present value by year $\bar{t}$, relative to the private equilibrium,

$$dW = \underbrace{\int_0^{\bar{t}} e^{-r_g t}\,\mathbb{E}\left[-dR_t\,H_t^0 - \tfrac{1}{2}\,dR_t\,dH_t\right]dt}_{\text{renters}} + \underbrace{\int_0^{\bar{t}} e^{-r_g t}\,\mathbb{E}\left[dR_t\,H_t^0\right]dt}_{\text{incumbent landlords}} + \underbrace{d\Pi^O}_{\text{selling landowners}} + \underbrace{d\Pi^D}_{\text{developers}} + \underbrace{dG}_{\text{government}}. \tag{26}$$

H_t^0 is the stock of the private equilibrium, and dR_t and dH_t are the changes in the rent and in the stock, in level. $d\Pi^O$ is the change in the value of the unsold parcels to their landowners, $d\Pi^D$ the change in developer rents $(1+\tau^M)(1-\bar{\theta}_m)V_i^D$, and dG the change in the government's receipts net of outlays, by $\bar{t}$ and after it, at fair value. Both private values include the value at $\bar{t}$ of the parcels not yet sold, so a policy that only brings a sale forward is credited with the gain from selling earlier, not with the whole sale. These values capitalise every later tax and subsidy. The government term values the same flows at fair value (Lucas, 2012). Each public flow is discounted at the rate at which the private side values it. Taxes and outlays at the start or at delivery, and the property tax of the new building, are discounted at ρ, as the developer value V^D. Taxes on standing buildings are discounted at $y^w + \delta$, as the standing value V^w, and taxes on commercial flows at y^C. The government borrows at $r_g < \rho$, but a lower borrowing rate does not make risky receipts worth more. Rent risk is aggregate, so spreading it across taxpayers does not remove it (Arrow and Lind, 1970; Lucas, 2014). A transfer between the government and landowners or developers therefore nets out, and welfare changes only through redevelopment: its timing, the floorspace it adds, and the surplus on the additional housing. The net cost (24) remains a cash measure at r_g, which is how a ministry judges whether a policy pays for itself. The rent change on the existing stock appears twice with opposite signs. It is a transfer. The surplus on the additional housing, $-\frac{1}{2}\,dR_t\,dH_t$, is the efficiency gain on the demand side. By (20), a share $s_H\varepsilon^L\alpha = 53\%$ of additional floorspace, and of this surplus, goes to new residents. In a fully open city rents do not fall and the whole gain of new supply is capitalised into land (Roback, 1982; Kline and Moretti, 2014). In a closed city

renters gain most. I report the renters' share at the calibrated α and in a closed city, its upper bound.

6.2. Relaxing land-use regulations

The first policy removes the density cap. Every cap is lifted to 1.5 times the 99th percentile of the inferred caps, a floor-area ratio of 12.47. The inferred envelope has no support beyond its 99th percentile, so this bound extrapolates it by half. Lifting the cap without any bound is well posed only if $\beta > \zeta + 1$, since otherwise the value of waiting for a taller building diverges. The calibration satisfies it, with $\beta = 2.59$ against $\zeta + 1 = 2.17$. The cap binds at the baseline rent for 74% of parcels. As a milder reform, I raise every cap to the median floor-area ratio across municipalities, in municipalities whose median cap is below it.

Lifting the caps changes the stock by -4.29% after twenty years and by 6.72% after fifty (Table 9 and Appendix Table F1). The reform involves no outlay, and its fiscal effect comes only from the taxes of the projects it brings forward or delays, -21.3 billion euros by year twenty.

At twenty years, the milder reform does more. It raises the stock by 5.07% and saves 6.4 years. At fifty years, lifting the caps turns from a loss into a gain, 6.72%, but the milder reform still does more, 9.63%. Proposition B.8 explains why. A taller allowance raises the value of waiting as well as the value of building. Redevelopment is a single choice of density, so a landowner who builds now gives up the option to build taller later. Below the cap, the surplus grows like $R^{\zeta+1}$, and the discount factor to the date the rent reaches R falls like $R^{-\beta}$. The closer β is to $\zeta + 1$, the longer landowners wait for a taller building. Lifting caps raises the threshold of 88% of the parcels it reaches. These parcels are redeveloped later and denser. Removing the cap is therefore a long-run supply policy, and in the first twenty years part of its effect is a delay. This is the intensity effect of Capozza and Li (1994).

Relaxing caps does not correct the markdown. On a built parcel, a higher cap raises the floorspace gained relative to the floorspace given up, which lowers the markdown distortion (15) at the new cap. Measured against the constrained-efficient thresholds of the status quo, however, the delay dominates, and the distortion on built parcels rises (Appendix Table E7).

Table 9: Relaxing land-use regulations

Scenario	dH (%)	dR (%)	Net cost (m€)	Net cost/unit (k€)	Time saved	Add. floorspace (m m2)	Loc. premium
Density: caps raised to the median	5.07	-2.60	-20,270.34	-77.88	6.42	17.95	1.02
Density: caps lifted	-4.29	2.40	21,313.87	–	-8.75	-13.91	1.12

Data: model estimation. Columns: change in the quality-adjusted stock and in the rent after 20 years, in percent of the private equilibrium, net cost to the consolidated budget by year 20, total and per additional unit of 60 square metres of living area (negative: a gain), years saved to reach the baseline stock of year 20, equation (25), additional physical floorspace by year 20, and mean location premium of new floorspace.

6.3. How far can supply policy lower rents?

Single instruments do little on their own. As designed in Appendix E, the self-financing transfer for the fixed cost and the reduced VAT rate on new housing lower market rents by 1.58% and 1.56% on average over twenty years, the rent metric of the mixes below. Their effects interact. A higher cap raises the base of a tax on under-used land, and a lower development tax raises the value of a transfer for the fixed cost. I therefore search for the combination of instruments that lowers rents the most under a balanced budget.

A mix x sets 10 instruments: a floor on the density cap in floor-area ratio, $\bar{h}_i \geq x^{\text{far}} T_i$, multiples of the development tax and of the property tax, the length of the exemption, the transfer tax, the VAT on new housing, the tax on under-used land τ^U, the transfer for the fixed cost $\bar{s}$, the inclusionary-zoning share ι (Appendix F.3), and the subsidy to sellers. The subsidy to sellers pays a share $z \in [0,1]$ of the rate $1/\bar{\theta}_m - 1$. The tax on under-used land is levied at a rate τ^U on the forgone rent of the gap $(\chi \bar{h}_i - h_i^{\text{old}})^+$, with $\chi = 0.5$ (Appendix F.2). Each instrument lies in a box $\mathcal{X}$ (Appendix Table G2). Not every bound is statutory. VAT and the development tax cannot rise above their current levels, but the transfer tax can rise to 10%, above its current 6.32%, and the property tax to twice its current level, so the box lets the two taxes on land exceed what current law allows. The tax on under-used land is at most 10% of the forgone rent, and the floor on the cap is at most the bound of Section 6.2. The government solves

$$\min_{x \in \mathcal{X}} \frac{\int_0^{\bar{t}} \mathbb{E}\left[R_t^{x,\text{paid}}\right] dt}{\int_0^{\bar{t}} \mathbb{E}\left[R_t^0\right] dt} - 1 \qquad \text{s.t.} \qquad NC^x(\bar{t}) \leq 0, \tag{27}$$

where R_t^0 is the rent of the private equilibrium and $R_t^{x,\text{paid}} = R_t^x\left[1 - (1-v)\,\iota\,(H_t^x - \bar{H})/H_t^x\right]$ is the average rent paid when a share ι of new floorspace lets at v times the market rent. The net cost adds to (24) the revenue of the tax on under-used land and the change in the property tax of standing buildings. A derivative-free search solves the problem (Appendix G.1).

The best rent-minimising mix lowers market rents by 14.02% on average over twenty years, and by 17.76% at year twenty. It raises the quality-adjusted stock by 45.58% and saves 14.6 years (Table 10). Alone, at its value in the mix and with the budget ignored, the best instrument, the fixed-cost subsidy, lowers the average rent over twenty years by 7.13%. The instruments are complements. The mix beats the sum of their solo effects by 5.43 points (Figure 4).

The mix removes most of the taxes that fall on the new building. VAT falls slightly, to 18.7%, and the

development tax and the property tax go to zero. The mix finances these cuts with the two taxes on land that the box allows, the transfer tax, raised to 9.7%, and the tax on under-used land, close to its upper bound. Within twenty years, the transfer tax collects revenue on the many projects that the other instruments bring forward, which is why a mix that minimises rents raises the tax that public finance ranks lowest. The mix also raises the floor on the cap to a floor-area ratio of 3.23. 3 of the 10 instruments end at a natural limit, and trade-offs within the box set the others. No instrument ends at a bound that could be widened, so the mix without bounds is the main mix (Appendix G.2).

Lower rents are mostly a transfer. On the existing stock, they move 120.5 billion euros from incumbent landlords to renters by year twenty, against a surplus of 28.8 billion on the additional housing. Selling landowners gain 200.3 billion euros and developers gain 58.0 billion, the part of the gain capitalised into land. Of the gains of renters, selling landowners, and developers, renters receive 37%, and at most 57% in a closed city. Appendix Table G3 reports the fiscal accounts.

The mix does not restore efficiency. It lowers thresholds where frictions and taxes bite, on built and empty parcels alike, and almost every project it induces starts below its constrained-efficient threshold (Appendix Table E7). The subsidy to sellers, the only instrument of the box that acts on the markdown, pays 12.5% of its full rate. It is paid on every sale, including sales that would happen anyway, and the objective values its effect on rents only.

6.4. Restoring efficiency

By Proposition 3, a policy restores the constrained-efficient allocation if and only if it raises what landowners receive relative to the standing value they give up by the factor $1/\bar{\theta}_m$. Two instruments do so: a subsidy to sellers at rate $1/\bar{\theta}_m - 1$ on the land price, and a tax at rate $1 - \bar{\theta}_m$ on the standing flows of a parcel until it is redeveloped. Both rates depend only on the local Herfindahl index and the two elasticities. The rent-minimising mix uses neither, because its objective counts rents only.

I therefore solve the same problem with welfare as the objective:

$$\max_{x \in \mathcal{X}^w} dW^x \qquad \text{s.t.} \qquad NC^x(\bar{t}) \leq 0, \tag{28}$$

where dW^x is the welfare change (26), with the government term weighted by $1 + \lambda_g$ and $\lambda_g = 0$. The transfer from landlords to renters nets out, and so does any transfer between the government and the private side, since public flows are valued at fair value. The objective values the surplus on additional

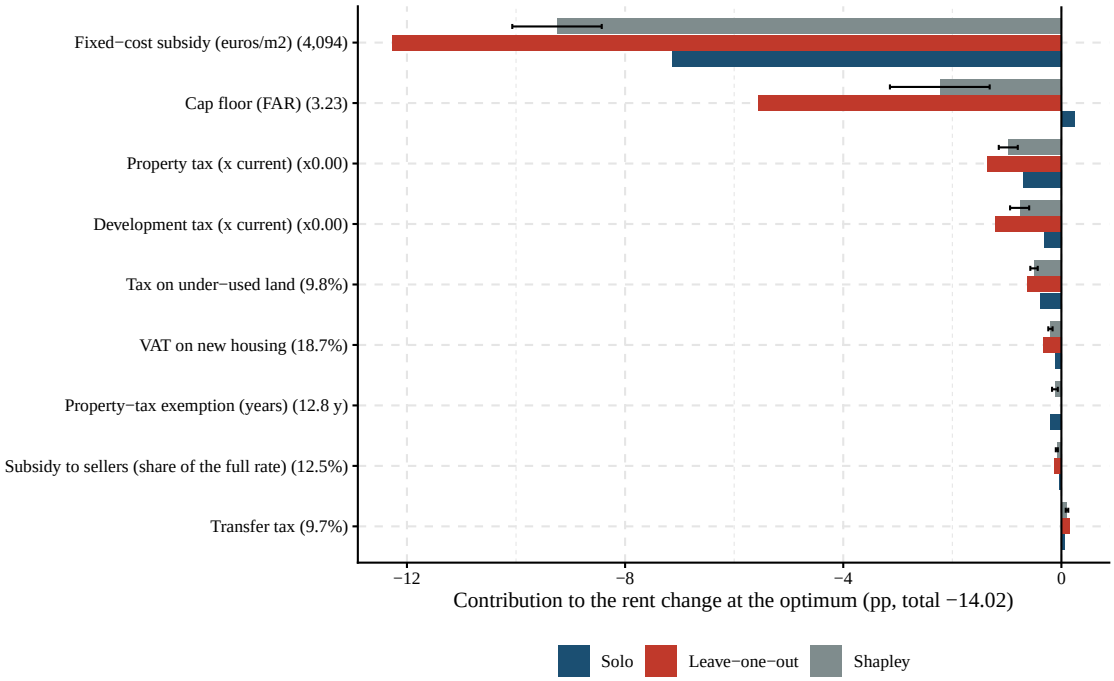


Figure 4: Contribution of each instrument to the rent decrease

Data: model estimation. Solo: the instrument alone at its optimal value. Leave-one-out: the optimum minus the optimum without the instrument. Shapley: average marginal contribution over every order of the instruments, which sums to the total. Budget ignored.

housing and the gains of selling landowners, developers, and the government, so it rewards an instrument only through the redevelopment it induces. The budget constraint stays the cash balance (24). The box $\mathcal{X}^w$ adds one instrument to $\mathcal{X}$: the tax on standing value, levied on parcels whose standing housing floorspace is below their cap, at a share $t \in [0,1]$ of its efficient rate $1-\bar{\theta}_m$. Along (14), a seller subsidy z and a tax $t = 1 - z$ restore efficiency together.

The welfare-maximising mix raises welfare by 414.5 billion euros by year twenty, against 202.1 billion for the rent-minimising mix. It changes the stock by -0.88% at year twenty and leaves market rents slightly higher (Table 10). It sets the tax on standing value at 29% of its efficient rate, 5% of the standing flow on average over the parcels of its base, and pays 0.0% of the seller-subsidy rate. The two mixes differ in the following instruments: the floor on the density cap, the development tax, the property tax, the property-tax exemption, the transfer tax, the VAT on new housing, the tax on under-used land, the fixed-cost subsidy, the subsidy to sellers and the tax on standing value. Table 10 compares them.

The tax on standing value alone isolates what correcting the markdown buys. At its efficient rate, 17% of the standing flow on average over the parcels of its base, it implements the constrained-efficient threshold on every parcel of its base, and leaving parcels at their cap out of the base changes no efficient

threshold (Proposition 3(ii)). It raises the stock by 0.96% after twenty years and lowers rents by 0.50%. It brings the stock of year twenty forward by 1.9 years, whereas lifting the caps delays it. The tax raises 23.3 billion euros by year twenty, and the consolidated balance improves by 28.4 billion, because the induced projects pay taxes. The capitalised value of unsold parcels to their landowners falls by 18.3 billion euros today (Appendix Table E6). Valued at year twenty, as in the welfare accounts, the loss of selling landowners is larger (Appendix Table E8). A subsidy to sellers implements the same thresholds but costs 158.5 thousand euros per additional unit, since 86% of its outlay goes to parcels that sell within twenty years anyway (Appendix E.4).

The tax is large, since its rate is several times the current property tax on the same flow and most parcels stand below their cap (Appendix Table A2). The model holds three margins of the standing building fixed. Landowners could cut maintenance or leave units empty, part of the tax could be passed on to tenants where rents are not controlled, and the tax is capitalised at once into the price of every parcel in the base. A tax on land is efficient (George, 1879; Arnott and Stiglitz, 1979), and land taxes speed up development (Oates and Schwab, 1997; Banzhaf and Lavery, 2010), but such taxes are among the least popular (Cabral and Hoxby, 2012; Fischel, 2001). Two features help. The rate follows a single public statistic, local developer concentration. And the revenue can replace the transfer tax, which falls on transactions rather than value. This fiscal swap raises the stock by 1.22% and improves the balance by 25.9 billion euros (Appendix E.4). Concentration is under-measured, the slope of the land share is imprecise, and the parcels where the markdown bites most fall outside the unsold stock. The rates follow the local Herfindahl index, so they rise with measured concentration without recalibration.

6.5. Taking stock

Table I5 puts every policy side by side, and Figure 5 plots the net cost per additional unit against the change in rent. Three lessons follow. First, relaxing land-use regulations is a long-run policy, because a taller allowance also raises the value of waiting. Second, supply policy can lower rents within a balanced budget, by cutting the taxes on new buildings and financing the cut with taxes on land. Most of the fall in rents is a transfer from incumbent landlords to renters. Third, restoring efficiency calls for an instrument on the land market. A tax on the standing value of under-used parcels, at a rate set by local concentration, corrects the markdown and raises revenue. Its gain is an efficiency gain. The rent-minimising mix lowers rents mostly through a transfer, but it also raises welfare, because it brings forward projects that taxes hold back.

Table 10: The rent-minimising and the welfare-maximising mixes

Item	Rent-minimising	Welfare-maximising
Cap floor (FAR)	3.23	12.47
Development tax (x current)	x0.00	x0.97
Property tax (x current)	x0.00	x0.59
Property-tax exemption (years)	12.8 y	13.4 y
Transfer tax	9.7%	8.6%
VAT on new housing	18.7%	12.6%
Tax on under-used land	9.8%	3.3%
Fixed-cost subsidy (euros/m2)	4,094	0
Inclusionary zoning (social-housing share)	0.0%	0.0%
Subsidy to sellers (share z of the rate)	12.5%	0.0%
Tax on standing value (share of the rate $1 - \bar{\theta}_m$)	0.0%	29.1%
Stock change at 20 years (%)	45.58	-0.88
Market rent, change at 20 years (%)	-17.76	0.45
Time saved (years)	14.6	-1.2
Net public cost (m€)	-728	-902
Renters, lower rents on the stock (bn€)	120.54	-3.68
Renters, surplus on additional housing (bn€)	28.76	0.05
Incumbent landlords (bn€)	-120.54	3.68
Selling landowners (bn€)	200.33	230.05
Developers (bn€)	58.01	54.57
Government, fair value (bn€)	-85.03	129.83
Welfare, total (bn€)	202.08	414.50

Data: model estimation. Instruments and outcomes of the mix that solves (27) and of the mix that solves (28), under a balanced budget by year 20. The welfare-maximising mix can also use the tax on standing value. Outcomes: change in the quality-adjusted stock and in the market rent after 20 years, years saved, net public cost by year 20 at r_g (negative: a gain), and the welfare decomposition (26) by group, billion euros, with the government at fair value.

To compare policies on one metric, Table I5 also reports the marginal value of public funds (Hendren and Sprung-Keyser, 2020), MVPF $=$ WTP/net cost, with willingness to pay, outlays, and revenue counted until year 50. In the narrow version, those affected are landowners and developers. The broad version adds the renters' surplus on additional housing, since lower rents on the existing stock are a transfer. A policy that gains for the budget and raises the welfare of those affected has an infinite MVPF. A policy that gains for the budget and lowers it, such as the tax on standing value, has none, and (26) is then the relevant metric.

7. Conclusion

Large cities build most new housing by replacing what already stands, and the private rate of replacement is not efficient. This paper brings buyer power on the land market into the real-options model of redevelopment. Developers mark down the price of land, and the markdown distorts the timing of

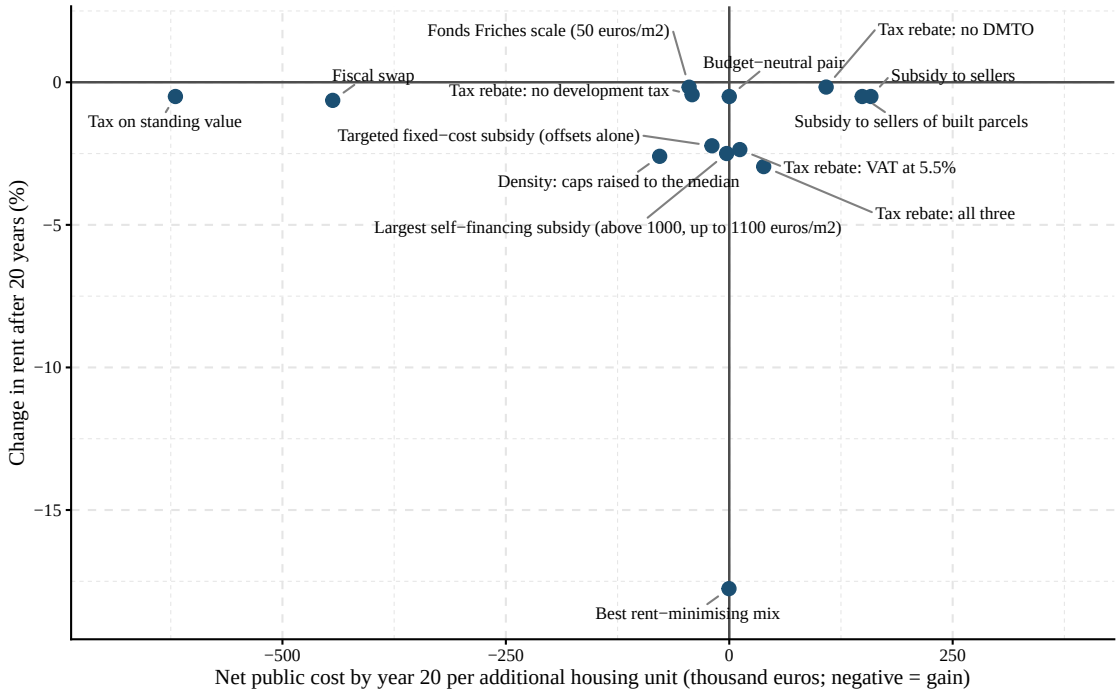


Figure 5: Rent change and net cost per additional unit

Data: model estimation. Each point is a policy. Horizontal axis: net cost to the consolidated budget by year 20 per additional housing unit of 60 square metres of living area, in thousand euros. A negative value is a gain for the budget. Vertical axis: change in the market rent after 20 years. Policies that lower the physical floorspace delivered by year 20 have no cost per additional unit and are not shown.

redevelopment only where the landowner gives up a standing building. It delays these parcels, blocks some of them for ever, and breaks the optimality of competitive exercise on a set of parcels that the data identify. Frictions and regulation raise the cost of every project, but a planner would face the frictions too. Separating the two requires the cost of redevelopment, which I recover parcel by parcel from the price of land, net of the markdown, taxes, the lag, and the value of waiting.

On the parcels whose prices the model rationalises, the fixed cost accounts for 30.9% of the twenty-year supply shortfall, taxes and land-use regulations for 45.3%, and the markdown for 0.9%. The markdown also blocks 0.3% of unsold parcels for ever, outside the decomposition.

The policy conclusions follow the line between efficiency and redistribution. Removing density caps costs no outlay, but a taller allowance also raises the value of waiting, so its effect builds over decades. A mix that cuts the taxes on new construction and taxes land instead lowers market rents by 14.02% over twenty years without costing the budget, mostly through a transfer from incumbent landlords to renters. Neither corrects the market failure. A tax on the standing value of under-used parcels, at a rate set by local buyer concentration, restores the constrained-efficient allocation and raises fiscal revenue.

The measurement leaves room for refinement. The model rationalises 63.1% of prices, mostly on lightly built parcels, where the markdown bites least. Concentration is under-measured, and the slope of the markdown is imprecisely estimated.

The framework opens two lines of future research. The first is what the fixed cost contains. It is a residual, net of every wedge the model measures. Land assembly is a natural candidate, since a project on several parcels needs every owner to sell, and hold-outs or assembly premia raise its cost. Attachment to land is another, since owners may value a standing building above what it earns. If either drives the residual, part of the fixed cost is a market failure too, and policy should act on it directly. The second is why the markdown exists. The model takes the number of developers as given. Entry costs or increasing returns to scale in development could each sustain concentration, and each calls for a different remedy. Both questions fit the framework of this paper, since any new mechanism enters as a change in one of its wedges, and its welfare cost follows from the same benchmarks.

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APPENDIX

A. Data and cleaning procedure

A.1. Sources

The main text names each source in plain English. This section gives the original name of each source and what I use it for.

Land tax records The *Fichiers Fonciers* are the fiscal record of every taxed built unit, produced by the tax administration (*Direction générale des finances publiques*, DGFiP) and redistributed for research by Cerema. They report each unit's use (housing, office and service, industrial, or outbuilding), floorspace, and construction date. They map a unit to a parcel through its postal address. This fails whenever a mailbox sits on a small contiguous parcel or several contiguous parcels share a front door. The rental values of business premises come from the same administration. The municipal property-tax rates come from its file of local tax rates (*Répertoire des impôts locaux*, REI).

Building database The *BD Topo*, produced by the national geographic institute (IGN), gives the footprint and height of every building, in successive vintages, with no address-matching problem but no use or ownership information.

Permit register *Sitadel* is the national register of authorised building permits, maintained by the statistical service of the ministry responsible for housing (SDES). It reports authorised floorspace and unit counts, the applicant's identity, and the permit date, but not reliably the use of the parcel before the permit. I impute it from the last land tax vintage preceding the permit. Applicants that are firms carry their business identifier (SIREN), assigned by the business register (*Sirene*) of the national statistical institute (INSEE).

Transaction register *Demande de valeurs foncières* (DVF) is published by the tax administration as open data. It records every registered property sale, with the parcel identifier, the price, and the date. I use its enriched version, DV3F, produced by Cerema. It links each sale to the land tax records and reports the names of buyers and sellers that are legal entities.

Cost survey The *Prix de revient des logements neufs* survey (PRLN) is a stratified survey of developers, drawn from the permit register, that asks for realised project costs separately by construction and demolition component. It is accessed through the secure data centre (CASD).

Other sources Rents come from the *Observatoires locaux des loyers*, a municipal index of market rents. Zoning comes from the Paris urban planning agency (Apur). Population and income come from INSEE's *Fidéli* files. Ground conditions come from the national geological survey (BRGM), and utility networks from network operators.

A.2. Sample restrictions

Permits I retain permits that add housing floorspace. An applicant is a private developer if it carries a SIREN and its legal category in the business register is a commercial company or a property civil company, such as an SCI or an SCCV. Social landlords, mixed-economy companies, public bodies, non-profits, and natural persons are not private developers. Private developers file 88.8% of the floorspace of permits with a SIREN. Social landlords file 3.9%, mixed-economy companies 1.5%, public bodies 4.6%, and non-profits 1.1%. Purchase dates, flows of redevelopment, realised densities, construction lags,

the hedonic price of parcels, and the samples of land sales use the permits of private developers only. Market shares use every applicant with a SIREN. A landowner can sell to a social landlord or to a public body, so leaving them out of the shares would overstate the concentration a landowner faces.

Parcels I keep privately owned land. Owners are classified with the owner typology of the land tax records, produced by Cerema. Land owned by the State and local authorities, social landlords, public land agencies, mixed-economy companies, network operators, universities, hospitals, religious bodies, and other public or non-profit legal persons is left out. Natural persons, co-ownerships, civil companies, and private firms are kept. For a parcel bought by a developer, the owner is the seller of the purchase deed. When the deed does not report the seller, it is the owner in the land tax records of the year before the purchase. For other parcels, it is the owner in 2025. This drops 12.4% of parcels. The owner is unknown for 0.3% of parcels, which I keep. I also drop the 0.1% of the remaining parcels that a social landlord, a mixed-economy company, a public body, or a non-profit redeveloped. Finally, I keep parcels within the size range of the land that private developers redevelop. The lower bound, 54 square metres, is the 1st percentile of the area of parcels that carry a permit on their own. A smaller plot cannot host a project without assembly. The upper bound, 9,829 square metres, is the 99th percentile of the area of every parcel that carries a permit. Larger parcels are mostly held by public bodies and social landlords. The bounds drop 4.8% and 0.2% of parcels.

Parcels and units The unit of analysis is the cadastral parcel, the smallest continuous unit of land with homogeneous property rights recorded by the cadastre. I aggregate taxed units to the parcel and call a parcel mixed when no single use accounts for 90% of its units. A parcel labelled empty carries no building and no taxed unit, so it has no standing floorspace and no rental value. A parcel labelled vacant, in Fact 1 only, carries a building whose taxed units are not occupied. Parcels without any building are kept. Matching building footprints to parcels can assign a neighbour's floorspace to a sliver of an empty parcel. I set these residues, below half a square metre, to zero. Table A1 reports the counts at each step of the estimation sample, and Table A2 compares the parcels kept with the parcels left out.

A.3. What land prices say about each parcel

The inversion of Proposition 5 places every parcel in one of six classes. Figure A1 shows their shares by initial use, and Table A3 reports them. Each class has a distinct economic reading.

A parcel is *ok* when its price lies between its standing value and its standing value plus the land share of the option at a zero fixed cost, $V_i^w < p_i \leq V_i^w + \theta_m O_i(F = 0)$. A unique fixed cost then reconciles the price with the model. These 63.1% of parcels form the estimation sample. Most are empty or lightly built. Empty parcels have no standing value, and 83% of them are ok.

A parcel is *above* when its price exceeds the value of redevelopment even at a zero cost. Only a negative fixed cost would fit (Section 5.1). This concerns 19% of parcels. Their price capitalises something the model leaves out. Candidates are assembly value, public subsidies, amenities of the existing use, or error in the hedonic price.

A parcel is *below* when its price does not exceed its standing value, $p_i \leq V_i^w$. This concerns 13% of parcels, mostly dense and co-owned buildings. For 5% of them, the option is worthless at any fixed cost: the density cap or the markdown leaves no gain from redevelopment. The model then predicts a price equal to the standing value, which the data confirm. For the others, the option is small. It is worth less than 5% of the standing value for 25% of below parcels. The fixed cost of these parcels is not identified. It barely matters for supply, since they would not redevelop at current rents.

The median price of below parcels is 30% under their standing value. Noise alone does not explain this gap. The yield y^w captures the average discount of whole tenanted buildings, not its dispersion. Buildings with long-standing tenants or in need of renovation sell for less. Commercial leases vary widely around y^C, and 30% of mixed-use parcels are below.

Table A1: Sample construction: counts and attrition

Step	N	% of previous step	% of the model sample
Parcels in the base data	625,660	–	–
With the model variables and a positive density cap	599,306	–	–
Privately owned (before the developer purchase for bought parcels)	525,220	87.6	–
Not redeveloped by a social landlord, a SEM, a public body or a non-profit	524,472	99.9	–
Within the size range of private redevelopment	498,257	95.0	100.0
Priced above the standing value	432,444	86.8	86.8
... and with a positive option value at a zero fixed cost	432,061	99.9	86.7
... and priced at or below that option value	338,252	78.3	67.9
... and consistent with no sale by 2024 (non-exercise bound)	316,698	93.6	63.6
... and a fixed cost below 100,000 euros per m2: estimation sample	314,271	99.2	63.1

Parcel count at each restriction, at the calibrated parameters and land shares: parcels in the base data, with complete data and a positive density cap, privately owned, not redeveloped by a social landlord, a mixed-economy company, a public body, or a non-profit, within the size range of private redevelopment, priced above their standing value (lower bound of (16)), with a positive option value at a zero fixed cost, priced at or below that option value (upper bound of (16)), consistent with no sale by 2024 (non-exercise bound, Section 3.7), and with a fixed cost inside the bracket of the bisection. The last row is the estimation sample of 314,271 parcels (Section 3.7). Columns: count, share of the previous step, and share of the model sample, the parcels left after the private-sector and size restrictions. The last share is the fit statistic of the inversion, the share of prices the model rationalises, and it is the share reported in Section 5.1 and Table A3.

Table A2: Who is in the estimation sample

Statistic	All parcels	Unsold	Sold by 2025	Out: price below standing value	Out: price above the zero-cost bound	Out: unsold, cost below the non-exercise bound	Out: option worthless or cost at the bracket
Parcels	498,257	310,489	3,782	65,813	93,809	21,554	2,810
Housing (%)	73	86	60	47	45	83	57
Mixed (%)	23	9	20	52	52	5	39
Non-housing (%)	1	0	2	1	2	1	2
Empty (%)	4	5	17	0	1	11	2
Empty (number)	18,315	14,470	653	0	874	2,271	47
Area (m2), median	323	350	414	333	216	291	328
Standing FAR, median	0.39	0.30	0.14	1.24	1.10	0.30	0.77
Cap FAR, median	1.74	1.51	2.17	3.16	5.68	1.14	5.27
Location premium, median	1.00	1.00	1.00	1.00	1.38	1.00	1.00
Price per m2 of land, median	1,873	1,342	1,541	2,265	11,208	2,286	2,973
Paris (%)	12	2	3	16	45	1	23

Data: model estimation. Columns: all parcels; the estimation sample, split into unsold parcels (the stock of the counterfactuals) and parcels bought by a developer by 2025; and the parcels left out, by reason (Proposition 5). Rows: count, initial use (shares, and the number of empty parcels), median parcel area, median floor-area ratio of the standing building and of the cap, median location premium, median predicted price per square metre of land, and share in Paris.

A parcel is *unsold below the bound* when its price lies within (16) but implies a fixed cost below the non-exercise bound $F_i^{\min}$ (Section 3.7). At that cost the parcel would have sold by 2024, and it did not. The bound bites most on empty parcels, which have no standing value and would be ripe at current rents unless their fixed cost were high (Table A3). A parcel is *blocked* when its price exceeds its standing value but the option is worthless at a zero cost. A parcel is *top* when its cost would exceed the bracket of the inversion. Both classes are small.

The counterfactuals use the ok parcels only. Below parcels redevelop little whatever their cost, so leaving them out understates supply only slightly. Above parcels give the main bound. Table F6 reports the results when they enter the unsold stock at a zero fixed cost.

A.4. The hedonic price of parcels

The transaction register is used twice. The hedonic model below gives the price $\hat{p}_i^L$ of every parcel, which identifies the fixed cost. The sales followed by a permit on the same parcel identify the slope of

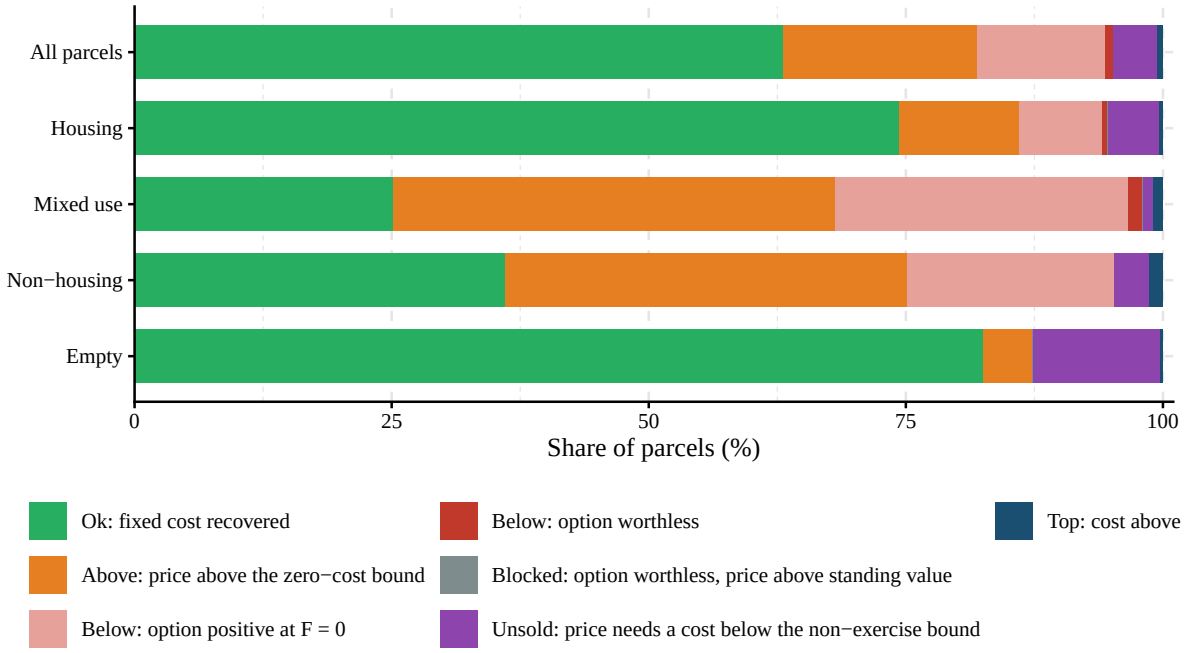


Figure A1: Classification of the parcels by the inversion

Data: model estimation. Share of parcels by class and initial use. Ok: price between the standing value and the option value at a zero fixed cost. Above: price above the option value at a zero fixed cost. Below: price at or below the standing value, split by the option value at a zero fixed cost. Blocked: price above the standing value, option worthless at a zero fixed cost. Unsold below the bound: fixed cost below the non-exercise bound. Top: fixed cost above the bracket of the inversion.

Table A3: Classification of the parcels by initial use (%)

Class	All parcels	Housing	Mixed use	Non-housing	Empty
Ok: fixed cost recovered	63.1	74.3	25.2	36.1	82.6
Above: price above the zero-cost bound	18.8	11.7	43.0	39.0	4.8
Below: option positive at F = 0	12.5	8.0	28.4	20.1	0.0
Below: option worthless	0.7	0.5	1.5	0.0	0.0
Blocked: option worthless, price above standing value	0.1	0.1	0.0	0.0	0.0
Unsold: price needs a cost below the non-exercise bound	4.3	5.0	1.0	3.4	12.4
Top: cost above the bracket	0.5	0.3	0.9	1.3	0.3

Data: model estimation. Share of parcels in each class of the inversion, over the model sample of Table A1, and by initial use. Classes as in Figure A1.

the markdown (Appendix D).

Operations, not deeds The price that identifies F_i is what a developer pays for a whole site. A single deed often records only part of it. A developer may buy two flats of a building in one deed, and the remaining flats with the land in a second deed. The second deed alone understates the price of the site, while the land tax records describe the whole building. Following Chapelle et al. (2026), I aggregate deeds into operations. I map every parcel sold or covered by a permit to its 2025 cadastral parcel, by identifier when it survives and by spatial overlay otherwise. Parcels sold in the same deed, covered by the same permit, or merged into the same 2025 parcel form a site. For each housing permit, the anchor is the last purchase on the site between three years before and one year after the permit. The operation sums every deed on the site in the five years before the permit whose buyer also buys in the anchor deed. Buyers are matched on the names of legal entities and on land-tax owner accounts. Intermediaries who

resell to the developer drop out by construction. Each deed is deflated to the year of the anchor with the regional price index. I keep the permits of private developers (Appendix A.2) and leave out the 16.4 % of operations in which a deed is sold by a non-private owner.
I use arm's-length sales (*ventes* and *ventes de terrain à bâtir*) in Paris and the three inner départements between 2010 and 2025. I exclude sales off plan (*vente en l'état futur d'achèvement*, VEFA), which price a future dwelling rather than an existing parcel. I also exclude portfolio sales of more than ten parcels and sites of more than 25 parcels. The register yields 3601 operations. Of these, 16.5 % combine several deeds. Such an operation costs a median 2.73 times its last deed. An operation is complete when the land sold covers at least 90% of the site and the units bought cover at least 90% of the units standing before the first deed. I estimate the model on complete operations, 55.8 % of the total. Multi-family buildings in co-ownership, which rarely sell in one deed, enter this sample.

Specification The regressors describe the site in the land tax records of the year of the first deed. They are the land area T_i, the habitable floorspace F_i^{hab}, the number of units of each type N_i^k, with $k \in \{\text{apartments}, \text{houses}, \text{business premises}, \text{annexes}\}$, and the number of cadastral parcels assembled n_i. For a site i in municipality m bought in year t, the conditional mean of the total price P_{it} is

$$\mathbb{E}\left[P_{it} \mid T_i, F_i^{\text{hab}}, \mathbf{N}_i, n_i\right] = \exp\left(\gamma_m + \gamma_t + b_T \ln T_i + b_F \ln\left(1 + F_i^{\text{hab}}\right) + \sum_k b_k \ln\left(1 + N_i^k\right) + b_n \ln n_i\right), \tag{A1}$$

where γ_m and γ_t are municipality and year fixed effects. I estimate three versions of (A1). The first uses sites that held only housing. The second pools all built sites. The third uses empty sites, with $\ln T_i$ and $\ln n_i$ as the only regressors. The transformation $\ln(1+x)$ keeps the sites without housing or without units of a type in the sample, and the third model prices empty sites separately. Unit prices are trimmed at the 1st and 99th percentiles of the price per square metre of land within each use. Municipalities without a training site take a département effect instead. Five-fold cross-validation by use of origin, with folds drawn by site, selects the version for each use. Housing parcels are priced by the housing model. Mixed and non-housing parcels are priced by the pooled model, because their own models predict worse out of sample. Empty parcels are priced by the model of empty sites. Table A5 reports the estimates.
All models target the total price in levels, the object that enters the landowner's value. I estimate them by Poisson pseudo-maximum likelihood (PPML). OLS on log prices would require a retransformation to recover prices in levels, which is biased whenever the variance of the error depends on the regressors. PPML only requires the conditional mean to be correctly specified. With municipality and year fixed effects, its first-order conditions also imply that predicted prices sum to observed prices within each municipality and year. Standard errors are clustered by municipality. Municipality effects absorb the land share $\bar{\theta}_m$ and the location premium ϕ_m, which are both measured at the municipal level. I do not control for the type of buyer. The target is the price developers pay, markdown included, and the inversion nets out the markdown with $\bar{\theta}_m$.

Assembly At constant area, a housing site assembled from more parcels sells for more. For housing sites, the elasticity of the price to the number of parcels assembled is 0.33 (s.e. 0.09 , Table A5, column 1), a premium consistent with holdout in land assembly (Brooks and Lutz, 2016). Pooling all built sites, it is 0.01 (s.e. 0.1), so the evidence of a premium is limited to housing sites. I score every parcel as a single-parcel site. The premium therefore stays out of $\hat{p}_i^L$ and loads on the fixed cost, as a cost of assembly.

Whole-parcel sales The operations are few and contain few dense buildings, so the operation model pins down poorly how the price varies with the standing building. The price of every parcel therefore comes from a second model, estimated on the 142,177 sales of a whole parcel: one parcel, its land, and at

least 90% of the units it held the year before, in one deed, whoever the buyer. A whole parcel sold in one deed is the asset whose value is W_i in (11). The specification replaces $\ln(1+x)$ by $\ln x$ and an indicator of $x = 0$ for the floorspace and every unit count, so that the elasticity of the price to the standing building is not forced through the origin. Empty parcels are priced by a model of whole-parcel sales of empty land, with $\ln T_i$ as the only regressor. Out of sample, in five folds by parcel, the root mean squared error of log prices on whole-parcel sales is 0.46, against 0.64 for the operation model. On buildings with a floor-area ratio above two, the ratio of predicted to observed prices is 0.93, against 0.71 (Table A4). The operation model is kept for the premium of assembly and for comparison. Co-owned buildings rarely sell whole, so their scores extrapolate from smaller buildings.

Table A4: Hedonic variants: fit on whole-parcel sales, out of sample

Variant	Sample	N	Calibration	Median abs. error	RMSE (log)
farpoly	All	132,375	0.94	0.31	0.69
log1p	All	132,375	0.87	0.33	0.64
logzero	All	132,375	0.90	0.33	0.64
sales	All	142,177	0.99	0.18	0.47
sales_logzero	All	142,177	1.00	0.17	0.46
farpoly	Built, FAR 0	4,644	0.64	0.59	0.98
farpoly	Built, FAR 0-0.5	95,450	1.01	0.28	0.46
farpoly	Built, FAR 0.5-1	18,277	0.87	0.30	0.51
farpoly	Built, FAR 1-2	3,867	0.92	0.38	0.69
farpoly	Built, FAR 2+	1,104	1.41	0.76	3.24
log1p	Built, FAR 0	4,644	0.61	0.56	0.96
log1p	Built, FAR 0-0.5	95,450	1.01	0.29	0.47
log1p	Built, FAR 0.5-1	18,277	0.75	0.36	0.58
log1p	Built, FAR 1-2	3,867	0.77	0.45	0.78
log1p	Built, FAR 2+	1,104	0.71	0.58	1.18
logzero	Built, FAR 0	4,644	0.63	0.63	1.00
logzero	Built, FAR 0-0.5	95,450	1.03	0.29	0.47
logzero	Built, FAR 0.5-1	18,277	0.81	0.34	0.55
logzero	Built, FAR 1-2	3,867	0.83	0.42	0.71
logzero	Built, FAR 2+	1,104	0.85	0.50	1.13
sales	Built, FAR 0	4,852	0.96	0.68	1.00
sales	Built, FAR 0-0.5	102,711	1.04	0.15	0.32
sales	Built, FAR 0.5-1	19,808	0.98	0.20	0.38
sales	Built, FAR 1-2	4,228	1.00	0.26	0.54
sales	Built, FAR 2+	1,182	0.86	0.40	1.05
sales_logzero	Built, FAR 0	4,852	0.99	0.84	1.03
sales_logzero	Built, FAR 0-0.5	102,711	1.01	0.15	0.31
sales_logzero	Built, FAR 0.5-1	19,808	1.00	0.19	0.37
sales_logzero	Built, FAR 1-2	4,228	1.02	0.24	0.53
sales_logzero	Built, FAR 2+	1,182	0.93	0.37	1.05

Data: transaction register, land tax records. Each variant is evaluated on whole-parcel sales in five folds by parcel. Variants trained on operations are trained once on operations, which never contain these sales. log1p: operations, $\ln(1+x)$. logzero: operations, $\ln x$ and an indicator of $x = 0$. farpoly: operations, $\ln(1+x)$ and a quadratic in the standing floor-area ratio. sales: whole-parcel sales, $\ln(1+x)$. sales_logzero: whole-parcel sales, $\ln x$ and an indicator of $x = 0$, the variant used. Calibration: sum of predicted over sum of observed prices. Rows: all sales, then built sales by standing floor-area ratio.

Scoring and fit I score every 2025 cadastral parcel in its 2025 state at the 2023 year effect. The variables are built from the same land tax records as in estimation. I then bring the predicted price to the year of the rents, 2024, with the regional transaction-based price index. The predicted price $\hat{p}_i^L$ is the price of the whole parcel, inclusive of any building. For the operation model, out of sample on operations, the ratio of predicted to observed total prices is 0.97 for housing sites, 0.91 for mixed sites, 0.84 for non-housing sites, and 1.23 for empty sites. Non-housing sites are the least precisely priced. Their prices are too low on average, which raises the fixed cost the inversion recovers. Their fixed costs are therefore an upper bound in this respect. 16.7 % of empty parcels lie outside the support of the training sample, mostly large parcels. Figure A2 compares predicted and observed prices.

Table A5: Hedonic model of the price of redevelopment operations

	price		
	Housing	All built (pooled)	Empty
	(1)	(2)	(3)
Log parcel area	0.573***	0.886***	0.963***
	(0.051)	(0.043)	(0.182)
Log(1 + habitable surface)	0.202***	-0.115**	
	(0.052)	(0.051)	
Log(1 + apartments)	-0.014	0.249***	
	(0.075)	(0.095)	
Log(1 + houses)	0.130	0.348***	
	(0.120)	(0.130)	
Log(1 + annexes)	0.126***	0.026	
	(0.043)	(0.044)	
Log number of parcels assembled	0.331***	0.014	-0.183
	(0.085)	(0.100)	(0.238)
Log(1 + activity units)		0.164	
		(0.146)	
Observations	977	1,442	179
Pseudo R^2	0.755	0.854	0.888
Commune fixed effects	✓	✓	✓
Year fixed effects	✓	✓	✓

Notes: PPML estimates of equation (A1) on complete redevelopment operations. The dependent variable is the total price of the operation, in euros of the year of the anchor deed. Column (1): sites that held only housing. Column (2): all built sites. Column (3): empty sites. Standard errors clustered by municipality in parentheses. Source: transaction register and land tax records, 2010–2025, départements 75, 92, 93 and 94.

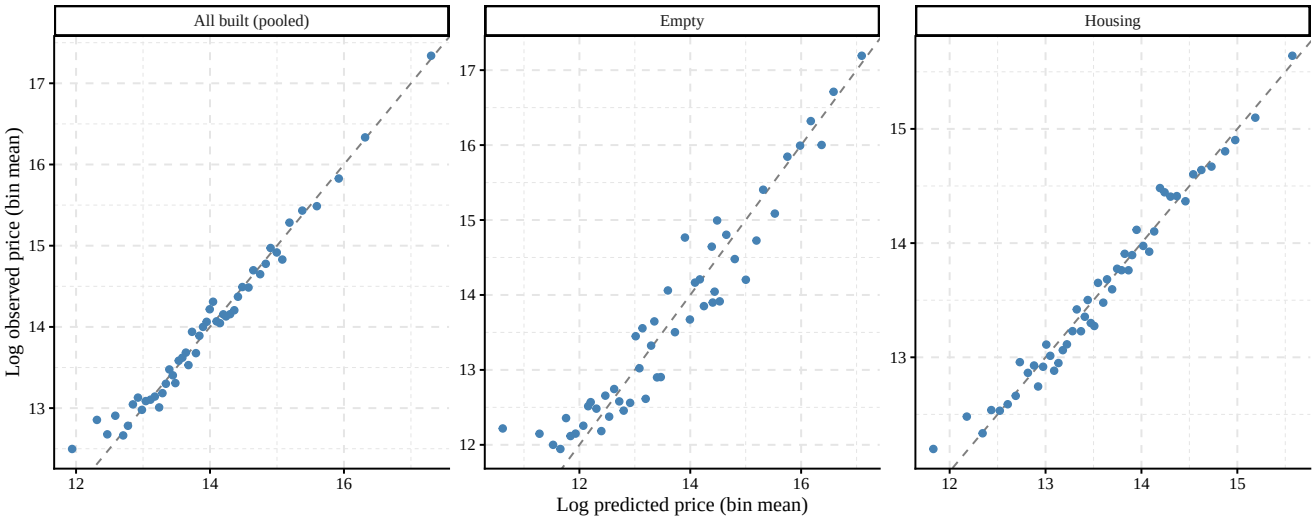
A.5. The density cap $\bar{h}_i$

Each municipality writes a zoning plan (*plan local d'urbanisme*, PLU). In Paris, the city adopts it. In the three other départements, the territories (*établissements publics territoriaux*, EPT) of the *Métropole du Grand Paris* have drafted intercommunal plans (PLUi) since 2016. No complete, parcel-level digitisation of every plan's coverage and height limits exists for the sample period. The cap $\bar{h}_i$ is therefore inferred: the 90th percentile of built coverage and height within each cadastral section, with a floor-to-ceiling height of 2.5 metres, checked against realised construction in Figure A3.

A.6. Developer identifiers and concentration

Developers are identified by their SIREN, matched to permit applicants. Many developers incorporate a project-specific company for each operation, often a *société civile de construction-vente* (SCCV), so shares computed at the SIREN level understate concentration. I consolidate these companies into their parent group where the parent can be identified. Under-consolidation biases the Herfindahl index down where

Figure A2: Predicted and observed prices of redevelopment operations



Notes: Binned scatter plot of observed against predicted total prices of complete redevelopment operations, in logs. Within each panel, operations are grouped into 50 equal-sized bins of predicted price. Each dot plots the log of the mean observed price against the log of the mean predicted price in the bin. Predictions are in-sample, from equation (A1) estimated by PPML with municipality and year fixed effects. The dashed line is the 45-degree line. Panels: housing sites, all built sites, empty sites. Source: transaction register and land tax records, 2010–2025, départements 75, 92, 93 and 94.

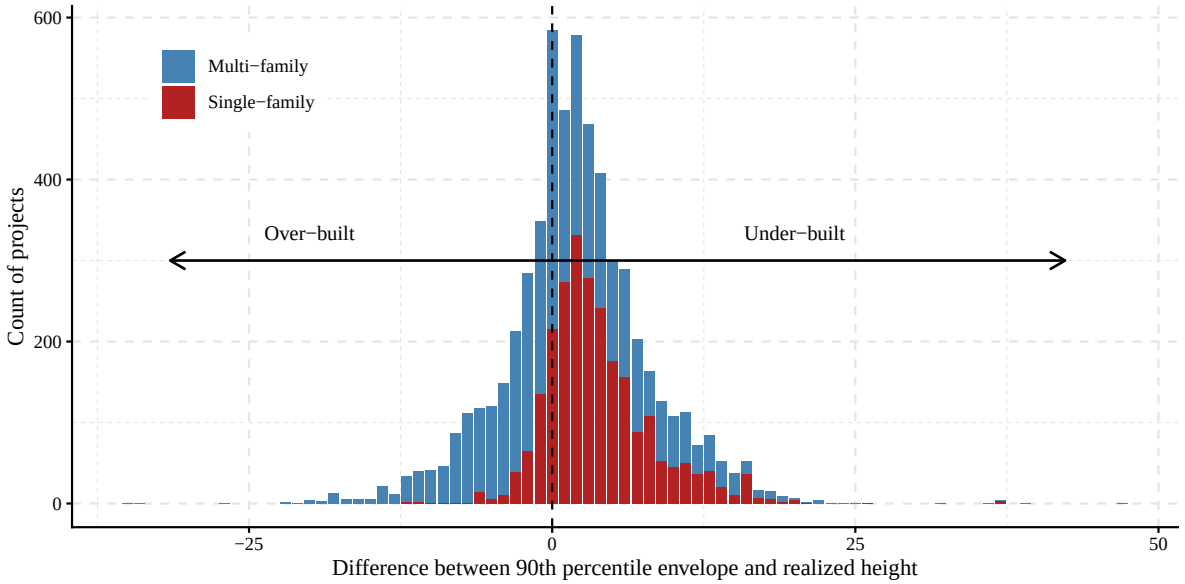


Figure A3: New construction against the inferred density cap

Data: land tax records, permit register, building database. Distribution, across new projects, of the difference between the inferred cap (90th percentile of building height in the cadastral section) and the realised height of the new building, for multi-family and single-family projects. Positive values are projects built below the inferred cap.

a group buys through several project companies, which raises the measured land share there and biases the estimated cost of concentration towards zero.

The municipal share of group j in year t is its share of the purchases of municipality m over the three years up to t, where each permit is valued at its floorspace times the local price of new housing, a proxy for the unobserved land payment: w_{jmt}. The municipal Herfindahl index is $HHI_{mt} = \sum_j w_{jmt}^2$.

The region-wide index of Fact 3 uses floorspace shares over 2013–2024, and N^{tot} counts consolidated applicants with at least one permit over the same window. Shares cover every applicant with a SIREN, including social landlords, mixed-economy companies, and public bodies, because a private landowner can sell to them (Appendix A.2). The parcels and the flows of the model are those of private developers.

Table A6: Matching permits to developer groups

Statistic	SIREN level	Parent group
Collective permits matched to a SIREN (%)	73.4	73.4
Floorspace of collective permits matched to a SIREN (%)	83.4	83.4
DVF land sales matched to a later private permit (%)	0.9	0.9
Floorspace of parent groups with several SIREN (%)	0.0	32.2
Region-wide HHI	0.005	0.007
Median municipal HHI	0.333	0.333
Median municipal HHI, municipalities with at least 10 projects	0.265	0.265
Top-7 share (%)	13.7	16.1
N^{tot}	4,421	3,857

Match rate of collective permits to a SIREN (count and floorspace), share of DVF land sales followed by a permit of a private developer, share of floorspace re-attributed from a project-specific company to its parent group, region-wide and median municipal Herfindahl index, overall and among municipalities with at least ten projects, top-7 share, and N^{tot} before and after consolidation. The median municipal index is the same before and after consolidation, overall and among municipalities with at least ten projects (0.265 and 0.265). Overall, it sits at 1/3, the index of a municipality with three projects of equal value, so small municipalities set it. A median moves only if consolidation changes the index of the municipalities near it. Consolidation raises the index where a group buys through several project companies, which concerns few of the projects of a typical municipality, so the median municipality is unaffected. The bias from under-consolidation shows in the region-wide index and the top-7 share.

A.7. Donut difference-in-differences

The sample of projects is every permit that creates housing, with no other project of at least 50 dwellings within 1000 metres and 3 years. The event is the year T_p in which the permit of project p is authorised. For each project, I collect every resale of an existing house or flat within 500 metres of its centroid, between $T_p - 4$ and $T_p + 4$. Each sale is assigned to its nearest project of the size class, and dropped if that project is contaminated. The samples of Table A7 are therefore not nested. With every project, the nearest project of a sale is often a small project next to a large one, hence contaminated, and the sale is dropped. With larger projects only, the same sale is assigned to a farther clean project and kept. This is why the column of projects with at least 10 dwellings counts more transactions than the column of all projects. Transactions within 20 metres are dropped, since they are exposed to construction nuisance and may include the project's own units. The inner ring runs from 20 to 100 metres. Transactions between 100 and 150 metres form a buffer and are dropped. The outer ring, the comparison group, runs from 150 to 500 metres, as in Asquith et al. (2023). I stack the project samples and estimate

$$\log p_{jpt} = \alpha_{p,r(j)} + \gamma_{p,t} + \sum_{k \neq -1} \xi_k \, \mathbb{1}\{t - T_p = k\} \, \mathbb{1}\{r(j) = \text{inner}\} + \varepsilon_{jpt}, \tag{A2}$$

where p_{jpt} is the price per square metre of transaction j in stack p and year t, $\alpha_{p,r}$ are project-by-ring fixed effects, and $\gamma_{p,t}$ project-by-year fixed effects. The event-time coefficients ξ_k measure the change in the price gap between the two rings relative to the year before the permit. Standard errors are clustered by project. Stacking by project compares each inner ring only with its own outer ring, which avoids comparisons across cohorts that bias two-way fixed-effects estimators (Cengiz et al., 2019). The pre-period coefficients test for parallel trends. The pooled estimate replaces the event-time dummies by a single post-permit indicator.

Rents are not observed at the dwelling level, so the test uses the price per square metre of existing

dwellings resold nearby. For projects of at least 50 dwellings, the pooled effect on nearby prices is -0.056 (s.e. 0.169, p-value 0.74). The estimates for other project sizes have either sign, and none is significant (Table A7). Figure J4 plots the event study by year relative to the permit. The design has little power. The minimum effect detectable with 80% power at the 5% level is about 47.4 log points, so the estimates do not rule out the effects of a few percent found in dense US cities (Asquith et al., 2023; Li, 2022; Pennington, 2021). A new project also has two opposite effects on its neighbours. It adds supply, which lowers rents. It also changes the amenities of the neighbourhood, upwards or downwards depending on the quality of the project relative to its surroundings (Diamond, 2016; Diamond and McQuade, 2019). A null average effect is consistent with both forces at work. Assumption 2 therefore rests on the size of projects (Fact 5), not on this test.

Table A7: Donut difference-in-differences around new projects

	Project size (dwellings created)			
	All	≥ 10	≥ 50	> 100
Treated $\times$ Post	-0.279	-0.073	-0.056	0.352
	(0.200)	(0.082)	(0.169)	(0.376)
Projects	1373	160	40	12
Transactions	4270	4978	2648	878

Notes: Treated ring 20–100 m, buffer 100–150 m (excluded), control ring 150–500 m from the project centroid. Each sale is assigned to its nearest project of the size class, so the columns are not nested samples.

Data: transaction register, permit register, building database. Inner ring 20–100 m, buffer 100–150 m (excluded), outer ring 150–500 m from the project centroid. Pooled post-permit estimate of (A2). Columns: all projects, projects creating at least 10, at least 50, and more than 100 dwellings. Each sale is assigned to its nearest project of the column's size class, so the columns are not nested samples (Appendix A.7). Standard errors clustered by project in parentheses. Number of projects and transactions reported below each column.

B. Proofs and the interior regime

B.1. The interior regime

Below the cap, the density of a project adjusts to the rent. Let $X \equiv \kappa_i R/(\rho-\delta)$ be the value at the start date of a square metre of new floorspace, with κ_i from (2), and $Y \equiv \lambda_i^C c_0 T_i^{-1/\zeta}$ the discounted cost parameter. The developer chooses the density h to maximise the payoff of the project,

$$\pi_i(R) = \max_{h \le \bar{h}_i} \; X h - Y h^{1+1/\zeta}. \tag{B1}$$

The objective is strictly concave in h. Ignoring the cap, the first-order condition with respect to h is $X = (1+1/\zeta)\, Y h^{1/\zeta}$. It gives the density the developer would choose at rent R, which increases with R and exhausts the cap at the saturation rent $\bar{R}_i$:

$$h_i^\star(R) = \left[\frac{\zeta X}{(\zeta+1)Y}\right]^\zeta = \left[\frac{\zeta\,\kappa_i R}{(\zeta+1)(\rho-\delta)\lambda_i^C c_0}\right]^\zeta T_i, \qquad \bar{R}_i = \frac{(\zeta+1)(\rho-\delta)\lambda_i^C c_0}{\zeta\,\kappa_i}\left(\frac{\bar{h}_i}{T_i}\right)^{1/\zeta}. \tag{B2}$$

Taxes and the lag lower density like a higher construction cost, since $h_i^\star$ is proportional to $(\kappa_i/\lambda_i^C)^\zeta$. At the interior optimum, $Y(h_i^\star)^{1+1/\zeta} = X h_i^\star\, \zeta/(\zeta+1)$, so the payoff is $\pi_i = X h_i^\star/(\zeta+1)$. In both regimes,

$$\pi_i(R) = \begin{cases} b_i^0\, R^{\zeta+1}, \quad b_i^0 \equiv \dfrac{Y}{\zeta}\left[\dfrac{\zeta\,\kappa_i}{(\zeta+1)(\rho-\delta)Y}\right]^{\zeta+1} & \text{if } R < \bar{R}_i \quad \text{(interior)}, \\[2ex] \kappa_i \dfrac{R\,\bar{h}_i}{\rho-\delta} - C_i & \text{if } R \ge \bar{R}_i \quad \text{(constrained)}, \end{cases} \tag{B3}$$

where $C_i = Y\bar{h}_i^{1+1/\zeta}$ is the construction cost at the cap. The two branches and their derivatives coincide at $\bar{R}_i$.

The lag enters both branches in the same way. Let $\Lambda_i \equiv \lambda_i^R/\lambda_i^C$, and let $\pi_i^{0,\tau}$ denote the payoff without lag, with the same taxes.

Lemma B.1 (The lag rescales the payoff)**.** *For every parcel and every rent,* $\pi_i(R) = \lambda_i^C\, \pi_i^{0,\tau}(\Lambda_i R)$.

Proof. Using $\lambda_i^R = \Lambda_i \lambda_i^C$, write (B1) as $\pi_i(R) = \lambda_i^C \max_{h\le\bar{h}_i}\{[(1-\tau^F)/(1+\tau^V)]\,\Lambda_i R h/(\rho-\delta) - c_0 T_i^{-1/\zeta} h^{1+1/\zeta}\}$. The bracket is the payoff without lag at the rent $\Lambda_i R$, and the constraint set depends neither on R nor on ℓ_i. □

The development tax In the interior regime, I evaluate the development tax at the density $h_i^\star(R)$ chosen without it, and neglect its feedback on that choice. At the cap, density does not respond to the tax, so the treatment is exact there. Below the cap, the tax would slightly lower density, which I neglect.

The threshold in the interior regime Consider a parcel without standing housing that stays in the interior regime, and abstract from the development tax. Its private surplus is $S_i^\theta(R) = b_i R^{\zeta+1} - K_i^{\text{int},\theta}$, with $b_i \equiv b_i^0/(1+\tau^M)$ and $K_i^{\text{int},\theta} \equiv F_i/(1+\tau^M) + \tilde{V}_i^C/\bar{\theta}_m$. The condition of Proposition B.5 reads $\beta(b_i R^{\zeta+1} - K_i^{\text{int},\theta}) = (\zeta+1) b_i R^{\zeta+1}$. Provided $\beta > \zeta+1$,

$$R_i^\star = \left(\frac{\beta}{\beta-\zeta-1}\right)^{\frac{1}{\zeta+1}} \left(\frac{K_i^{\text{int},\theta}}{b_i}\right)^{\frac{1}{\zeta+1}}. \tag{B4}$$

At the calibrated parameters, the premium $[\beta/(\beta-\zeta-1)]^{1/(\zeta+1)}$ equals 2.32. The condition $\beta > \zeta+1$ is necessary. Since $\mathbb{E}[e^{-\rho t}R_t^n]$ is bounded if and only if $n < \beta$, a payoff that grows like $R^{\zeta+1}$ has a finite option value only if $\zeta+1 < \beta$. Otherwise the value of waiting for a taller building diverges, and only parcels that build at their cap are ever redeveloped. The constrained payoff is affine in R, so its option value is finite for any $\beta > 1$.

B.2. Proofs

This subsection states the results that the main text uses without stating them formally, and proves every result in the order in which the main text uses it. Lemma B.2 and Proposition B.3 adapt Berger et al. (2022) to the land market. Proposition B.5 applies the standard exercise rule of Dixit and Pindyck (1994, ch. 5) to the private surplus.

Lemma B.2 (Parcel supply, adapted from Berger et al., 2022)**.** *The solution of the landowners' problem* (6) *is the supply of parcels* (7)*.*

Proof. The first-order condition of the landowners' problem with respect to L_{jm} is $\theta_{jm} = C'(\mathcal{L}_m)\,\partial\mathcal{L}_m/\partial L_{jm}$, where $C(\mathcal{L}) = \bar{\Theta}\bar{L}_m^{-1/\eta^S}\mathcal{L}^{1+1/\eta^S}/(1+1/\eta^S)$. Since $C'(\mathcal{L}_m) = \bar{\Theta}(\mathcal{L}_m/\bar{L}_m)^{1/\eta^S}$ and $\partial\mathcal{L}_m/\partial L_{jm} = (L_{jm}/\mathcal{L}_m)^{1/\eta^D}$,

$$\theta_{jm} = \bar{\Theta}\left(\frac{\mathcal{L}_m}{\bar{L}_m}\right)^{\frac{1}{\eta^S}}\left(\frac{L_{jm}}{\mathcal{L}_m}\right)^{\frac{1}{\eta^D}}. \tag{B5}$$

Let $P_m \equiv \bar{\Theta}(\mathcal{L}_m/\bar{L}_m)^{1/\eta^S}$. Then $L_{jm} = \mathcal{L}_m(\theta_{jm}/P_m)^{\eta^D}$. Raising to the power $(1+\eta^D)/\eta^D$ and summing over j gives $\mathcal{L}_m^{(1+\eta^D)/\eta^D} = \mathcal{L}_m^{(1+\eta^D)/\eta^D}\sum_j \theta_{jm}^{1+\eta^D}/P_m^{1+\eta^D}$, so $P_m = \Theta_m$. Hence $\mathcal{L}_m = \bar{L}_m(\Theta_m/\bar{\Theta})^{\eta^S}$ and $L_{jm} = (\theta_{jm}/\Theta_m)^{\eta^D}(\Theta_m/\bar{\Theta})^{\eta^S}\bar{L}_m$. □

Proposition B.3 (Markdown, adapted from Berger et al., 2022)**.** *Under Assumption 4, with developers that choose quantities and post one offer, developer j pays the share* (8)*, and the land share of municipality m is* (9)*.*

Proof. Developer j takes the purchases of its rivals as given and pays the share given by the inverse supply (B5). Taking logs, $\partial\ln\theta_{jm}/\partial\ln L_{jm} = 1/\eta^D + (1/\eta^S - 1/\eta^D)\,\partial\ln\mathcal{L}_m/\partial\ln L_{jm}$. The last derivative is $L_{jm}^{(1+\eta^D)/\eta^D}/\sum_k L_{km}^{(1+\eta^D)/\eta^D}$. Since $\theta_{jm}L_{jm} \propto L_{jm}^{(1+\eta^D)/\eta^D}$, it equals the payment share w_{jm}. Developer j earns $(1-\theta_{jm})L_{jm}$, since one euro of developer value is worth one euro to it. The first-order condition is $1-\theta_{jm}(1+\partial\ln\theta_{jm}/\partial\ln L_{jm}) = 0$, so $1/\theta_{jm} = 1 + w_{jm}/\eta^S + (1-w_{jm})/\eta^D$, which is (8). For the aggregate,

$$\frac{1}{\bar{\theta}_m} = \frac{\sum_j L_{jm}}{\sum_k \theta_{km}L_{km}} = \sum_j \frac{w_{jm}}{\theta_{jm}} = \sum_j w_{jm}\left(1 + \frac{w_{jm}}{\eta^S} + \frac{1-w_{jm}}{\eta^D}\right) = 1 + \frac{HHI_m}{\eta^S} + \frac{1-HHI_m}{\eta^D},$$

which is (9). □

Lemma B.4 (The landowner's problem)**.** *The value of a parcel to its landowner is* (11)*.*

Proof. Under the perceived process, and with the commercial value held at $\tilde{V}_i^C$ (Section 3.2), so that it yields $\rho\tilde{V}_i^C$ in the landowner's problem, the standing value satisfies $V_i^w(R) = \mathbb{E}[\int_0^\tau e^{-\rho t}(R_t\tilde{h}_i^{\text{old}} + \rho\tilde{V}_i^C)\,dt] + \mathbb{E}[e^{-\rho\tau}V_i^w(R_\tau)]$ for every stopping time τ. A landowner who sells at τ collects the standing flows until τ and receives $\bar{\theta}_m V_i^D(R_\tau)$ at τ. The landowner's value is therefore $V_i^w(R) + \mathbb{E}[e^{-\rho\tau}(\bar{\theta}_m V_i^D - V_i^w)(R_\tau)] = V_i^w(R) + \bar{\theta}_m\mathbb{E}[e^{-\rho\tau}S_i^\theta(R_\tau)]$. Taking the supremum over τ gives (11). □

Proposition B.5 (Redevelopment threshold, after Dixit and Pindyck, 1994)**.** *The landowner sells, and the parcel is redeveloped, the first time the rent reaches $R_i^\star$, the smallest root of $\beta S_i^\theta(R) = R\,S_i^{\theta\prime}(R)$ with $S_i^\theta(R) > 0$. In the constrained regime, $S_i^\theta(R) = a_i^\theta R - K_i^\theta$ with a_i^θ and K_i^θ as in Section 3.5, and, provided $a_i^\theta > 0$, the threshold is* (12)*.*

Proof. Let $\mathcal{Q}(x) \equiv \frac{1}{2}\sigma^2 x(x-1) + \delta x - \rho$, with roots $\beta_- < 0 < 1 < \beta$. In the continuation region, the option value solves $\frac{1}{2}\sigma^2 R^2 \mathcal{O}'' + \delta R \mathcal{O}' - \rho \mathcal{O} = 0$, so $\mathcal{O}_i^\theta = C_1 R^\beta + C_2 R^{\beta_-}$, and boundedness at zero imposes $C_2 = 0$. Value matching, $C_1 R^{\star\beta} = S_i^\theta(R^\star)$, and smooth pasting, $\beta C_1 R^{\star\beta-1} = S_i^{\theta\prime}(R^\star)$, give the condition $\beta S_i^\theta = R S_i^{\theta\prime}$. In the constrained regime, $V_i^D(R) = q_i R/(\rho-\delta) - (F_i + C_i + \tau^A \bar{h}_i)/(1+\tau^M)$ by (3) and (4), and $V_i^w(R)/\bar{\theta}_m = [R\tilde{h}_i^{\text{old}}/(\rho-\delta) + \tilde{V}_i^C]/\bar{\theta}_m$. This gives a_i^θ and K_i^θ. The condition reads $\beta(a_i^\theta R - K_i^\theta) = a_i^\theta R$, which gives (12). Appendix B.1 gives the interior regime. □

Proof of Proposition 1 (i) In the constrained regime, $K_i^\theta/a_i^\theta = (\rho-\delta)[(F_i + C_i + \tau^A \bar{h}_i)/(1+\tau^M) + \tilde{V}_i^C/\bar{\theta}_m]/[q_i - \tilde{h}_i^{\text{old}}/\bar{\theta}_m]$. The numerator decreases with $\bar{\theta}_m$, strictly if $\tilde{V}_i^C > 0$. The denominator increases with $\bar{\theta}_m$, strictly if $\tilde{h}_i^{\text{old}} > 0$. The threshold therefore decreases with $\bar{\theta}_m$, strictly if $V_i^w > 0$, and does not depend on $\bar{\theta}_m$ if $V_i^w = 0$. In the interior regime without standing housing, $K_i^{\text{int},\theta}$ decreases with $\bar{\theta}_m$, strictly if $\tilde{V}_i^C > 0$. Since $\bar{\theta}_m \leq 1$, $R_i^\star \geq R_i^{\text{CE}}$. Parcels whose threshold straddles the saturation rent are solved numerically, and the property holds there by the same argument, since $S_i^\theta = S_i - (1/\bar{\theta}_m - 1)V_i^w$ falls pointwise as $\bar{\theta}_m$ falls. (ii) The constrained-efficient allocation redevelops the parcel if $a_i = (q_i - \tilde{h}_i^{\text{old}})/(\rho-\delta) > 0$, that is if $\tilde{h}_i^{\text{old}} < q_i$. Privately, $a_i^\theta \leq 0$ if $\bar{\theta}_m \leq \tilde{h}_i^{\text{old}}/q_i$. Then $S_i^\theta(R) = a_i^\theta R - K_i^\theta < 0$ at every rent, since $K_i^\theta > 0$, and the landowner never sells. □

Proof of Proposition 2 Total private surplus on parcel i under a threshold $r \geq R_0$ is $\text{TS}_i(r) = V_i^w(R_0) + S_i(r)(R_0/r)^\beta$. The developer's rent is a transfer within this sum. R_i^{CE} maximises $S_i(r) r^{-\beta}$, which is single-peaked on the region where $S_i > 0$. (i) At $\bar{\theta}_m = 1$, the private surplus S_i^θ equals S_i, so $R_i^\star = R_i^{\text{CE}}$. For $\bar{\theta}_m < 1$, Proposition 1 gives $R_i^\star > R_i^{\text{CE}}$ if $V_i^w > 0$ and $R_i^\star = R_i^{\text{CE}}$ otherwise. If $V_i^w > 0$, $\partial \text{TS}_i/\partial r < 0$ at $R_i^\star$. Since $R_i^\star$ decreases with $\bar{\theta}_m$, $d\text{TS}_i/d\bar{\theta}_m = (\partial \text{TS}_i/\partial r)(\partial R_i^\star/\partial \bar{\theta}_m) > 0$, a first-order gain. For a parcel blocked by concentration, raising $\bar{\theta}_m$ above $\tilde{h}_i^{\text{old}}/q_i$ yields a discrete gain. (ii) The fixed cost enters S_i and S_i^0 through the developer value, the lag through κ_i and C_i, and the option premium through β, which is common to every threshold. If $\bar{\theta}_m = 1$ and taxes are zero, the private threshold is $R_i^{\text{CE}} = R_i^{\text{FB}}$. A subsidy s moves it to $r(s) = R_i^{\text{CE}}(F_i - s)$, while the resource cost remains F_i. Since $r(0)$ maximises TS_i, $d\text{TS}_i/ds|_{s=0} = (\partial \text{TS}_i/\partial r)\, r'(0) = 0$, and the change in total surplus is $O(s^2)$. The outlay moves resources from taxpayers to the parcel and changes welfare only through the shadow cost λ_g. (iii) Welfare is $W = \Pi + (1+\lambda_g)G$, where Π is the value of landowners and developers and G the public balance. The landowner chooses the threshold that maximises the landowner's value, so by the envelope theorem a marginal change in the threshold has no first-order effect on it. The developer's value changes to first order with the threshold only through the markdown: Π moves with the threshold as TS_i does, and $\partial \text{TS}_i/\partial r = 0$ at $R_i^\star$ whenever $R_i^\star = R_i^{\text{CE}}$. Hence $d\Pi = X + M$, with $M = -\sum_i (\partial \text{TS}_i/\partial r)\, dr_i \geq 0$ on parcels with $R_i^\star > R_i^{\text{CE}}$ and $dr_i \leq 0$, and $M = 0$ if $\bar{\theta}_m = 1$. The public balance changes by $dG = T - X$. Therefore $dW = X + M + (1+\lambda_g)(T - X) = (1+\lambda_g)T - \lambda_g X + M$. If $T \geq X$, $dW \geq T + M \geq 0$, with strict inequality whenever the policy moves any project. □

Proof of Proposition 3 (i) Under the pair $(\tau_m^{\text{sell}}, t_m^w)$, the landowner collects the standing flows net of the tax until the sale, so the standing value becomes $(1 - t_m^w)V_i^w$, and the landowner receives $(1+\tau_m^{\text{sell}})\bar{\theta}_m V_i^D$ at the sale. By the argument of Lemma B.4, the landowner exercises on

$$(1+\tau_m^{\text{sell}})\bar{\theta}_m \left[V_i^D - \frac{1 - t_m^w}{(1+\tau_m^{\text{sell}})\bar{\theta}_m} V_i^w \right],$$

which is proportional to S_i for every parcel and every rent if and only if (14) holds. A positive factor does not change the optimal stopping rule, so the threshold is then R_i^{CE}. Conversely, if (14) fails, the bracket differs from S_i on every parcel with $V_i^w > 0$, and the threshold differs from R_i^{CE} by Proposition 1. The subsidy multiplies every offer by $1 + \tau_m^{\text{sell}}$, so parcel supply (7) depends on $(1+\tau_m^{\text{sell}})\theta_{jm}$ and $(1+\tau_m^{\text{sell}})\Theta_m$. With η^S and η^D fixed, the elasticity $\partial \ln L_{jm}/\partial \ln \theta_{jm}$ is unchanged, and so are the first-order condition, the offer θ_{jm}, and the developer's rent per unit of developer value. The tax does not enter

offers. A parcel blocked by concentration has $a_i^\theta \leq 0 < a_i$. Under (14) the landowner exercises on a multiple of S_i, whose slope is $a_i > 0$, so the parcel has a finite threshold. Thresholds depend on the pair only through (14), so supply and the fiscal revenue from projects are the same along it. Moving along (14) towards a higher t_m^w lowers τ_m^{sell}, which lowers the outlay and raises the revenue of the tax. The net balance is therefore continuous and strictly increasing along (14). If it is negative at the pure subsidy and positive at the pure tax, it crosses zero exactly once. (ii) Every parcel that the constrained-efficient allocation redevelops has $a_i > 0$, that is $\tilde{h}_i^{\text{old}} < q_i$. Then $h_i^{\text{old}} = \tilde{h}_i^{\text{old}}/(1 - t_m^F \psi^{\text{new}}) < \kappa_i \bar{h}_i/[(1+\tau^M)(1 - t_m^F \psi^{\text{new}})] \leq \bar{h}_i$ under the stated condition, so the parcel is in the base. A parcel outside the base has $a_i \leq 0$, is never redeveloped in the constrained-efficient allocation, and has no threshold to correct. Parcels without housing have $\tilde{h}_i^{\text{old}} = 0$. If the base is restricted to them, part (i) applies to them, and parcels with housing keep the private threshold $R_i^\star > R_i^{\text{CE}}$. (iii) For the transfer for the fixed cost, see the next paragraph. For the gap tax, the standing value becomes $V_i^w - \tau^U G_i R/(\rho - \delta)$, with $G_i \equiv (\chi \bar{h}_i - h_i^{\text{old}})^+$. The landowner exercises on $\bar{\theta}_m V_i^D - V_i^w + \tau^U G_i R/(\rho - \delta)$. This equals $\bar{\theta}_m S_i$ at every rent if and only if $\tau^U G_i R/(\rho - \delta) = (1 - \bar{\theta}_m) V_i^w(R)$ at every rent, that is if $\tilde{V}_i^C = 0$ and $\tau^U G_i = (1 - \bar{\theta}_m)\tilde{h}_i^{\text{old}}$. On a parcel without standing value, the landowner exercises on $\bar{\theta}_m V_i^D + \tau^U G_i R/(\rho - \delta)$. In the constrained regime, this raises the slope of the private surplus and leaves its intercept unchanged, so the threshold falls below R_i^{CE}. Finally, $HHI_m \to 0$ gives $\bar{\theta}_m \to \theta^{\text{atom}}$, which is below one unless $\eta^D \to \infty$ (Section 3.4), and part (i) of Proposition 2 applies. □

The fixed-cost subsidy A lump-sum subsidy s_i to the fixed cost raises the developer value (4) by $s_i/(1+\tau^M)$, so the private surplus becomes $S_i^\theta + s_i/(1+\tau^M)$. In the constrained regime, the threshold becomes $\frac{\beta}{\beta-1}[K_i^\theta - s_i/(1+\tau^M)]/a_i^\theta$. It equals R_i^{CE} only for $s_i = (1+\tau^M)(K_i^\theta - K_i a_i^\theta/a_i)$, a function of the fixed cost and the standing value of each parcel. At $\bar{\theta}_m < 1$, a parcel with a standing value has $R_i^\star > R_i^{\text{CE}}$, so $\partial \text{TS}_i/\partial r < 0$ at $R_i^\star$, and a small uniform subsidy that lowers its threshold raises total surplus to first order. A parcel without one has $R_i^\star = R_i^{\text{CE}}$, which maximises TS_i, so the same subsidy lowers total surplus to second order. If $a_i^\theta \leq 0$ and $s_i \leq F_i$, the private surplus is $a_i^\theta R - [K_i^\theta - s_i/(1+\tau^M)]$, with $K_i^\theta - s_i/(1+\tau^M) \geq (C_i + \tau^A \bar{h}_i)/(1+\tau^M) + \tilde{V}_i^C/\bar{\theta}_m > 0$, so it is negative at every rent. □

Proof of Proposition 4 (i) If $r_i = (1+\omega) r_i^b$ for every parcel, the set $\{i : r_i \leq r\}$ equals $\{i : r_i^b \leq r/(1+\omega)\}$ for every regional rent r. The private economy converts at rent r exactly the parcels benchmark b converts at $r/(1+\omega)$, so the order of conversion and the set of converted parcels for any quantity of new floorspace coincide. (ii) In the constrained regime, Proposition B.5 gives $R_i^\star/R_i^{\text{CE}} = (K_i^\theta/K_i)(a_i/a_i^\theta)$, with $K_i^\theta = K_i + (1/\bar{\theta}_m - 1)\tilde{V}_i^C$, $a_i^\theta = a_i - (1/\bar{\theta}_m - 1)\tilde{h}_i^{\text{old}}/(\rho - \delta)$, and $a_i = (q_i - \tilde{h}_i^{\text{old}})/(\rho - \delta)$. Substituting gives (15). Both the numerator and the inverse of the denominator equal one if $\tilde{V}_i^C = \tilde{h}_i^{\text{old}} = 0$, and both increase with $1/\bar{\theta}_m - 1$. □

Proof of Proposition 5 For a fixed stopping time τ, $\mathbb{E}[e^{-\rho\tau} S_i^\theta(R_\tau; F)]$ is affine and strictly decreasing in F, with slope $-\mathbb{E}[e^{-\rho\tau}]/(1+\tau^M) < 0$ whenever τ is finite with positive probability. The supremum of a family of continuous decreasing functions is continuous and decreasing, and strictly so because the optimal stopping time is finite with positive probability for every finite F. As $F \to \infty$, the threshold diverges. Since the payoff is eventually affine, $\mathcal{O}_i^\theta(R_{i0}; F) \leq a_i^\theta R_{i0}^\beta (R_i^\star)^{1-\beta} \to 0$ because $\beta > 1$. The map $F \mapsto \mathcal{O}_i^\theta(R_{i0}; F)$ is therefore a continuous, strictly decreasing bijection from $[0, \infty)$ onto $(0, \mathcal{O}_i^\theta(R_{i0}; 0)]$. The observed price of an unsold parcel is its value to the landowner, $\hat{p}_i^L = V_i^w + \bar{\theta}_m \mathcal{O}_i^\theta(R_{i0}; F_i)$ by Lemma B.4, so a unique $F_i \geq 0$ exists if and only if (16) holds. □

Proposition B.6 (Concentration lowers supply at every horizon). *Holding the demand path fixed, the number of parcels redeveloped by any date is weakly lower under the markdown than in the constrained-efficient allocation. Delivered floorspace is weakly lower whenever parcels build at the cap. If the running maximum of rents grows without bound, the gap converges to the floorspace of parcels blocked by concentration (Proposition 1), and to zero if no parcel is blocked.*

Proof. By Proposition 1, $R_i^\star \geq R_i^{\mathrm{CE}}$ for every parcel. On every demand path, the running maximum is nondecreasing, so the start date $\inf\{t : \phi_m \mathcal{M}_t \geq R_i^\star\}$ is weakly later under the markdown. The set of parcels started by any date is therefore weakly smaller. Parcels that build at the cap deliver $\bar{h}_i - h_i^{\mathrm{old}}$ whenever they start, so delivered floorspace is weakly lower. If $\mathcal{M}_t \to \infty$, every parcel with a finite threshold eventually starts under both regimes, and the gap converges to the floorspace of parcels with $a_i^\theta \leq 0 < a_i$. □

Proposition B.7 (Where each wedge acts)**.** *Parcel i starts at the first date τ_i at which the running maximum reaches its threshold r_i, and it delivers at $\tau_i + \ell_i$.*

(i) Extensive margin. *The threshold r_i increases with the fixed cost F_i, the standing value ($\tilde{h}_i^{old}$ and $\tilde{V}_i^C$), the option premium $\beta/(\beta-1)$, the taxes τ^V, τ^F, and τ^M, and the development tax τ^A in the constrained regime. It increases with the lag ℓ_i if $\delta < \rho/2$. It decreases with the land share $\bar{\theta}_m$ for every parcel with a standing value.*

(ii) Intensive margin. *The density built at a given rent, $h_i(R)$, depends on the lag ℓ_i, the taxes τ^V and τ^F, and the cap $\bar{h}_i$. It does not depend on the fixed cost, the standing value, the option premium, the land-market taxes, or the land share.*

(iii) Delay after the start. *The time between start and delivery is the lag ℓ_i alone.*

Proof. (i) In the constrained regime,

$$r_i = \frac{1}{\phi_m} \frac{\beta}{\beta - 1} \frac{(\rho - \delta)\left[F_i + C_i + \tau^A \bar{h}_i + (1 + \tau^M)\tilde{V}_i^C/\bar{\theta}_m\right]}{\kappa_i \bar{h}_i - (1 + \tau^M)\tilde{h}_i^{\mathrm{old}}/\bar{\theta}_m},$$

with a positive denominator. The numerator increases with F_i, τ^A, τ^M, and $\tilde{V}_i^C$. The denominator decreases with τ^V, τ^F, τ^M, and $\tilde{h}_i^{\mathrm{old}}$. Both move against redevelopment as $\bar{\theta}_m$ falls (Proposition 1). The premium $\beta/(\beta-1)$ enters multiplicatively. For the lag, divide numerator and denominator by λ_i^R. The denominator, $(1-\tau^F)\bar{h}_i/(1+\tau^V) - (1+\tau^M)\tilde{h}_i^{\mathrm{old}}/(\bar{\theta}_m \lambda_i^R)$, falls with ℓ_i. The numerator rises, since F_i/λ_i^R rises and $C_i/\lambda_i^R = c_0 T_i^{-1/\zeta} \bar{h}_i^{(\zeta+1)/\zeta}/\Lambda_i$, where Λ_i decreases with ℓ_i when $\delta < \rho/2$: indeed $d \ln \Lambda_i / d\ell_i = -(\rho - \delta) + 1/\ell_i - \rho/(e^{\rho \ell_i} - 1) \leq \delta - \rho/2$. In the interior regime without standing housing, $(\phi_m r_i)^{\zeta+1} = \frac{\beta}{\beta - \zeta - 1}[F_i + (1+\tau^M)\tilde{V}_i^C/\bar{\theta}_m]/b_i^0$, where b_i^0 is proportional to $[(1-\tau^F)/(1+\tau^V)]^{\zeta+1}(\lambda_i^R)^{\zeta+1}/(\lambda_i^C)^\zeta$. The same monotonicity follows, and $d \ln b_i^0 / d\ell_i \leq -(\zeta+1)(\rho-\delta) + \zeta\rho/2 < 0$ when $\delta < \rho/2$. (ii) By (B2), $h_i^\star(R)$ depends on R, T_i, c_0, ζ, $\rho - \delta$, and $\kappa_i/\lambda_i^C = \Lambda_i(1-\tau^F)/(1+\tau^V)$, and increases with the latter. None of F_i, $\tilde{h}_i^{\mathrm{old}}$, $\tilde{V}_i^C$, β, τ^M, or $\bar{\theta}_m$ enters it, and τ^A does not by the treatment of the development tax. (iii) Delivery occurs ℓ_i years after the start, with ℓ_i deterministic by Assumption 3. □

Proposition B.8 (Relaxing the cap)**.** *Consider a parcel with $a_i^\theta > 0$ that stays in the constrained regime. Its threshold satisfies (19). Since $\varepsilon_i^a \geq 1$, a higher cap brings redevelopment forward whenever $C_i/\zeta \leq F_i + (1+\tau^M)\tilde{V}_i^C/\bar{\theta}_m$. On a parcel without standing value, it delays redevelopment if and only if $C_i/\zeta > F_i$.*

Proof. In the constrained regime, $\ln R_i^\star = \ln[\beta/(\beta-1)] + \ln K_i^\theta - \ln a_i^\theta$. By Proposition B.5, $(1+\tau^M)K_i^\theta = F_i + C_i + \tau^A \bar{h}_i + (1+\tau^M)\tilde{V}_i^C/\bar{\theta}_m$. Since $C_i \propto \bar{h}_i^{(\zeta+1)/\zeta}$, $\bar{h}_i \, dC_i/d\bar{h}_i = (1 + 1/\zeta)C_i$, so $d \ln K_i^\theta / d \ln \bar{h}_i = \varepsilon_i^K$. Since $q_i \propto \bar{h}_i$, $d \ln a_i^\theta / d \ln \bar{h}_i = q_i/(q_i - \tilde{h}_i^{\mathrm{old}}/\bar{\theta}_m) = \varepsilon_i^a$, which is at least one, and equal to one if and only if $\tilde{h}_i^{\mathrm{old}} = 0$. Finally, $\varepsilon_i^K \leq 1$ if and only if $C_i/\zeta \leq F_i + (1+\tau^M)\tilde{V}_i^C/\bar{\theta}_m$. On a parcel without standing value, $\varepsilon_i^a = 1$ and $\tilde{V}_i^C = 0$, so the threshold rises if and only if $C_i/\zeta > F_i$. □

C. Calibration appendix

C.1. Demand

I reconstruct $\log \hat{A}_t = \log R_t + \alpha \log H_t$ annually over 2015–2025, with R_t the regional hedonic rent index and H_t the quality-adjusted housing stock in square metres. Rents are observed over 2015–2024. Table C1 flags the years in which the rent is interpolated or the stock extrapolated. Under Assumption 1, $\Delta \log A_t = (\mu - \sigma^2/2) + \sigma\epsilon_t$ with independent standard normal draws ϵ_t. The volatility of the rent-based shifter is $\hat{\sigma}_R^2 = \widehat{\mathrm{Var}}(\Delta \log \hat{A}_t)$ over the 7 consecutive years in which both rent and stock are observed, with a block-bootstrap standard error. Interpolated years are left out because interpolation removes variance. The drift is estimated from the long difference between the first and the last year in which both are observed, 2015–2024: $\hat{\mu} = [\log \hat{A}_{t_1} - \log \hat{A}_{t_0}]/(t_1 - t_0) + \hat{\sigma}_R^2/2 = 1.54\%$, with a standard error of $\hat{\sigma}_R/\sqrt{t_1 - t_0} = 0.21$ points. I set $\mu = 1.5\%$. The mean of the observed annual changes, 1.9%, is not a good estimate of the drift here: it drops the annual changes next to the interpolated years, in which the rent index barely moved, and so overweights the years of fast growth. The series is short, so $\hat{\sigma}_R = 0.64\%$ is imprecise. Smoothed rent indices also bias it down, which raises β and lowers every threshold. I therefore set σ to the volatility of the transaction-based price index, 3.3%. Appendix Table F5 uses $\hat{\sigma}_R$ as a lower bound. The rent-based volatility depends on α through $\hat{A}_t$, and is estimated at the calibrated $\alpha = 0.52$.

Table C1: Demand shifter reconstruction

Year	Rent index	Stock (million m2)	Quality-adjusted stock (million m2)	log A	dlog A	Data	Used for sigma
2015	17.042	244.249	307.598	13.031	–	observed	no
2016	17.315	245.931	309.399	13.050	0.019	observed	yes
2017	17.316	247.344	310.914	13.053	0.003	rent interpolated	no
2018	17.317	248.816	312.480	13.055	0.003	observed	no
2019	17.540	250.401	314.189	13.071	0.016	observed	yes
2020	17.923	252.043	315.951	13.096	0.025	observed	yes
2021	18.173	253.982	317.980	13.113	0.017	observed	yes
2022	18.252	256.292	320.360	13.121	0.008	observed	yes
2023	18.713	258.115	322.224	13.149	0.028	observed	yes
2024	19.039	259.989	324.125	13.169	0.020	observed	yes
2025	19.337	261.740	325.943	13.188	0.018	rent interpolated	no
mu, long difference (%)	–	–	–	–	1.537	–	–
s.e. (%)	–	–	–	–	0.213	–	–
mu, mean of the observed changes (%)	–	–	–	–	1.900	–	–
s.e. (%)	–	–	–	–	0.097	–	–
mu, used (%)	–	–	–	–	1.500	–	–
sigma, rent index (%)	–	–	–	–	0.638	–	–
s.e. (%)	–	–	–	–	0.138	–	–
sigma, price index: used (%)	–	–	–	–	3.269	–	–

Data: rent index (*Observatoires locaux des loyers*), land tax records, transaction register. Annual $\hat{A}_t$ at $\alpha = 0.52$. Data: whether the rent and the stock of the year are observed, interpolated, or extrapolated. Used for σ: the annual change is between two consecutive years in which both are observed. Bottom rows: drift estimated from the long difference over the observed years, with its standard error; mean of the observed annual changes plus $\hat{\sigma}^2/2$, with its bootstrap standard error; the drift used; $\hat{\sigma}_R$ with its bootstrap standard error; and the volatility of the transaction-based price index, used for σ.

C.2. Discounting

The net residential yield y is annual rent net of property tax, non-recoverable charges, and vacancy, divided by price. I set $y = 2.6\%$, as developers report. The model applies the same deductions to the flows it capitalises. Rents are market rents, charges included, times 90%, and the standing building is valued net of its property tax. The ratio of these rents to transaction prices per square metre averages 4.0%, before the property tax. Capitalising them net of the property tax reproduces transaction prices per square metre with a mean absolute deviation of 42.1% at y and 33.7% at y^w. Transaction prices are those of standing dwellings, sold one by one. The commercial rental value v_i^C is capitalised at the yield $y^C = 6.5\%$, and its value $V_i^C = v_i^C/y^C$ is held constant (Section 3.2).

One yield per asset The model values three assets, and each trades in a different market. New units are sold vacant, one by one, to owner-occupiers and small investors. Their yield y is the one developers report. The standing building is sold whole, with its tenants. Sitting tenants can be asked to leave only at the end of their lease, and only to sell, to occupy, or for a serious reason. Their rents are capped at renewal, and by rent control in Paris since 2019 and in Plaine Commune and Est Ensemble since 2021. Since 2025, the least energy-efficient dwellings can no longer be let on a new lease, so many old buildings must be renovated before their units can be let again (Law 2021-1104 of 22 August 2021, *Climat et Résilience*). Each of these lowers the price of a whole tenanted building relative to its rent, so it trades at a yield y^w above y. Shops and offices carry commercial leases and a higher vacancy risk, and I set $y^C = 6.5\%$, the yield of secondary retail and offices. The capitalisation check above uses transaction prices of dwellings sold one by one. It fits slightly better at y^w than at y, but its deviations are large at both yields, so I do not use it to set y^w.

Calibration of y^w No market statistic reports the yield of whole tenanted buildings in the area, so I calibrate y^w on a restriction of the model rather than on a target. A landowner sells only if the price covers the standing value she gives up, the lower bound of (16). Every parcel a developer bought must therefore satisfy $\hat{p}_i^L \geq V_i^w$. I use the parcels bought by a developer since 2023 that are still in their pre-sale state in 2025, so that the standing building at the sale is observed. At $y^w = y$, 20.5% of them violate the restriction. The share falls with y^w, steeply at first and slowly beyond 5.5%. It cannot reach zero, since the hedonic price is measured with error: out of sample, the root mean squared error of log prices on whole-parcel sales is 0.46 (Appendix A.4). I set y^w at the smallest yield at which at most 10% of recent purchases violate the restriction, $y^w = 5.5\%$ (9.8%), 2.9 points above y. The tolerance is a choice, so Appendix Table F5 moves y^w by 50 basis points in each direction. The markdown then accounts for 0.9% and 1.0% of the shortfall, against 0.9% in the baseline, and the fixed cost for 30.4% and 31.6%.

Regional rather than municipal yields The restriction identifies one yield, not one per municipality: recent purchases are too few in most municipalities. Municipal ratios of rents to prices exist, but they measure the yield of dwellings sold one by one, not of whole buildings. Their variation across municipalities mostly reflects expected rent growth and risk. In the model, local rent levels enter through the location premium ϕ_m, and expected growth is common to the area. A municipal yield built from these ratios would load the same location variation a second time, on the yield as well as on the rent. I therefore keep the three yields regional, so that the cross-section of fixed costs reflects parcels rather than yields. The cost of this choice falls on Paris, where yields are lowest: the inversion attributes part of the location value of Paris parcels to their fixed cost (Section 4.1).

The perceived drift $\delta = 1.5\%$ is set close to the drift of the simulated rent. Along the simulated private equilibrium, the regional rent grows at 1.5% a year over twenty years. The discount rate follows, $\rho = y + \delta = 4.1\%$. It is the required return of the landowner of the parcel, who holds the option. For comparison, the average yield of 10-year French government bonds (*obligations assimilables du Trésor*, OAT) is 1.27% over 2015–2024 and 2.99% over 2023–2024. The cost of capital that Nexity, the main listed French developer, discloses for its residential segment in its IAS 36 impairment tests (7.9% in 2023 and 8.4% in 2024) lies 5.16 points above the OAT yield of the same years. Against the same benchmark years, ρ embeds a premium of 1.11 points. A developer's cost of capital remunerates development risk, which the landowner of a standing building does not bear, so the landowner's premium is lower. Appendix Table F5 reports how β and the main results move with y, δ and y^w ± 50 basis points. Because β is close to ρ/δ when σ is small, the option premium $\beta/(\beta-1)$ is close to ρ/y, and thresholds depend more on the two rates than on the volatility of rents.

C.3. Construction

PRLN reports, for each surveyed project j, its construction cost net of demolition and land. Taking logs of the variable-cost function of Assumption 3,

$$\log C_j^{\text{var}} = \log c_0 - \tfrac{1}{\zeta} \log T_j + \left(1 + \tfrac{1}{\zeta}\right) \log h_j + X_j' b + e_j, \tag{C1}$$

with project-type and year controls X_j. The coefficients on log parcel area and log floorspace sum to one, a testable restriction of the Cobb-Douglas technology. I impose ζ from Combes et al. (2021) and estimate $c_0 = 1,186$ from the intercept, at the mean of the controls, in a separate estimation on the PRLN microdata, accessed through the CASD. Costs are contemporaneous with the works and measured net of deductible VAT, so the lag factor λ_i^C is not applied a second time.

C.4. Construction lag

The lag ℓ_u uses the medians of Figure J1 (Fact 4), without a separate estimation. For each initial use, it sums the median time from the sale of the parcel to the permit, from the permit to the start of works, and from the start of works to completion (Table 3). The model has four initial uses. Empty parcels take the column of new buildings on empty land. Housing parcels average the demolitions of single-family houses, multi-family buildings, and outbuildings. Mixed parcels take the demolition of mixed-use buildings. Non-housing parcels average the demolitions of office and service, industrial, and vacant buildings. A developer usually signs a conditional sale agreement (*promesse de vente*) before the permit and the deed after it. The model dates the start of a project at the landowner's commitment, so the sale-to-permit step is measured on projects whose deed precedes the permit. The lags are $\ell_{\text{empty}} = 4.6$, $\ell_{\text{housing}} = 5.4$, $\ell_{\text{mixed}} = 6.7$, and $\ell_{\text{non-housing}} = 6.2$ years. The non-housing and empty-land lags rest on few permits, but the differences across initial uses are significant (Table I2).

C.5. Taxes

This appendix details the four taxes of Sections 3.3 and 3.4 and the levies left out of the model. Rates are those in force in 2025.

Value-added tax New dwellings sold by a developer are subject to VAT at the standard rate of 20% on the full sale price (article 278 of the *Code général des impôts*, CGI). The developer deducts the VAT paid on construction works, so VAT is a proportional wedge on the value of the completed building. Reduced rates apply to owner-occupied housing sold in urban renewal zones (5.5%) and to intermediate rental housing (10%). These programmes are a small share of private permits, and I apply the standard rate to every project. When the land is bought from a landowner who is not subject to VAT, no deductible VAT is attached to it, so VAT also falls on the land component of the price. The wedge $1/(1+\tau^V)$ on the value of the completed building captures this.

Transfer tax The transfer tax (*droits de mutation à titre onéreux*, DMTO) is paid by the buyer on the price of the land and of any building on it. It combines a departmental rate, a municipal rate of 1.2%, and a state levy of 2.37% of the departmental amount. The 2025 finance law allows *départements* to raise their rate from 4.5% to 5% between 2025 and 2028. I apply 5%, which gives $\tau^M = 6.32\%$. A developer who buys land subject to VAT and commits to build within four years pays a fixed duty instead (article 1594-0 G of the CGI). Most acquisitions in the sample are purchases of existing buildings from households, which do not qualify. The full rate therefore applies to most of them, and it overstates τ^M for the others.

Development tax The *taxe d'aménagement* is due on the floorspace authorised by the permit. Its base is a statutory value per square metre, 1,036 € in Île-de-France in 2025. Three rates apply: a municipal rate

between 1% and 5%, which can reach 20% in designated sectors, a departmental rate of at most 2.5%, and a regional rate of at most 1% in Île-de-France. I apply the three standard maxima, 5%, 2.5%, and 1%, which gives $\tau^A = 88$ € per square metre of new floorspace. The tax is paid after the permit, in one or two instalments. I treat it as paid at the start date.

Property tax The *taxe foncière sur les propriétés bâties* is paid annually by the landowner. Its base is half the cadastral rental value of the building, and the municipality sets the rate. Since 2021, the municipal rate includes the former departmental share. Cadastral rental values date from 1970 and are revised only by statutory coefficients, so the base is a small fraction ψ^{new} of the market rent. Rates vary across municipalities. I apply the municipal rate t_m^F of the DGFiP file of local tax rates (REI), which covers 100% of parcels. Across municipalities, the median rate is 34.1%, and the 10th and 90th percentiles are 20.5% and 43.5%. The base share is $\psi^{\text{new}} = 0.25$. New dwellings are exempt for two years after completion (article 1383 of the CGI). Municipalities may limit the exemption on their share of the base. I apply the full exemption, $e = 2$ years. The standing building pays the property tax too, at the same rate and on the same base share, with no exemption. The model deducts it from the standing value (1), which is then on the same basis as the net yield y. It also enters the fiscal accounts of Section 6.1, as the revenue lost once the building is emptied.

Levies left out The household waste levy (TEOM) and the special equipment tax are assessed with the property tax. They fall on old and new buildings alike, so they change the level of both values but not, to first order, the comparison between them. The preventive archaeology levy is small. Connection charges for sewerage (PFAC) depend on the network, and they are part of the fixed cost. The capital gains tax on the landowner's sale is the main omission. Gains on a principal residence are exempt. Other gains are taxed, with an abatement that grows with the holding period and exempts the gain after 22 years for income tax and after 30 years for social contributions. The tax lowers the net price a recent landowner receives, so it raises the landowner's threshold. It acts like the markdown on parcels held for a short time, and it vanishes for parcels held long.

C.6. Details of the main calibration

The perceived drift δ splits y into a discount rate and a growth rate. I set δ close to the realised drift of rents, so that perceived and realised growth are consistent. The discount rate follows, $\rho = y + \delta = 4.1\%$. It is the required return of the landowner, who holds the option, not the cost of capital of a developer. Over 2023–2024, the years in which listed developers disclose their cost of capital, it lies 1.11 points above the average 10-year French government bond yield, against 5.16 points for the developers' cost of capital (Appendix C.2). The perceived drift is $\delta = 1.5\%$. Appendix Table F5 moves y, δ and y^w by 50 basis points each.

Together with σ, these two rates determine the root β of (10), and hence the option premium. The calibrated root is $\beta = 2.59$, so the premium is $\beta/(\beta - 1) = 1.63$. The root exceeds $\zeta + 1 = 2.17$, so the threshold in the interior regime is finite (Appendix B.1). When σ is small relative to δ, β is close to ρ/δ, and the premium reflects expected rent growth more than uncertainty (Lange and Teulings, 2024).

The construction lag ℓ_u runs from the sale of the parcel to completion. I take it directly from Fact 4: ℓ_u is the last column of Table 3, the sum of three median steps by initial use. It ranges from 4.6 to 6.7 years. The first step needs care. A developer rarely buys land outright. It first signs a conditional sale agreement at a fixed price, which lapses if no permit is obtained. The deed is signed once the permit is final, so the recorded sale usually follows the permit. The model dates the start of a project at the landowner's commitment, and I measure the first step on projects whose deed precedes the permit (Appendix C.4).

No digitised cap exists for every parcel, so I infer $\bar{h}_i$ from the 90th percentile of building height and coverage within each cadastral section, with a floor-to-ceiling height of 2.5 metres. Figure A3 shows that most new buildings stay below it. Parcel area T_i comes from the cadastre. Standing housing floorspace h_i^{old} is the habitable surface of dwellings in the land tax records, converted to floorspace, without outbuildings. The commercial rental flow v_i^C is the rental value of business premises recorded by the tax administration, without the rental value of dwellings.

The location premium ϕ_m is the ratio of the municipal rent to the regional median. Paris counts as 20 districts, so the centre, where premia vary most, is finely divided. The hedonic model of land prices uses municipal effects at the same level (Appendix A.4). Within a municipality, the model therefore treats parcels as identical in location and different in everything else. The cost of this choice is that misallocation within municipalities is not measured. The recovered fixed cost still absorbs location value through the yield. I apply the regional yield to every parcel, whereas yields are lower in central Paris than in the inner suburbs. Where the true yield differs from the regional one, the model misprices both the standing and the new building, and the inversion attributes the error to the fixed cost. The highest recovered fixed costs are in central Paris, where yields are lowest (Figure F2).

Concentration may respond to local land values. I instrument the municipal index with the average index of the other municipalities of the same group of municipalities. Developer groups operate across such groups, so the instrument shifts the buyer side of the market. The exclusion restriction is that concentration elsewhere in the group affects the price of a parcel only through this channel, given département and year effects. It is open to shocks to land values common to a group of municipalities, such as new stations of the Grand Paris Express or the adoption of an intercommunal zoning plan, which département effects do not absorb. Two further concerns apply. The sample is selected on ripeness, which itself depends on the land share. And g_i is computed at a zero fixed cost, so the fixed cost enters the error, and it may be correlated with concentration. At the mean index $\overline{HHI}$, the model implies $\varsigma = b/(1 + b\,\overline{HHI})$.

The slope is imprecisely estimated: $b = -0.28$ (s.e. 0.38), which implies $\varsigma = -0.29$ with a 95% confidence interval of $[-1.38, 0.50]$ (Appendix Table D1). With 12 clusters, I also report a wild cluster bootstrap of the Anderson-Rubin test of $b = 0$, whose p-value is 0.53. With territory instead of département effects, $b = -0.52$ (s.e. 0.52). The data cannot reject that concentration leaves the land share unchanged. I calibrate the model at the point estimate, $\varsigma = -0.29$. The local concentration of Fact 3, developer margins, and the evidence that homebuilding is concentrated (Quintero, 2023; Favilukis et al., 2026;

Lefebvre and Daly, 2026; Charreire et al., 2025) make a markdown plausible. They do not pin down its size. Quantitatively, the markdown of the model rests on this choice, and Appendix Table F6 reports the main results at the lower bound of the interval, $\varsigma = -1.38$, the largest markdown the data allow, and at $\varsigma = 0$.

At $\varsigma = -0.29$, the margin target gives $\theta^{\text{atom}} = 90.3\%$, and the median model margin is 8.0%, against 8.0% for listed developers. The calibration gives $\eta^D = 9.30$ and $\eta^S = 2.31$. At observed concentration, landowners receive 81% of the developer value on average, between 72% and 86% at the 10th and 90th percentiles across municipalities. At the lower bound of the slope, θ^{atom} is at its upper bound and the mean land share is 66% (Appendix Table F5).

The elasticities come from a static supply of ripe parcels, while the timing problem follows landowners without idiosyncratic preferences . The two are not confronted by construction. As a check, I raise every offer by 5% and compute how much more developer value is sold in the model. No unsold parcel is ripe at the current rent (Section 3.7). Counting sales over twenty years, the implied elasticity of sales to offers is 0.51. The calibrated η^S is 2.31 (Appendix Table I4, Panel E). Forward-looking landowners facing a posted share could also wait for a better offer, which would raise commitment issues for developers that the model ignores.

D. Estimating developer market power

This appendix details the calibration of the land share summarised in Section 4.2. The land market is defined at the municipality level. I compute each developer group's share of purchases from the universe of redevelopment permits, after consolidating project-specific companies into their parent group. Each permit is valued at its floorspace times the local price of new housing, a proxy for the value of the project to the developer, since land payments are not observed for every permit. I call these value-weighted shares, not shares of land payments. Shares are computed over a rolling three-year window, which smooths lumpy projects.

D.1. The slope: municipal concentration

Equation (21) uses sales of housing parcels followed by a permit on the same parcel, filed at most one year before and 3 years after the sale, with the municipal Herfindahl index of the year of the sale. Mixed and non-housing parcels are excluded. For them, the model value g_i misses the value of the existing use, and the price often exceeds g_i. The index is instrumented with its average over the other municipalities of the same territory (EPT, Appendix A.5). Standard errors are clustered by territory, with 12 clusters. With so few clusters, the confidence interval uses the Student critical value with one degree of freedom less than the number of clusters, 2.20, rather than 1.96. I also report a restricted wild cluster bootstrap of the Anderson-Rubin test of $b = 0$, run on the reduced form of the log price ratio on the instrument after partialling out the controls and fixed effects, with Webb weights and 9,999 draws (Cameron et al., 2008). It is robust to few clusters and to a weak first stage. Its p-value is 0.53. The first stage has a Wald statistic of 37. With territory instead of département effects, $b = -0.52$ (s.e. 0.52), and by OLS $b = -0.44$ (s.e. 0.20). Territory effects absorb the time-invariant land value of each territory, but not shocks that vary over time within a territory, such as new stations of the Grand Paris Express or the adoption of an intercommunal zoning plan.

Posted offers Under a posted offer, the ratio of the price to the developer value is the same for every parcel a developer buys in a municipality, up to the fixed cost. I match each sale of the slope sample to the developer group of the permit that follows it, residualise the log ratio on log parcel area and year effects, and keep developer-by-municipality cells with at least two sales: 768 sales in 283 cells. Cell effects explain 74% of the variance of the residual. Its standard deviation is 0.55 across all sales and 0.28 within cells. The fixed cost, which the ratio omits, varies within cells, so this dispersion bounds the

dispersion of offers from above. The test cannot distinguish posted offers from bilateral bargaining. The mapping $\varsigma = b/(1 + b\,\overline{HHI})$ uses the mean index of the estimation sample, $\overline{HHI} = 0.174$, and is exact to first order around it. The slope is $b = -0.28$ (s.e. 0.38), so $\varsigma = -0.29$ (s.e. 0.42) with a 95% confidence interval of $[-1.38, 0.50]$ (Table D1). The calibration sets ς at its point estimate, as in Section 4.2, and Appendix Table F6 reports the results at the lower bound of this interval, the largest markdown the data allow, and at $\varsigma = 0$. Table F4 discusses how this choice and the others move the measured effect of the markdown. Incomplete consolidation biases measured indices down, but not uniformly. A uniform rescaling of the indices would be undone by the estimation: it would push b away from zero and leave $\varsigma\, HHI_m$, and hence the land shares, unchanged. Project companies, however, split the purchases of a group only where the group buys repeatedly. The error is therefore small in municipalities with few projects, whose index is high, and largest in municipalities with many projects, whose index is low (Table A6). Measured indices are then too dispersed, which biases b towards zero. Correcting the error raises $|\varsigma|\, HHI_m$ in every municipality, since the slope rises everywhere and the index rises where it was understated. Better consolidation therefore lowers the implied land shares and raises the efficiency gains.

D.2. The level: developer margins

The fixed costs absorb the level of the land share in sale prices. I pin it down with the operating margins of listed developers. Given ς, the land share in municipality m is $\bar{\theta}_m = \theta^{\text{atom}}/(1 - \varsigma\, HHI_m)$, so a single scalar remains. I set θ^{atom} so that the median model-implied operating margin equals the median margin of listed French developers, 8.0%. The margin is computed on every parcel of the estimation sample with a finite threshold, at that threshold $R_i^\star$. The margin of parcel i is the developer's rent, $(1 + \tau^M)(1 - \bar{\theta}_{m(i)})V_i^D(R_i^\star)$, relative to the value of the completed building. Both elasticities then follow from (22). With $\varsigma = -0.29$, the margins give $\theta^{\text{atom}} = 90.3\%$, $\eta^D = 9.30$ and $\eta^S = 2.31$. Through (9), the observed indices imply a land share of 81% on average, between 72% and 86% at the 10th and 90th percentiles across municipalities.

D.3. Joint solution for the level and the fixed costs

The fixed costs F_i are recovered by inverting observed land prices, given the land share. The model margin depends on these fixed costs. I solve for θ^{atom} and the fixed costs with a damped fixed point. Given θ^{atom} and ς, (22) gives η^D and η^S, and (9) gives $\bar{\theta}_m$ in every municipality. I then recover F_i for every parcel, compute its threshold $R_i^\star$ and its developer value relative to the value of the completed building. Holding these ratios fixed, a new θ^{atom} sets the median margin equal to the target exactly. The update averages the old and new values with equal weights. The procedure runs first on a random sample of 60,000 parcels, then on all parcels, until θ^{atom} changes by less than 10^{-4}.

Table D1: The land market: slope and calibrated parameters

Panel A. Slope of the land share in municipal concentration, Eq. (21)	
Statistic	Value
Slope of log land share on HHI, b	-0.279
s.e.	0.377
First stage (Wald)	37.3
Mean HHI	0.174
Implied ς (calibration)	-0.293
s.e.	0.417
Implied ς, 95% lower bound	-1.375
Implied ς, 95% upper bound	0.504
Land sales	1,631
Clusters (EPT)	12
Wild cluster bootstrap p-value, Anderson-Rubin test of $b = 0$	0.534
Wild cluster bootstrap p-value, 2SLS	–
b with EPT and year effects	-0.518
s.e.	0.524
b by OLS	-0.442
s.e.	0.197

Panel B. Calibrated land-market parameters, ς at its point estimate	
Slope ς	-0.29
Atomistic land share θ^{atom} (%)	90.3
Elasticity η^D	9.30
Elasticity η^S	2.31
Payment-weighted regional index $\overline{HHI}$	0.321
Land share, mean (%)	81
Land share, 10th to 90th percentile (%)	72–86

Notes: Panel A estimates (21) on housing land sales followed by a permit on the same parcel, by two-stage least squares with département and year effects. The municipal Herfindahl index is instrumented with its average over the other municipalities of the same territory. Standard errors clustered by territory. Panel B sets ς at its point estimate and combines it with the median operating margin of listed developers through (22).

E. Policies one at a time

E.1. Fiscal accounts

Let t_i be the start date of parcel i, measured from today. Let $\Phi_i(x)$ be the tax revenue a project collects in its first x years, discounted to its start date, with time t measured from the start date:

$$\begin{aligned}\Phi_i(x) = \tau^A h_i + \tau^M p_i^L + \mathbb{1}\{\ell_i \le x\}\, e^{-r_g \ell_i}\, \frac{\tau^V}{1+\tau^V}\, P_i^{\text{new}} \\ - \int_0^{\min(x,\, \ell_i+e)} e^{-r_g t}\, t_m^F\, b_{it}^{\text{old}}\, dt + \int_{\ell_i+e}^{\max(x,\, \ell_i+e)} e^{-r_g t}\, t_m^F \big(b_{it}^{\text{new}} - b_{it}^{\text{old}}\big)\, dt,\end{aligned} \tag{E1}$$

with the property-tax bases

$$b_{it}^{\text{old}} = \psi^{\text{new}} \big(e^{\delta t} R_{t_i} h_i^{\text{old}} + v_i^C\big), \qquad b_{it}^{\text{new}} = \psi^{\text{new}}\, e^{\delta t} R_{t_i} h_i. \tag{E2}$$

The development tax and the transfer tax, on the land price $p_i^L = \bar{\theta}_m V_i^D$, are paid at the start. VAT is collected at delivery on the sale price of the new units, $P_i^{\text{new}} = (1-\tau^F)\, e^{\delta \ell_i} R_{t_i} h_i / (\rho - \delta)$, which capitalises the property tax. Most new units are sold off plan, so part of the VAT is collected before delivery. The property tax on the standing building is lost from the start date, when the building is emptied. The new building pays from the end of the exemption. Housing bases start from the rent at the start date and grow at the perceived drift δ. The commercial base stays constant, as the commercial value does in the standing value (1). Here h_i^{old} and v_i^C are gross quantities. VAT accrues to the State and the other taxes to local governments, so the balance (24) is consolidated.

The welfare decomposition (26) values the same flows at fair value. Each flow is discounted at the rate at which the private side values it (Lucas, 2012). In (E1), r_g becomes ρ for the taxes and outlays at the start, VAT at delivery, and the property tax of the new building, as in the developer value. The property tax of the standing building is discounted at $y^w + \delta$ on the housing base and at y^C on the commercial base, and its housing base starts from today's rent, as in the landowner's standing value (1). The tax on standing value and the tax on under-used land follow the same rule. With these rates, a tax or a subsidy that leaves thresholds and densities unchanged moves value between the government and the private side and leaves welfare unchanged.

E.2. Lower taxes on redevelopment

Policy (b) lowers the three taxes that fall on redevelopment. Each instrument already exists in French law for some projects. The first applies the reduced VAT rate of 5.5% to every new housing unit, instead of 20%. The reduced rate already applies to new housing in priority neighbourhoods and to social housing. The second removes the transfer tax on land bought for redevelopment. A buyer who commits to build within four years already pays a fixed duty instead (article 1594-0 G of the tax code). The third removes the development tax, from which some social housing and some public facilities are already exempt. The rebates involve no outlay. Their cost is the revenue forgone on every project, including the projects that would have started anyway, net of the revenue the induced projects bring in.

The VAT rebate raises the stock by 4.65% after twenty years, lowers rents by 2.36%, and saves 9.7 years, at a net cost of 11.8 thousand euros per additional unit. Removing the transfer tax raises the stock by 0.32%, at 108.4 thousand euros per additional unit. Removing the development tax raises it by 0.84%. It pays for itself by year twenty, with a gain of 41.6 thousand euros per additional unit, but not by year fifty (Table F1). The three combined raise the stock by 5.88%, lower rents by 2.96%, and save 12.3 years. Their net cost by year twenty is 10.62 billion euros, 38.5 thousand euros per additional unit (Table E1). VAT accrues to the State, the transfer tax to the départements, and the development tax to municipalities, départements, and the region.

The three taxes are not equally costly to cut. The transfer tax lowers the number of sales and the mobility of landowners (Dachis et al., 2012; Kopczuk and Munroe, 2015; Best and Kleven, 2018), and its 2014 increase in France was capitalised into prices (Bérard and Trannoy, 2018). Public finance recommends replacing it with a tax on land value (Mirrlees, 2011). Cuts in VAT are not always passed on to buyers (Benzarti and Carloni, 2019).

Table E1: Policy (b): lower taxes on redevelopment

Scenario	dH (%)	dR (%)	Net cost (m€)	Net cost/unit (k€)	Time saved	Add. floorspace (m m2)	Loc. premium
Tax rebate: VAT at 5.5%	4.65	-2.36	2,562.19	11.82	9.67	14.95	1.07
Tax rebate: no DMTO	0.32	-0.17	1,658.52	108.43	0.50	1.05	1.07
Tax rebate: no development tax	0.84	-0.44	-1,661.69	-41.57	1.58	2.76	1.07
Tax rebate: all three	5.88	-2.96	10,619.80	38.52	12.33	19.01	1.07

Data: model estimation. Same columns as Table 9. Rows: VAT on new housing at the reduced rate, no transfer tax on land bought for redevelopment, no development tax, and all three. The net cost is the change in fiscal revenue by year 20, all taxes included.

E.3. A transfer to developers for the fixed cost

Policy (c) pays developers the part of the fixed cost above a deductible of k euros per square metre of parcel area, up to $\bar{s}$ euros per square metre. The outlay on parcel i is $s_i = \min(\bar{s}T_i, \max(F_i - kT_i, 0))$, and its threshold is evaluated at the fixed cost $F_i - s_i$. Like the *Fonds Friches*, the transfer funds the deficit of an operation rather than its whole cost, which requires the administration to observe the fixed cost. I search a grid of deductibles from 0 to 1,500 euros and of scales from 10 to 3,000 euros per square metre (Tables E3 and E4), and keep the design that raises the stock most while being self-financing by year twenty.

Without a deductible, the subsidy is self-financing by year twenty only up to 600 euros per square metre. A uniform subsidy pays every parcel, including parcels whose fixed cost is low and which need little help to start. A deductible keeps the transfer away from these parcels and allows a larger scale. The selected design pays the fixed cost above 1,000 euros per square metre, up to 1,100 euros, and reaches 89% of unsold parcels. It raises the stock by 4.93%, lowers rents by 2.50%, and saves 9.6 years, and it is self-financing by year twenty (Table E2).

The transfer pays for a friction and does not restore efficiency (Proposition 3(iii)). 92% of its outlay goes to parcels with a standing building, and 99% of the induced projects start below their constrained-efficient threshold. Its case rests on the budget, with three caveats. Counted until year 50, it has a net cost of 8,235 million euros, because much of the revenue it collects by year twenty is brought forward from later years. Local governments, which would plausibly fund it, recover 24% of the outlay, because VAT accrues to the State. And if part of F_i is a hold-out rent or an error in the computed developer value, the transfer raises the price of land instead of supply. Targeted parcel by parcel, as the offsetting subsidy of Appendix F.3, it raises the stock by 4.38%, 0.9 times as much, but this requires the threshold of each parcel.

Table E2: Policy (c): a transfer to developers for the fixed cost

Scenario	dH (%)	dR (%)	Net cost (m€)	Net cost/unit (k€)	Time saved	Add. floorspace (m m2)	Loc. premium
Fonds Friches scale (50 euros/m2)	0.33	-0.17	-748.11	-44.93	0.58	1.15	1.07
Largest self-financing subsidy (above 1000, up to 1100 euros/m2)	4.93	-2.50	-695.03	-2.92	9.58	16.39	1.05

Data: model estimation. Same columns as Table 9. The first row is the uniform subsidy at the scale of the *Fonds Friches* programme. The second is the selected design: the fixed cost above 1,000 euros per square metre of parcel area, up to 1,100 euros per square metre, the design of the grid that raises the stock most while being self-financing by year 20.

Table E3: Fixed-cost subsidy: the largest self-financing scale by deductible

Deductible (euros/m2)	Eligible parcels	Largest self-financing scale (euros/m2)	dH (%)	Outlay (m€)	Net cost, 20y (m€)	Net cost, 50y (m€)
0	310,489	600	3.94	20,044.38	-433.60	8,103.30
250	306,242	700	4.47	22,791.94	-88.14	9,279.54
500	299,155	800	4.82	23,618.45	-362.33	9,346.12
750	288,938	900	4.84	22,255.62	-894.87	8,386.62
1,000	275,087	1,100	4.93	22,049.46	-695.03	8,234.63
1,250	259,111	1,300	4.63	20,120.39	-585.59	7,454.43
1,500	242,043	1,500	4.13	17,374.66	-597.18	6,421.57

Data: model estimation. The subsidy pays the fixed cost above k euros per square metre of parcel area, up to $\bar{s}$ euros per square metre. A deductible of 0 is the uniform subsidy. For each deductible: parcels with a positive outlay, the largest scale self-financing by year 20 on a grid from 10 to 3,000 euros (counted from zero without a gap), and at that scale the change in the quality-adjusted stock after 20 years, the outlay by year 20, and the net cost to the consolidated budget by year 20 and by year 50, from the same simulation (positive: a cost).

Table E4: Fixed-cost subsidy: grid of scales at the selected deductible

Scale (euros/m2)	dH (%)	Outlay (m€)	Induced fiscal revenue (m€)	Net cost, 20y (m€)	Net cost, 50y (m€)
0	0.00	0.00	0.00	-0.00	-0.00
10	0.05	112.81	222.59	-109.78	-15.12
20	0.09	227.74	438.88	-211.14	-24.26
30	0.14	344.66	658.53	-313.86	-32.30
40	0.19	463.94	878.05	-414.11	-40.05
50	0.24	585.67	1,102.86	-517.19	-46.25
100	0.47	1,226.76	2,204.24	-977.48	-51.55
200	0.95	2,669.28	4,400.29	-1,731.01	114.97
300	1.42	4,306.72	6,575.39	-2,268.67	491.66
400	1.88	6,116.17	8,723.72	-2,607.55	1,036.90
500	2.34	8,067.66	10,856.47	-2,788.81	1,726.44
600	2.79	10,157.81	12,958.90	-2,801.09	2,550.53
700	3.24	12,358.61	14,992.51	-2,633.90	3,506.61
800	3.67	14,662.53	17,007.78	-2,345.25	4,551.19
900	4.10	17,050.57	18,969.48	-1,918.91	5,685.39
1,000	4.52	19,520.18	20,881.63	-1,361.45	6,923.06
1,100	4.93	22,049.46	22,744.49	-695.03	8,234.63
1,200	5.32	24,638.45	24,555.40	83.06	9,623.70
1,300	5.71	27,270.21	26,325.66	944.54	11,069.37
1,400	6.08	29,942.95	28,047.52	1,895.43	12,570.96
1,500	6.45	32,632.42	29,707.83	2,924.59	14,108.50
1,750	7.33	39,408.69	33,666.47	5,742.22	18,137.10
2,000	8.13	46,223.02	37,304.55	8,918.47	22,321.63
2,250	8.87	52,985.96	40,677.62	12,308.34	26,609.53
2,500	9.56	59,614.13	43,788.20	15,825.93	30,916.54
2,750	10.19	66,097.71	46,653.84	19,443.88	35,216.61
3,000	10.78	72,394.63	49,324.43	23,070.21	39,428.69

Data: model estimation. Deductible of 1,000 euros per square metre of parcel area. Columns: change in the quality-adjusted stock after 20 years, outlay and fiscal revenue induced relative to the private equilibrium by year 20, and net cost to the consolidated budget by year 20 and by year 50, from the same simulation (positive: a cost).

E.4. The instruments that restore efficiency

Policy (d): a subsidy to sellers Every landowner who sells a parcel for redevelopment in municipality m receives a subsidy at the ad valorem rate $\tau_m^{\text{sell}} = 1/\bar{\theta}_m - 1$ on the land price. The rate averages 24% across municipalities. The outlay on parcel i is $(1 - \bar{\theta}_m)V_i^D$: the State buys back the developer's rent on every parcel sold. The subsidy implements the constrained-efficient thresholds, so it raises the stock by 0.96% after twenty years, with 3.2 million square metres of additional floorspace, lowers rents by

0.50%, and saves 1.9 years. At fifty years the stock is 1.03% higher, barely more than at twenty. On the unsold stock, concentration mostly delays supply, so the gain does not build up with the horizon (Proposition B.6). Most of the gain is direct: with the running maximum of rents held at its private-equilibrium path, the stock rises by 1.10%, and the fall in rents crowds out the rest. The subsidy is paid to every seller, and 86% of the outlay goes to parcels that would have sold within the horizon anyway. Its net cost is 158.5 thousand euros per additional unit, about 1.9 times the construction cost of a unit. Restricting it to parcels with a standing building, which is observable, removes the outlay on empty land without changing supply (Proposition 1). It saves 447.0 million euros by year twenty and lowers the net cost to 148.7 thousand euros per additional unit (Table E5).

Policy (e): a tax on the standing value of under-used parcels Every parcel whose standing housing floorspace is below its cap pays, every year until it is redeveloped, a tax at rate $t_m^w = 1 - \bar{\theta}_m$ on its standing flows, with new buildings exempt. Section 6.4 reports its effects. Restricted to parcels without housing, the definition closest to the existing French tax on unused commercial premises, the tax raises the stock by 0.01% and 0.2 billion euros (Table E6). Few unsold parcels carry no housing and a valuable standing building, so the restricted base barely moves supply. It restores efficiency on non-housing parcels only, and adds misallocation between housing and non-housing parcels (Proposition 3(ii)).

The budget-neutral pair Any pair of rates along (14) implements the same thresholds. The pair that balances the consolidated budget by year twenty pays 80% of the seller-subsidy rate and taxes standing value at 4% on average. It moves the cost of correcting the markdown from taxpayers to the landowners of under-used parcels.

A fiscal swap The transfer tax is paid when a parcel is sold, so it delays sales. The tax on standing value is paid while a parcel is not redeveloped, so it brings sales forward. I replace the transfer tax on land bought for redevelopment with the tax on standing value of policy (e). The revenue of the tax covers 618% of the transfer-tax revenue forgone. The swap raises the stock by 1.22%, lowers rents by 0.63%, saves 2.5 years, and improves the net balance by 25.9 billion euros by year twenty.

What the other instruments do instead Table E7 reports, for each policy, the markdown distortion after the policy and the share of induced projects that start below their constrained-efficient threshold. The efficiency-restoring policies set the distortion to zero on every parcel. Supply policies do not. The transfer for the fixed cost shrinks the distortion on built parcels with high fixed costs and brings other parcels forward of their efficient date. The gap tax on under-used land does the same with empty land. At the upper bound of the box, 51% of the parcels it reaches start below their constrained-efficient threshold, and 11% of the parcels it induces carry no building, where the markdown is zero (Proposition 3(iii)). These instruments can raise welfare, because taxes hold private thresholds above the first best. They do not correct the market failure.

Table E5: Policies (d) and (e): restoring efficiency

Scenario	dH (%)	dR (%)	Net cost (m€)	Net cost/unit (k€)	Time saved	Add. floorspace (m m2)	Loc. premium	Inframarginal (%)	Change in owners' value (m€)
Subsidy to all sellers	0.96	-0.50	7,264.69	158.49	1.92	3.16	1.07	86.40	31,632.47
Subsidy to sellers of built parcels	0.96	-0.50	6,817.73	148.74	1.92	3.16	1.07	85.89	30,459.54
Tax on standing value	0.96	-0.50	-28,407.96	-619.77	1.92	3.16	1.07	–	-18,266.43
Budget-neutral pair (z = 80%)	0.96	-0.50	-0.01	-0.00	1.92	3.16	1.07	86.40	21,470.60

Data: model estimation. Same columns as Table 9, plus the share of the outlay paid to parcels that start by year 20 without the policy, and the change in the capitalised value of parcels to their landowners. Rows: subsidy to all sellers at rate $1/\bar{\theta}_m - 1$, subsidy restricted to parcels with a standing building, tax on the standing value of under-used parcels at rate $1 - \bar{\theta}_m$, and the budget-neutral pair along (14). Supply columns are identical across rows by Proposition 3.

Table E6: Policy (e): bases of the tax on standing value

Scenario	dH (%)	dR (%)	Net cost (m€)	Net cost/unit (k€)	Time saved	Add. floorspace (m m2)	Loc. premium	Tax revenue (m€)	Change in owners' value (m€)
Quantity base: housing below the cap	0.96	-0.50	-28,407.96	-619.77	1.92	3.16	1.07	23,289.22	-18,266.43
Use base: parcels without housing	0.01	-0.00	-231.97	-877.88	0.00	0.02	1.07	208.81	-196.21
Fiscal swap: quantity base, no DMTO	1.22	-0.63	-25,882.96	-443.75	2.50	4.02	1.07	23,116.24	-8,650.80

Data: model estimation. Same columns as Table 9, plus the revenue of the tax by year 20 and the change in the capitalised value of unsold parcels to their landowners. Rows: base restricted to parcels whose standing housing floorspace is below the cap (quantity), base restricted to parcels without housing (use), and the fiscal swap in which the quantity-based tax replaces the transfer tax.

Table E7: The markdown distortion under each policy

Scenario	Built: median	Built: IQR	No building: median	No building: IQR	Induced below the CE threshold (%)
Private equilibrium	0.03	[0.02, 0.06]	0.00	[0.00, 0.00]	–
Density: caps raised to the median	0.01	[-0.10, 0.03]	0.00	[-0.06, 0.00]	92.91
Density: caps lifted	0.38	[0.20, 0.51]	0.35	[0.27, 0.41]	3.96
Tax rebate: all three	-0.14	[-0.15, -0.13]	-0.14	[-0.15, -0.14]	99.99
Largest self-financing subsidy (above 1000, up to 1100 euros/m2)	-0.08	[-0.16, -0.01]	-0.15	[-0.22, -0.07]	99.39
Subsidy to sellers	0.00	[0.00, 0.00]	0.00	[0.00, 0.00]	0.00
Tax on standing value	0.00	[0.00, 0.00]	0.00	[0.00, 0.00]	0.00
Budget-neutral pair	0.00	[0.00, 0.00]	0.00	[0.00, 0.00]	0.00
Fiscal swap	-0.01	[-0.02, -0.01]	0.00	[0.00, 0.00]	99.99
Gap tax on under-used land at 10%	0.00	[-0.02, 0.03]	-0.04	[-0.05, -0.04]	63.35
Best rent-minimising mix	-0.43	[-0.60, -0.27]	-0.97	[-0.98, -0.38]	99.96
Welfare-maximising mix	0.20	[0.05, 0.31]	0.19	[0.12, 0.23]	84.70

Data: model estimation. Unsold parcels with a finite threshold. Columns: median and interquartile range of the distortion relative to the constrained-efficient threshold of the status quo, $R_i/R_i^{\mathrm{CE}} - 1$, under each policy, on built parcels and on parcels without a building, and share of the parcels induced by year 20 that start below their constrained-efficient threshold. A negative distortion means that the policy brings a parcel forward of its constrained-efficient date.

E.5. Who gains, policy by policy

Table E8: Who gains: welfare decomposition by group

Scenario	Renters: lower rents	Renters: surplus, incumbents	Renters: surplus, new residents	Incumbent landlords	Selling landowners	Developers	Government	Total	Renters' share (%)	Renters' share, closed city (%)
Density: caps raised to the median	8.68	0.09	0.11	-8.68	34.05	9.11	43.50	86.87	17.06	32.21
Density: caps lifted	-8.06	0.07	0.08	8.06	175.84	35.83	184.41	396.24	n.a.	n.a.
Tax rebate: all three	17.34	0.22	0.25	-17.34	51.57	13.05	-53.26	11.82	21.61	39.12
Largest self-financing subsidy (above 1000, up to 1100 euros/m2)	13.39	0.14	0.15	-13.39	11.61	4.85	-13.80	2.94	45.38	72.01
Subsidy to sellers	2.90	0.01	0.01	-2.90	28.50	0.66	-25.80	3.37	9.08	17.84
Tax on standing value	2.90	0.01	0.01	-2.90	-27.11	0.66	29.81	3.37	n.a.	n.a.
Budget-neutral pair	2.90	0.01	0.01	-2.90	17.17	0.66	-14.47	3.37	14.04	27.23
Fiscal swap	3.72	0.01	0.01	-3.72	-18.34	0.78	21.67	4.13	n.a.	n.a.
Best rent-minimising mix	120.54	13.50	15.26	-120.54	200.33	58.01	-85.03	202.08	36.63	56.92
Welfare-maximising mix	-3.68	0.03	0.03	3.68	230.05	54.57	129.83	414.50	n.a.	n.a.

Data: model estimation. Present values by year 20, relative to the private equilibrium, billion euros, equation (26). Selling landowners and developers include the value at year 20 of the parcels not sold by then, which differs from the change in capitalised value today reported in Tables E5 and E6. Government: receipts net of outlays at fair value, each flow discounted at the rate at which the private side values it, including the receipts and outlays after year 20. Columns: renters (lower rents on the existing stock, and the surplus on additional housing, split between incumbent and new residents), incumbent landlords, selling landowners, developers, government, and total. Last two columns: renters' share of the gains of renters, selling landowners, and developers, at the calibrated α and at the closed-city $\alpha = 1/\varepsilon^h$. n.a.: the share is undefined when a group loses, so that the gains do not all have the same sign.

F. Additional results

F.1. Fifty-year horizon

Table F1: Counterfactuals at 20 and 50 years

Scenario	dH (%)	dH (50y, %)	dR (%)	dR (50y, %)	Net cost/unit (k€)	Net cost/unit, (50y, k€)
Density: caps raised to the median	5.07	9.63	-2.60	-4.71	-77.88	-63.71
Density: caps lifted	-4.29	6.72	2.40	-3.38	–	-11.57
Tax rebate: VAT at 5.5%	4.65	5.74	-2.36	-2.86	11.82	84.80
Tax rebate: no DMTO	0.32	0.35	-0.17	-0.18	108.43	293.53
Tax rebate: no development tax	0.84	0.73	-0.44	-0.37	-41.57	2.08
Tax rebate: all three	5.88	6.74	-2.96	-3.33	38.52	112.65
Fonds Friches scale (50 euros/m2)	0.33	0.22	-0.17	-0.11	-44.93	3.62
Largest self-financing subsidy (above 1000, up to 1100 euros/m2)	4.93	4.03	-2.50	-1.99	-2.92	34.15
Subsidy to sellers	0.96	1.03	-0.50	-0.53	158.49	364.63
Subsidy to sellers of built parcels	0.96	1.03	-0.50	-0.53	148.74	347.47
Tax on standing value	0.96	1.03	-0.50	-0.53	-619.77	-867.09
Budget-neutral pair	0.96	1.03	-0.50	-0.53	-0.00	113.79
Fiscal swap	1.22	1.31	-0.63	-0.68	-443.75	-587.28
Targeted fixed-cost subsidy (offsets alone)	4.38	5.22	-2.23	-2.61	-19.40	15.06

Data: model estimation. Change in the quality-adjusted stock and in the rent, and net cost per additional unit of 60 square metres of living area, with units and fiscal revenue both counted until year 20 and until year 50.

F.2. The tax on under-used land

The rent-minimising mix of Section 6.3 contains a tax on under-used land with a gap base. An annual tax at rate τ^U on the forgone rent of the shortfall between a fraction χ of the density cap and the standing floorspace, paid until the parcel is redeveloped, lowers the standing value by the present value of that flow:

$$V_i^{w,\tau}(R) = V_i^w(R) - \tau^U \frac{(\chi \bar{h}_i - h_i^{\text{old}})^+ R}{y^w}. \tag{F1}$$

It raises the surplus by the same amount, lowers the threshold of every under-used parcel, and does not distort the density choice. Its base is largest where the standing building is smallest. The markdown distortion (15) is largest where the standing building is largest. The gap tax therefore does not restore efficiency (Proposition 3(iii)). It brings empty land forward of its constrained-efficient date and corrects too little on heavily built parcels. The tax on standing value of Section 6.4 replaces the gap $(\chi \bar{h}_i - h_i^{\text{old}})^+ R$ by the standing flows $(1 - t_m^F \psi^{\text{new}})(R h_i^{\text{old}} + v_i^C)$ and sets the rate at $1 - \bar{\theta}_m$. With these two changes, the tax corrects the markdown on every parcel. French law already allows municipalities in tight housing markets to raise the property-tax base of undeveloped constructible land (article 1396 of the tax code), and to tax commercial premises left unused (article 1530). The two taxes considered here extend these ideas to built parcels whose floorspace falls short of the cap.

Table F2: The gap tax on under-used land

Tax rate (%)	dH, 20 y (%)	dR, 20 y (%)	Time saved (y)	Unsold parcels redeveloped by year 20	Revenue, years 1-10 (m€/yr)	Revenue, years 11-20 (m€/yr)
1.00	0.12	-0.06	0.17	662.97	173.37	189.51
2.00	0.24	-0.12	0.42	1,340.17	345.81	377.66
5.00	0.58	-0.30	1.00	3,346.00	857.33	934.63

Data: model estimation. Annual tax at rate $\tau^U \in \{1\%, 2\%, 5\%\}$ on $R_{it}(\chi \bar{h}_i - h_i^{\text{old}})^+$ with $\chi = 0.5$. Columns: change in the quality-adjusted stock and in the rent after 20 years, years saved, additional idle parcels started by year 20 (expected number), and annual revenue in the first and second decades.

F.3. Inclusionary zoning with an offsetting subsidy

Since 2000, municipalities whose stock falls short of a target share of social housing, 20 to 25% depending on local circumstances, must write inclusionary requirements into their zoning plan and pay a penalty for missing the target. 45% of the municipalities of the area are currently short of their target. Lebret et al. (2026) show that inclusionary zoning in New York crowds out market-rate units, and that who bears the cost depends on the incentive paired with it. I evaluate the closest instrument in the French setting: a share $\iota = 25\%$ of new floorspace delivered as social housing at a rent capped at $v = 50\%$ of the market rent, paired with a subsidy to the fixed cost. The project payoff becomes

$$\pi_i^{\text{cs}}(R) = \max_{h \le \bar{h}_i} \frac{1-\tau^F}{1+\tau^V} \lambda_i^R \frac{[(1-\iota)+\iota v]\, R\, h}{\rho - \delta} - \lambda_i^C c_0 T_i^{-1/\zeta} h^{1+1/\zeta}, \tag{F2}$$

and the fixed cost net of the subsidy is $F_i - s_i$. Inclusionary zoning acts as a proportional tax on the value of the building at rate $\tau^{\text{IZ}} = \iota(1-v)$. For a parcel without standing value in the constrained regime, it scales the slope a_i^θ of the private surplus by $1-\tau^{\text{IZ}}$ and leaves its intercept K_i^θ unchanged, so it raises the threshold by the factor $1/(1-\tau^{\text{IZ}})$. The subsidy that offsets inclusionary zoning exactly solves $K_i^\theta - s_i/(1+\tau^M) = (1-\tau^{\text{IZ}})K_i^\theta$:

$$s_i = (1+\tau^M)\, \tau^{\text{IZ}} K_i^\theta. \tag{F3}$$

For every parcel I solve numerically for the smallest subsidy in $[0, F_i]$ that restores the threshold without inclusionary zoning. Where no such subsidy exists, I set $s_i = 0$ and inclusionary zoning applies without compensation.

Inclusionary zoning alone lowers the quality-adjusted stock by 3.57% after twenty years. Paired with the offsetting subsidy, the stock changes by -0.50%, and the policy delivers 5,132 thousand square metres of social floorspace, at an implicit cost of 1.40 square metres of private floorspace forgone per square metre of social floorspace. The outlay is 8,529 million euros by year 20. For 5% of parcels, inclusionary zoning cannot be offset by a subsidy bounded by the fixed cost (Table F3). Induced revenue does not finance the combined policy. They fall by 58.8% of its outlay, since social units pay VAT and property tax on a lower rent and the stock falls. Paid without inclusionary zoning, the same subsidies raise the stock by 4.38% and are self-financing, with receipts at 123.4% of the outlay (Section E.3).

Table F3: Inclusionary zoning, the offsetting subsidy, and the two combined

Policy	dH (%)	dR (%)	Social floorspace (thousand m2)	Private m2 forgone/social m2	Outlay (m€)	Receipts / outlay (%)
Inclusionary zoning only	-3.57	1.96	1,920.57	8.76	0.00	–
Offsetting subsidy only	4.38	-2.23	0.00	–	17,207.57	123.40
Inclusionary zoning with an offsetting subsidy	-0.50	0.26	5,132.03	1.40	8,529.26	-58.77

Data: model estimation. Rows: inclusionary zoning only ($\iota = 25\%$, $s_i = 0$), subsidy only (the offsetting subsidies, without inclusionary zoning), and inclusionary zoning with an offsetting subsidy. Columns: changes in the quality-adjusted stock and the rent at 20 years, social floorspace delivered, private floorspace forgone per square metre of social floorspace, outlay by year 20, and induced change in fiscal revenue as a share of the outlay. In the last row, fiscal revenue falls although the subsidy is paid: social units are let below the market rent, which lowers the VAT and property-tax bases, and the stock falls, so the change in fiscal revenue is negative.

F.4. Robustness

Tables F5 and F6 report the main results under alternative calibrations. Every variant re-inverts the fixed costs of the unsold stock at its own parameters (Proposition 5), then solves the private equilibrium, the Shapley decomposition of Section 5.3, and the subsidy to sellers on the same demand draws. Land shares, α, σ, and μ keep their baseline values unless the variant changes them.

The first four variants move the two financial rates by 50 basis points. Two move the yield of new units y and hold the perceived drift δ and y^w, so that $\rho = y + \delta$ moves with y. Two move δ and hold y, so

Table F4: Direction of the main biases in the measured effect of the markdown

Choice	Where	Effect of the markdown
Whole developer margin attributed to the markdown	Section 4.2	Overstated
Incomplete consolidation of project companies	Appendix A.6	Understated
Parcels blocked by concentration left out	Section 5.3	Understated
Estimation sample almost unbuilt	Section 5.1	Understated
Groups buy outside their main municipality	Section 2.3	Overstated
Markdown lowers the recovered fixed costs	Section 3.7	Understated
No depreciation of standing buildings	Section 3.5	Ambiguous

Effect of each modelling or measurement choice on the share of the supply shortfall, the delay, and the misallocation attributed to the markdown.

that ρ moves with δ. The estimation sample is sensitive to y: a lower yield raises the value of the new building and brings more prices within the bounds (16) (Appendix Table F5). The next two move the yield of the standing building y^w by 50 basis points. The markdown then accounts for 0.9% and 1.0% of the shortfall, and the fixed cost for 30.4% and 31.6%.

Two variants change the slope of the land share. One sets it at the lower bound of its confidence interval, $\varsigma = -1.38$, the other at zero. In each, θ^{atom} is solved again with the fixed costs to match the median developer margin, on the random sample of the first stage of Appendix D.3. At the lower bound, θ^{atom} is at its upper bound, the mean land share is 66%, and the markdown accounts for 1.8% of the shortfall. With no slope, every municipality has the same land share, the markdown no longer depends on concentration, and its share of the shortfall is 0.9%.

The baseline volatility comes from the transaction price index, 3.3%. The volatility variant uses the rent index instead, 0.64%, a lower bound. In both cases σ is small relative to δ, so β is close to ρ/δ and the option premium is close to ρ/y. With the rent index, the root rises to 2.73, and the fixed cost accounts for 30.9% of the shortfall.

The last variant brings the parcels priced above the zero-cost bound into the unsold stock with a zero fixed cost (Section 5.1, Appendix A.3). It bounds their contribution from above. The fixed cost then accounts for 23.2% of the shortfall and the markdown for 2.0%. The subsidy to sellers raises the stock by 1.57% after twenty years.

Table F5: Robustness: parameters and samples

Variant	ρ (%)	y (%)	y_w (%)	δ (%)	σ (%)	β	Premium $\beta/(\beta-1)$	θ^{atom} (%)	Mean $\bar{\theta}_m$ (%)	Unsold parcels with a fixed cost	Blocked by concentration	Median F_i/T_i (€/m²)	Paris / suburbs, median F_i/T_i
Baseline	4.10	2.60	5.50	1.50	3.27	2.59	1.63	90.29	80.81	310,489	1,392	3,145.51	4.03
y + 50 bp, δ fixed	4.60	3.10	5.50	1.50	3.27	2.87	1.53	90.29	80.81	226,133	2,677	1,475.83	7.29
y - 50 bp, δ fixed	3.60	2.10	5.50	1.50	3.27	2.29	1.77	90.29	80.81	367,125	687	7,692.18	2.76
δ + 50 bp, y fixed	4.60	2.60	5.50	2.00	3.27	2.23	1.81	90.29	80.81	326,933	1,392	4,228.03	3.97
δ - 50 bp, y fixed	3.60	2.60	5.50	1.00	3.27	3.22	1.45	90.29	80.81	278,609	1,392	2,441.33	3.50
y_w + 50 bp	4.10	2.60	6.00	1.50	3.27	2.59	1.63	90.29	80.81	315,185	1,069	2,957.45	4.21
y_w - 50 bp	4.10	2.60	5.00	1.50	3.27	2.59	1.63	90.29	80.81	299,771	1,846	3,400.09	3.89
Slope ς at its 95% lower bound (-1.38)	4.10	2.60	5.50	1.50	3.27	2.59	1.63	99.85	66.16	271,294	4,973	2,714.42	4.31
Slope $\varsigma = 0$	4.10	2.60	5.50	1.50	3.27	2.59	1.63	83.63	83.63	311,989	1,154	3,197.73	3.98
σ = 0.6%, rent index	4.10	2.60	5.50	1.50	0.64	2.73	1.58	90.29	80.81	303,903	1,392	2,917.53	3.91
Parcels priced above the zero-cost bound at $F = 0$	4.10	2.60	5.50	1.50	3.27	2.59	1.63	90.29	80.81	402,951	1,392	3,145.51	4.03

Data: model estimation. Columns: discount rate, yield of new units, yield of the standing building, perceived drift, volatility, root β of the value of waiting and option premium, atomistic land share and mean municipal land share, unsold parcels whose fixed cost is recovered, unsold parcels blocked by concentration, median recovered fixed cost per square metre of parcel area, and its ratio between Paris and the inner suburbs.

F.5. Regimes, supply paths and maps

Table F6: Robustness: decomposition and policy (d)

Variant	Median threshold / rent	Shapley: markdown (%)	Shapley: fixed cost (%)	Shapley: frictions (%)	Markdown removed alone (%)	Policy (d): dH (%)	Policy (d): dR (%)	Policy (d): time saved (y)
Baseline	1.91	0.94	30.91	53.76	0.81	0.96	-0.50	1.92
y + 50 bp, δ fixed	1.57	1.25	25.37	49.79	1.82	1.19	-0.61	2.17
y - 50 bp, δ fixed	2.66	0.86	31.14	57.73	0.26	0.51	-0.27	1.58
δ + 50 bp, y fixed	2.49	0.69	25.95	68.15	0.47	0.62	-0.33	2.17
δ - 50 bp, y fixed	1.48	0.83	24.15	43.66	1.35	1.30	-0.67	1.58
y_∞ + 50 bp	1.82	0.89	30.44	53.41	0.80	0.96	-0.50	1.75
y_∞ - 50 bp	2.00	0.99	31.60	54.26	0.80	0.93	-0.49	2.08
Slope ζ at its 95% lower bound (-1.38)	1.79	1.83	28.65	52.07	1.84	1.89	-0.97	4.00
Slope $\zeta = 0$	1.92	0.87	31.17	53.95	0.72	0.86	-0.45	1.67
σ = 0.6%, rent index	1.77	0.92	30.93	52.45	0.92	1.08	-0.56	2.25
Parcels priced above the zero-cost bound at $F = 0$	1.88	1.99	23.17	54.92	1.02	1.57	-0.81	2.75

Data: model estimation. Columns: median threshold relative to the current rent, Shapley shares of the twenty-year supply shortfall of the markdown, of the fixed cost, and of frictions, gain from removing the markdown alone, and the change in the quality-adjusted stock and in the rent after 20 years and the years saved under policy (d).

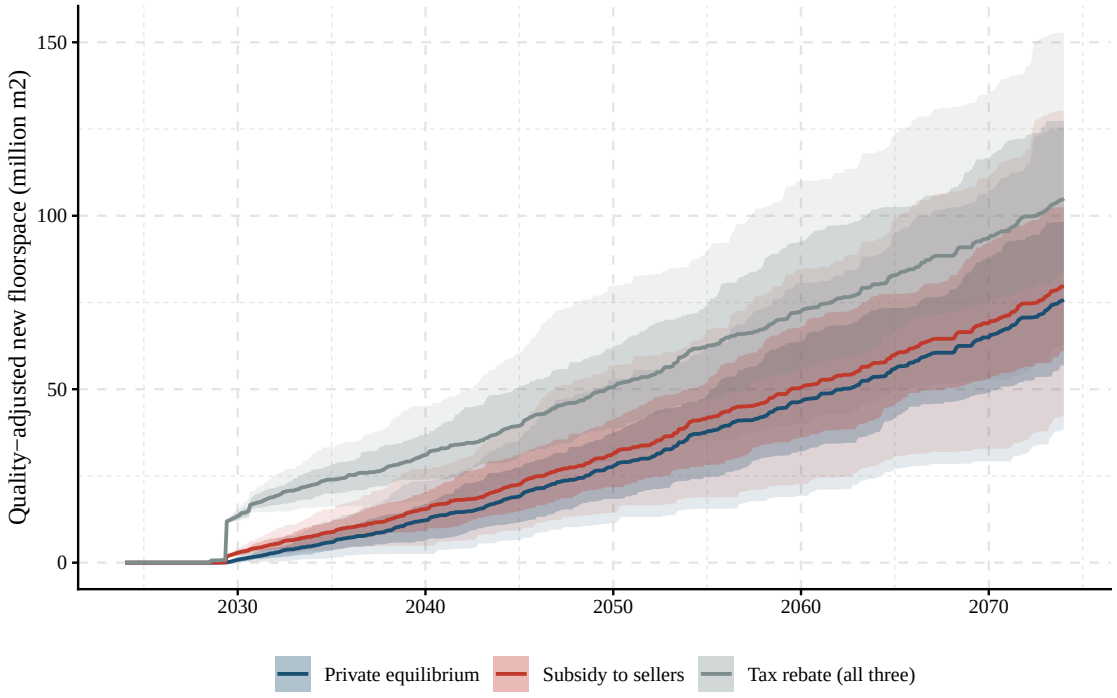


Figure F1: Simulated supply paths: private equilibrium, subsidy to sellers, tax rebate

Data: model estimation. Quality-adjusted new floorspace at the median demand path, with interquartile and interdecile ribbons across simulated draws of A_t, under the private equilibrium, the subsidy to sellers (Section 6.4), which reproduces the competitive land market, and the combined tax rebate (Section E.2). In the private equilibrium, no unsold parcel is ripe at the current rent (Section 5.2), so supply does not jump. Under the two policies, the parcels whose threshold the policy brings below the current rent start at once, which produces the jump once their construction lag has passed.

Table F7: Regime of parcels at their redevelopment threshold

Initial use	Blocked by concentration	Blocked by the cap	Never redeveloped	Constrained	Interior	No fixed cost recovered	Parcels
Housing	0.3	0.4	0.0	74.1	0.3	25.0	361,792
Mixed	0.5	0.9	0.0	24.8	0.3	73.4	114,055
Non-housing	0.0	0.0	0.0	34.0	2.1	63.9	4,095
Empty	0.0	0.0	0.0	80.5	2.1	17.4	18,315

Data: model estimation. All parcels of the base data. Share of parcels of each initial use, in %: blocked by concentration, blocked by the cap, with a recovered fixed cost but never redeveloped for another reason, in the constrained regime (density at the cap at the threshold), in the interior regime (density below the cap at the threshold), and without a recovered fixed cost (outside the bounds of (16)). Shares sum to 100 by row. Last column: number of parcels. Empty parcels carry no standing value, so none is blocked by concentration (Proposition 1).

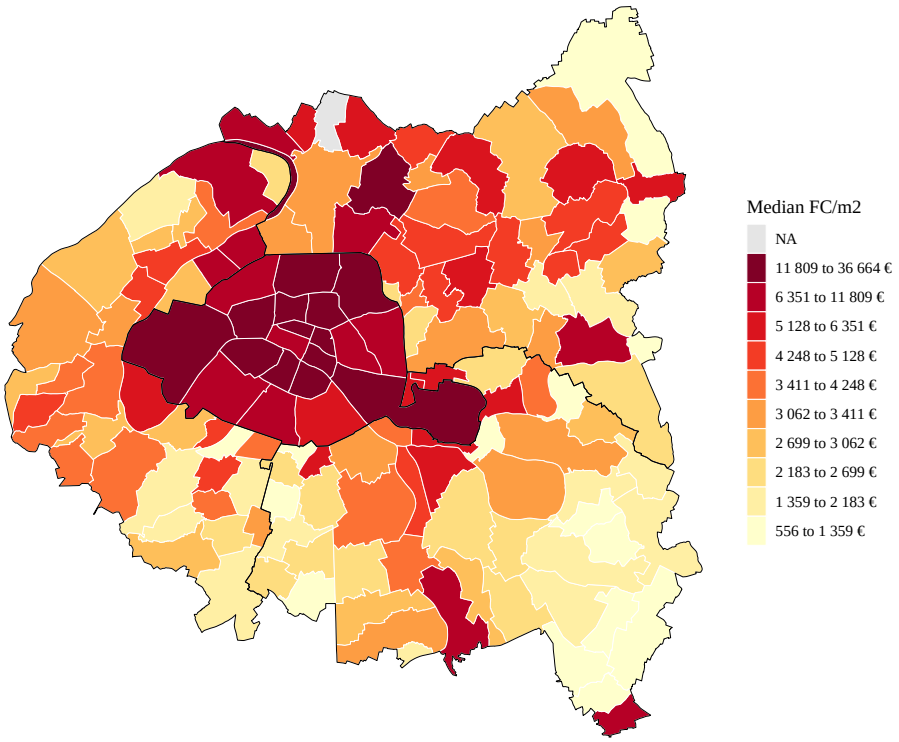


Figure F2: Fixed cost per square metre, by municipality

Data: model estimation. Median recovered fixed cost per square metre of parcel area, by municipality, parcels with a positive fixed cost.

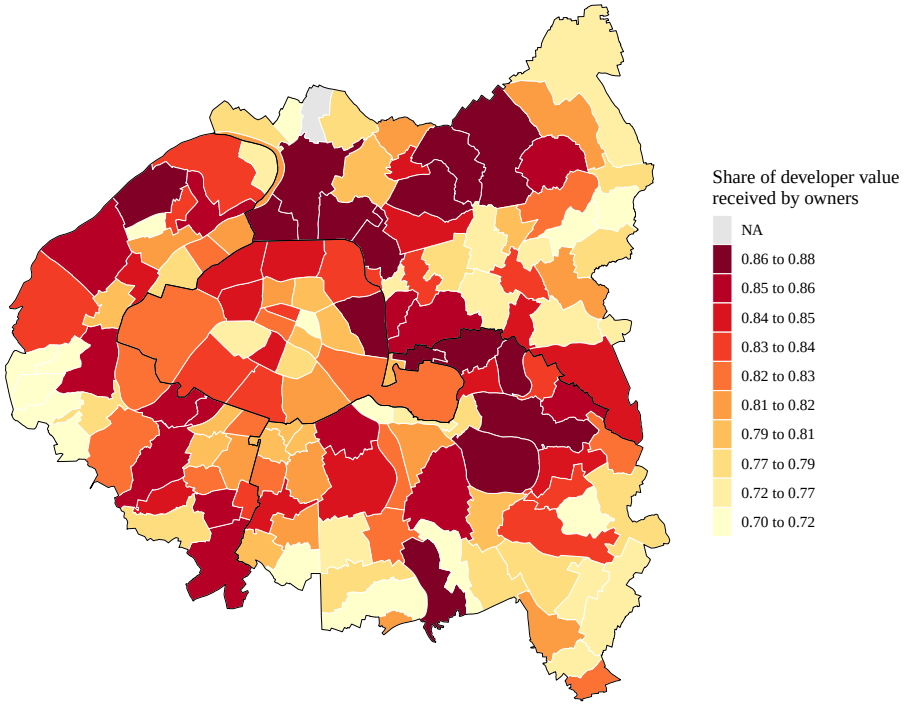


Figure F3: Land share $\bar{\theta}_m$, by municipality

Data: model estimation, permit register. Share of developer value received by landowners, from equation (9) at the estimated elasticities and the municipal Herfindahl index.

G. The policy mixes

G.1. Algorithm

Each instrument is mapped to the unit interval between the bounds of the reference box. The objective of (27) is evaluated by solving the thresholds of every parcel, simulating supply on the common demand paths, and computing fiscal revenue and outlays. A mix within budget always ranks above a mix that exceeds it. Mixes above budget rank by their violation. The search has three stages. The first is a differential evolution of 10 dimensions on the unsold parcels of département 93. Its stock is the regional stock times its share of potential floorspace, 26.6%, so that new supply moves its rent as much as it would move the regional rent. The population starts from a Latin hypercube, the status quo, and the last optimum. Each generation mutates every member towards the best, with a random weight, and keeps the better of the two. The evolution stops after eight generations without gain. The second stage refines the two best distinct solutions by a pattern search. Each iteration tries a step along every instrument and, in alternate iterations, along random orthogonal directions. A successful direction is repeated while it pays. A failed iteration halves the step, down to 0.1% of each range. The third stage repeats the pattern search on the whole region, with thirty steps in the search for each threshold instead of twenty.

G.2. Lifting the bounds

The main mix stays within the reference box. To see which bounds matter, a second pass widens them in rounds. At the start of each round, every instrument within 1% of a bound that is not a natural limit has that side of its range doubled, up to the natural limit. Natural limits are zero for a tax or a subsidy, one for a share, and the largest F_i / T_i in the unsold stock, 99,883 euros per square metre, for the transfer for the fixed cost. The incumbent keeps its natural values, so it lies inside the new box. A differential evolution of at most twenty generations on the département, seeded with the incumbent, looks for other basins in the widened box. A regional pattern search then starts from the incumbent and from the best solution of the evolution, if it differs. The pattern search never accepts a worse point, so the objective can only improve from one round to the next. Rounds stop when no instrument ends at a bound that can still be widened, or after eight rounds. An instrument whose range reaches 16 times its reference range is no longer widened. If it still ends at its bound, it is divergent: within the model, more of it always lowers rents within budget. Table G1 reports each round.
Without bounds, the mix lowers market rents by 14.02% on average over twenty years and raises the stock by 45.58%. No instrument ends at a bound that can be widened, so the search stops at round 0 and the mix without bounds is the main mix (Table G1). The objective still omits construction capacity, the shadow cost of public funds, and the welfare of landowners, which would all argue for a smaller mix.

Table G1: Expansion of the bounds, round by round

Round	Bounds widened	Average rent paid (%)	Market rent (%)	Stock (%)	Net cost (m€)	Evaluations
0	Reference bounds	-14.02	-14.02	45.58	-728	2,922

Data: model estimation. Round 0: optimum within the reference box. Each later round: instruments whose bound is widened, and the optimum after the regional search in the widened box. Changes relative to the private equilibrium: average rent paid and market rent, mean over 20 years, and quality-adjusted stock at year 20. Net public cost by year 20. Evaluations: cumulative number of mixes evaluated.

G.3. The rent-minimising mix: instruments and accounts

Table G2 reports each instrument of the rent-minimising mix against its status quo and its bounds. Table G3 reports the fiscal accounts and the incidence of the mix by year 20. Fiscal revenue rises by 183.9 billion euros. The tax on under-used land raises 18.5 billion, and the transfer tax 27.0 billion. Outlays are 183.2 billion euros, of which 174.9 billion go to the transfer for the fixed cost and 8.3 billion to

sellers. A transfer for the fixed cost raises the developer value, and the land market splits it: landowners receive their land share of it and developers keep the rest. A subsidy to sellers goes to landowners alone (Proposition 3). 43% of the additional floorspace is built in Val-de-Marne, and ten municipalities account for 32%.

Table G2: The best rent-minimising mix

Lever	Status quo	Reference box	Optimum	Status
Cap floor (FAR)	none	none – 12.47	3.23	interior
Development tax (x current)	x1.00	x0.00 – x1.00	x0.00	natural limit
Property tax (x current)	x1.00	x0.00 – x2.00	x0.00	natural limit
Property-tax exemption (years)	2.0 y	0.0 y – 15.0 y	12.8 y	not identified
Transfer tax	6.3%	0.0% – 10.0%	9.7%	interior
VAT on new housing	20.0%	5.5% – 20.0%	18.7%	interior
Tax on under-used land	0.0%	0.0% – 10.0%	9.8%	interior
Fixed-cost subsidy (euros/m2)	0	0 – 70,287	4,094	interior
Inclusionary zoning (social-housing share)	0.0%	0.0% – 50.0%	0.0%	natural limit
Subsidy to sellers (share z of the rate)	0.0%	0.0% – 100.0%	12.5%	interior
Average rent paid, change, mean over 20 years (%)	0.00		-14.02	
Market rent, change, mean over 20 years (%)	0.00		-14.02	
Market rent, change at 20 years (%)	0.00		-17.76	
Stock change at 20 years (%)	0.00		45.58	
Net public cost (m€)	0		-728	
Time saved (years)	0.0		14.6	
Social floorspace (thousand m2)	0		0	

Data: model estimation. Upper panel: status quo, bounds of the reference box, and optimum. Status: interior, at a bound of the box, or at a natural limit (zero for a tax or a subsidy, one for a share). Lower panel: outcomes relative to the private equilibrium, problem (27). The average rent paid counts social units at v times the market rent; the market rent does not. Net public cost by year 20 (negative: a gain). Bound of the transfer for the fixed cost: the 99.5th percentile of F_i/T_i in the unsold stock, 70,287 euros per square metre.

G.4. Extended results

Table G4 reports the contributions of Figure 4. With more than six active instruments, the Shapley values are averages over fifteen random orders and their reverses, thirty orders in all, and the table reports their standard errors. Figure G2 places each instrument within its bounds. Figure G3 plots the rent path of the optimum against the best instrument alone. Additional floorspace is positive in 140 of 142 municipalities. 95% of the outlays go to parcels with a standing building. Each euro of outlay raises the value of unsold parcels to their landowners by 1.09 euros.

Table G3: Fiscal accounts and incidence of the best rent-minimising mix

Account	Item	Billion euros
Public budget	Development tax	-2.17
Public budget	DMTO	27.02
Public budget	VAT	152.83
Public budget	Property tax, redeveloped parcels	-0.43
Public budget	Property tax, standing structures	-11.85
Public budget	Tax on under-used land	18.55
Public budget	Tax on standing value	0.00
Public budget	Change in fiscal revenue	183.94
Public budget	Fixed-cost subsidy	174.90
Public budget	Subsidy to sellers	8.32
Public budget	Outlays	183.22
Public budget	Net balance	0.73
Unsold parcels	Landowners	200.33
Unsold parcels	Developers	58.01
Housing market	Renters, lower rents on the stock	120.54
Housing market	Renters, surplus on additional housing	28.76
Housing market	Landlords of the existing stock	-120.54

Data: model estimation. Present values by year 20, relative to the private equilibrium, billion euros. Public budget at $r_g = 3.0\%$: revenue by tax, equation (E1), plus the tax on under-used land, the tax on standing value, and the change in the property tax of standing buildings. Unsold parcels at ρ: landowners' value, seller subsidy included, and developer rents $(1+\tau^M)(1-\bar{\theta}_m)V_i^D$, with the value at year 20 of the parcels not yet sold. Housing market at r_g: lower rents on the private-equilibrium stock, and the surplus $\frac{1}{2}\,dH_t\,dR_t$ on additional housing.

Table G4: Contribution of each instrument

Lever	Optimum	Solo (pp)	Leave-one-out (pp)	Shapley (pp)	s.e.
Cap floor (FAR)	3.23	0.25	-5.56	-2.23	0.47
Development tax (x current)	x0.00	-0.31	-1.22	-0.77	0.09
Property tax (x current)	x0.00	-0.70	-1.36	-0.97	0.09
Property-tax exemption (years)	12.8 y	-0.20	-0.00	-0.12	0.03
Transfer tax	9.7%	0.06	0.15	0.10	0.01
VAT on new housing	18.7%	-0.12	-0.34	-0.20	0.02
Tax on under-used land	9.8%	-0.40	-0.62	-0.50	0.03
Fixed-cost subsidy (euros/m2)	4,094	-7.13	-12.27	-9.25	0.42
Subsidy to sellers (share z of the rate)	12.5%	-0.05	-0.14	-0.08	0.01

Data: model estimation. Change in the average rent over 20 years, in percentage points. Solo, leave-one-out, and Shapley values as in Figure 4. Standard errors of the sampled Shapley values in the last column (zero when exact).

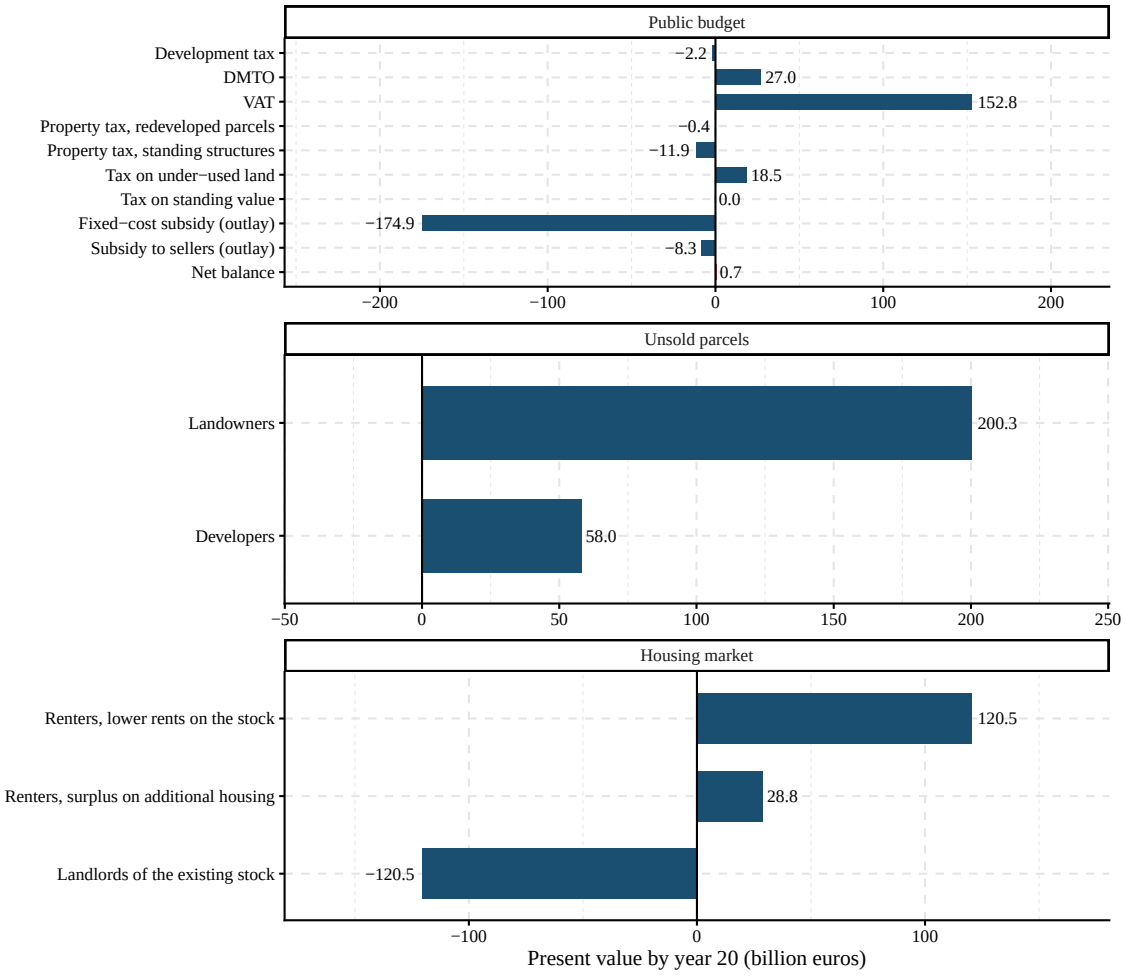


Figure G1: Fiscal accounts and incidence of the best rent-minimising mix

Data: model estimation. Present values by year 20, relative to the private equilibrium, billion euros. Outlays enter with a negative sign. Red: net balance of the consolidated budget. See Table G3.

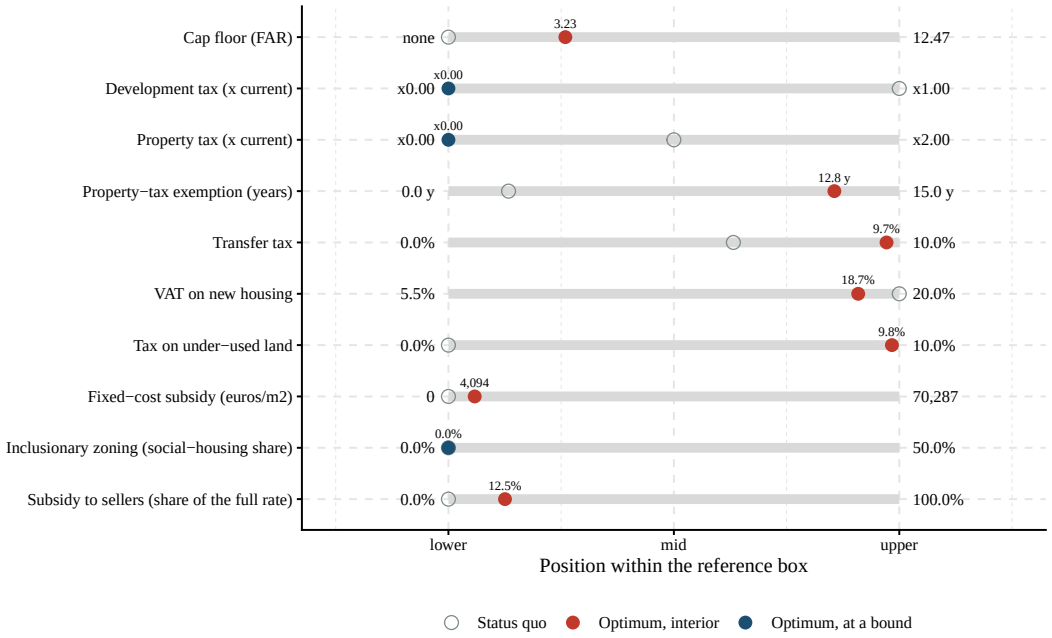


Figure G2: The instruments within their bounds

Data: model estimation. Position of the status quo and of the optimum within the reference box of each instrument.

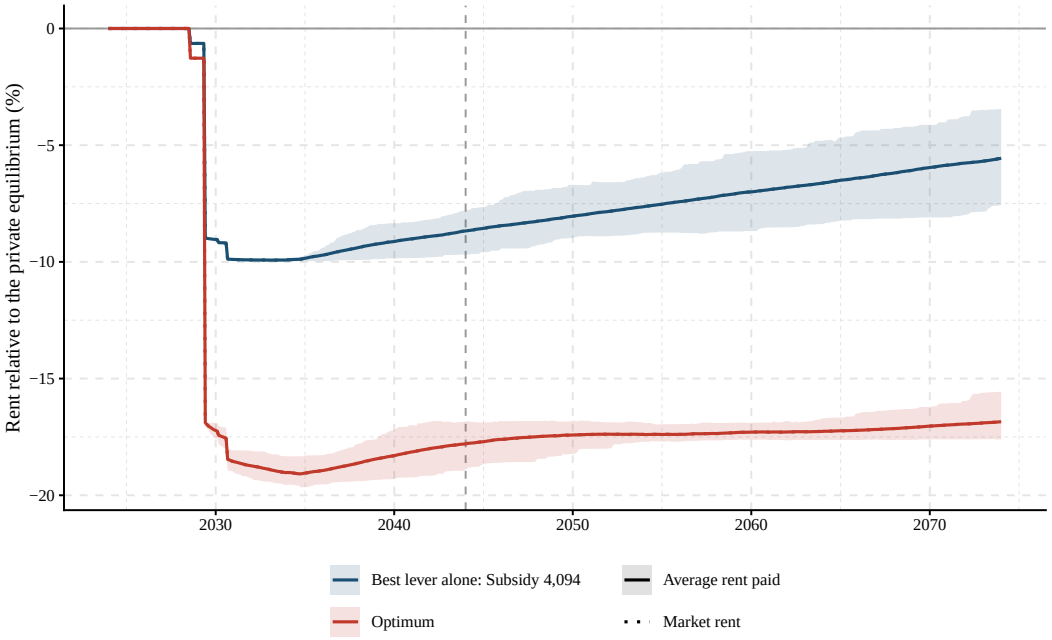


Figure G3: Rents under the best rent-minimising mix

Data: model estimation. Rent relative to the private equilibrium, path by path, mean across demand draws. Solid: average rent paid $R_t^{x,\text{paid}}$, which counts social units at v times the market rent, with its interdecile band. Dotted: market rent R_t^x. The text reports the market rent. Dashed vertical line: year 20.

H. Composition of the fixed cost

The fixed cost F_i is a residual (Section 3.7). This appendix asks how much of it observable characteristics explain. The sample contains the parcels of the estimation sample with a positive recovered cost. I regress the log fixed cost per square metre of parcel area on eight blocks of observables. *Demolition*: demolished floorspace per square metre of parcel area, the demolition cost predicted from the cost survey per square metre, and the height and age of the building. *Ground*: the ground class of the geological survey. Networks: connection to the gas or underground electricity network within 100 metres. *Neighbours*: the number of contiguous parcels and their mean height. *Ownership*: the number of landowners and a co-ownership indicator. *Area*: log parcel area. *Regulation*: the municipal refusal rate of permits and architectural protection. *Municipal*: population density, income, and distance to Paris. The predicted demolition cost accounts for a median 2.9% of the recovered fixed cost, and 1.1% in aggregate.

I apportion the R^2 of the regression, 0.17, across the eight blocks with a grouped Shapley decomposition: each block receives its average marginal contribution to the R^2 over every order in which blocks could be added (Table H1). The second column demeans every variable by municipality and drops the municipal block. Table H2 reports the share of the fixed cost that each block accounts for, relative to a reference parcel with nothing to demolish, the most frequent ground class, one landowner, and every other characteristic at its median.

Most of the variation in the fixed cost lies within municipalities: the municipal means explain 22% of the variance of $\log(F_i/T_i)$, and the observables explain 0.10 of the variance within municipalities. Predicted demolition costs are a small part of the recovered cost. The fixed cost is lower on parcels with more to demolish (Table H2), which is consistent with an overstated standing value of old buildings (Section 3.5): the inversion attributes part of the gap between price and developer value to a lower fixed cost.

Two features of the residual matter for policy. First, the recovered fixed cost depends on the land share. Given a price, a lower land share means the landowner keeps a smaller share of the option, so the option must be worth more and the fixed cost is lower. Setting the land share to one raises the aggregate fixed cost by 35% (Section 3.7). The comparison is on the same parcels: 0.4% of the parcels of the estimation sample lose their fixed cost when the land share is set to one, because their price then leaves the bounds of (16). Second, if the share of variance left unexplained is large, part of the residual may reflect a failure inside the fixed cost itself, such as co-ownership hold-outs or the premium of multi-parcel assembly, rather than a technological cost.

Table H1: Fixed cost: grouped Shapley shares of the explained variance

Block	All blocks (%)	Within municipality (%)
Demolition	9.10	20.29
Ground	21.72	11.75
Networks	0.01	0.00
Neighbours	11.93	19.71
Ownership	4.67	6.27
Area	7.93	28.57
Regulation	7.51	13.41
Municipal	37.13	–
R^2	0.17	0.10

Data: model estimation, land tax records, building database, geological survey, cost survey. Dependent variable: $\log(F_i/T_i)$. Each cell is the share of the R^2 attributed to a block (%). Last row: R^2. Within municipality: all variables demeaned by municipality.

Table H2: Fixed cost: decomposition of the level relative to a reference parcel

Block	Median share of F_i (%)	Aggregate share (%)	Median euros per m^2
Demolition	-0.22	-4.70	-3.86
Ground	1.92	-3.17	10.19
Networks	0.00	0.01	0.00
Neighbours	0.29	13.32	4.59
Ownership	0.00	5.71	0.00
Area	0.00	19.20	0.00
Regulation	0.55	9.18	15.04
Municipal	-4.49	-10.94	-82.04

Data: model estimation. For each block, the share of F_i the parcel would save with the characteristics of the reference parcel, $1 - \exp(-\hat{b}'(X_i - X^{\text{ref}}))$: median across parcels, aggregate share of total fixed costs, and median in euros per square metre of parcel area.

I. Tables

Table I1: Density of new projects by type of project and initial use

Initial use	Floorspace (m^2)	Units	Floor-area ratio	Parcels
New building on empty land				
Empty land	152.0	1	0.55	1
Conversion				
Outbuilding	149.0	1	0.62	1
Mixed use	103.0	2	0.21	1
Multi-family building	117.0	2	0.39	1
Industrial	260.0	5	0.30	1
Single-family house	141.0	1	0.43	1
Offices and services	263.0	7	0.58	1
Vacant building	154.0	1	0.56	1
Demolition and rebuilding				
Outbuilding	153.0	1	0.54	1
Mixed use	1141.5	19	1.18	1
Multi-family building	253.5	3	0.78	1
Industrial	2330.0	35	1.24	1
Single-family house	191.0	1	0.53	1
Offices and services	1439.0	26	1.76	1
Vacant building	191.5	1	0.58	1

Data: land tax records, building database, permit register. Median new floorspace, number of units, floor-area ratio (new floorspace over parcel area), and number of parcels, by type of project (new building on empty land, conversion, demolition and rebuilding) and initial use.

Table I2: Construction time across initial uses: Kruskal-Wallis and pairwise Wilcoxon tests

	Empty land	Outbuilding	Mixed	Multi	Industrial	Single	Offices
Outbuilding	0.170						
Mixed	0.050^{**}	0.617					
Multi	0.617	0.156	0.034^{**}				
Industrial	0.034^{**}	0.700	0.248	0.034^{**}			
Single	0.248	0.352	0.034^{**}	0.110	0.077^{*}		
Offices	0.007^{***}	0.975	0.050^{**}	0.004^{***}	0.617	0.001^{***}	
Vacant	0.110	0.617	0.617	0.050^{**}	0.189	0.248	0.050^{**}
Kruskal–Wallis: $\chi^2 = 32.7$, df $= 7$, $p = <0.001$, 2,910 permits							

Data: permit register. Kruskal-Wallis test of the joint null that construction time is equally distributed across initial uses. Benjamini-Hochberg adjusted p-values from pairwise Wilcoxon rank-sum tests between each pair of uses.

Table I3: Developer concentration

Panel	Statistic	Value
A. Region	Top 1 share (%)	4.7
A. Region	Top 4 share (%)	11.5
A. Region	Top 7 share (%)	16.1
A. Region	Top 20 share (%)	27.2
A. Region	HHI	0.007
A. Region	1/HHI	153.1
A. Region	N^{tot}, all developers	3857
A. Region	N^{tot}, at least 2 projects	1918
A. Region	N^{tot}, at least 5 projects	478
B. Municipalities (P10 / median / P90)	Municipal HHI	0.17 / 0.33 / 0.89
B. Municipalities (P10 / median / P90)	Effective number of developer groups	1.13 / 3.01 / 5.79
B. Municipalities (P10 / median / P90)	Share of the largest group (%)	27.70 / 49.89 / 94.08
B. Municipalities (P10 / median / P90)	Municipalities	142
B. Municipalities (P10 / median / P90)	... with a permit in the last window	137

Data: permit register, business register, 2013–2024. Panel A, region: cumulative share of redeveloped floorspace by the top 1, 4, 7, and 20 developers, Herfindahl index, effective number of developers, and number of active developers, in total and restricted to developers with at least two and at least five projects. Panel B, municipalities: distribution of the municipal Herfindahl index, of the effective number of developer groups, and of the share of the largest group.

Table I4: Validation

Panel	Statistic	Value
A	Hit rate among parcels predicted to start in 2019-2024, bound at 2019 (%)	0.61
A	Base rate (%)	0.73
A	Parcels predicted to start (%)	2.59
B	Model supply elasticity	0.39
B	Supply elasticity, Paris area, Chapelle et al. (2023)	0.35
B	Model floorspace started, unsold stock, first 10 years (million m2/y)	1.13
B	... without the parcels ripe at the current rent (million m2/y)	1.13
B	Sitadel floorspace authorised, estimation sample (million m2/y)	0.32
B	Sitadel floorspace authorised, all parcels of the base data (million m2/y)	0.80
B	Sitadel floorspace authorised, all parcels of the area (million m2/y)	1.32
B	Model supply elasticity, scaled to observed flow	0.46
B	Unsold parcels ripe at the current rent (%)	0.00
B	Perceived drift delta = rho - y (%)	1.50
B	Realised drift mu - alpha g_Q, observed stock (%)	1.20
B	Realised drift of rents, simulated private equilibrium (%)	1.49
C	Model operating margin (%), aimed at the target	8.00
C	Listed developers (%)	8.00
D	Reduced-form slope of log land share on HHI	-0.28
D	s.e.	0.38
D	Model-implied slope	-0.28
E	Elasticity of sales to offers, parcels ripe at the current rent	–
E	Elasticity of sales to offers, starts by year 20	0.51
E	Elasticity of selling eta_S (calibrated)	2.31
F, HHI quintile 1	Priced above the zero-cost bound (%)	11.86
F, HHI quintile 2	Priced above the zero-cost bound (%)	17.27
F, HHI quintile 3	Priced above the zero-cost bound (%)	32.87
F, HHI quintile 4	Priced above the zero-cost bound (%)	30.29
F, HHI quintile 5	Priced above the zero-cost bound (%)	16.27

Data: model estimation, transaction register, permit register, building database. Panel A: hit rate, base rate, and share of parcels predicted to start of the timing prediction over 2019–2024, with the fixed costs inverted under the non-exercise bound at 2019. Panel B: model and external supply elasticities; model flow on the unsold stock, with and without the parcels ripe at the current rent (none, by the non-exercise bound), and Sitadel flows on the estimation sample, on all parcels of the base data, and on all parcels of the area, permits with a recorded developer only; share of the unsold stock ripe at the current rent; perceived drift δ against the realised drift of rents, on the observed stock and along the simulated private equilibrium. Panel C: model-implied and observed developer margins (the target). Panel D: reduced-form and model-implied slope of the land share in the municipal Herfindahl index. Panel E: elasticity of sales, in euros of developer value, to a 5% rise in every offer, implied by the thresholds, against the calibrated η^S. Panel F: share of parcels priced above the zero-cost bound by quintile of the municipal Herfindahl index.

Table I5: The policies side by side

Scenario	dH (%)	dR (%)	Net cost/unit (k€)	Time saved	Loc. premium	MVPF, narrow	MVPF, broad
Density: caps raised to the median	5.07	-2.60	-77.88	6.42	1.02	∞	∞
Density: caps lifted	-4.29	2.40	–	-8.75	1.12	∞	∞
Tax rebate: VAT at 5.5%	4.65	-2.36	11.82	9.67	1.07	1.94	1.99
Tax rebate: no DMTO	0.32	-0.17	108.43	0.50	1.07	1.22	1.23
Tax rebate: no development tax	0.84	-0.44	-41.57	1.58	1.07	56.51	56.85
Tax rebate: all three	5.88	-2.96	38.52	12.33	1.07	1.67	1.71
Fonds Friches scale (50 euros/m2)	0.33	-0.17	-44.93	0.58	1.07	35.37	35.46
Largest self-financing subsidy (above 1000, up to 1100 euros/m2)	4.93	-2.50	-2.92	9.58	1.05	3.25	3.36
Subsidy to sellers	0.96	-0.50	158.49	1.92	1.07	1.14	1.14
Subsidy to sellers of built parcels	0.96	-0.50	148.74	1.92	1.07	1.16	1.16
Tax on standing value	0.96	-0.50	-619.77	1.92	1.07	–	–
Budget-neutral pair	0.96	-0.50	-0.00	1.92	1.07	2.32	2.33
Fiscal swap	1.22	-0.63	-443.75	2.50	1.07	–	–
Best rent-minimising mix	45.58	-17.76	-0.32	14.58	1.01	8.00	9.85
Welfare-maximising mix	-0.88	0.45	–	-1.17	1.06	∞	∞
Targeted fixed-cost subsidy (offsets alone)	4.38	-2.23	-19.40	9.08	1.06	6.98	7.19

Data: model estimation. Change in the quality-adjusted stock and in the rent after 20 years, net cost by year 20 per additional unit of 60 square metres of living area (thousand euros, negative is a gain), years saved, mean location premium of new floorspace, narrow and broad MVPF. Rows: relaxing caps (Section 6.2), the policies of Appendix E, and the rent-minimising and welfare-maximising mixes. Last row: the offsetting subsidies of Appendix F.3 paid without inclusionary zoning, a transfer for the fixed cost targeted parcel by parcel. The MVPF of a policy that gains for the budget and lowers the welfare of those affected is undefined.

J. Figures

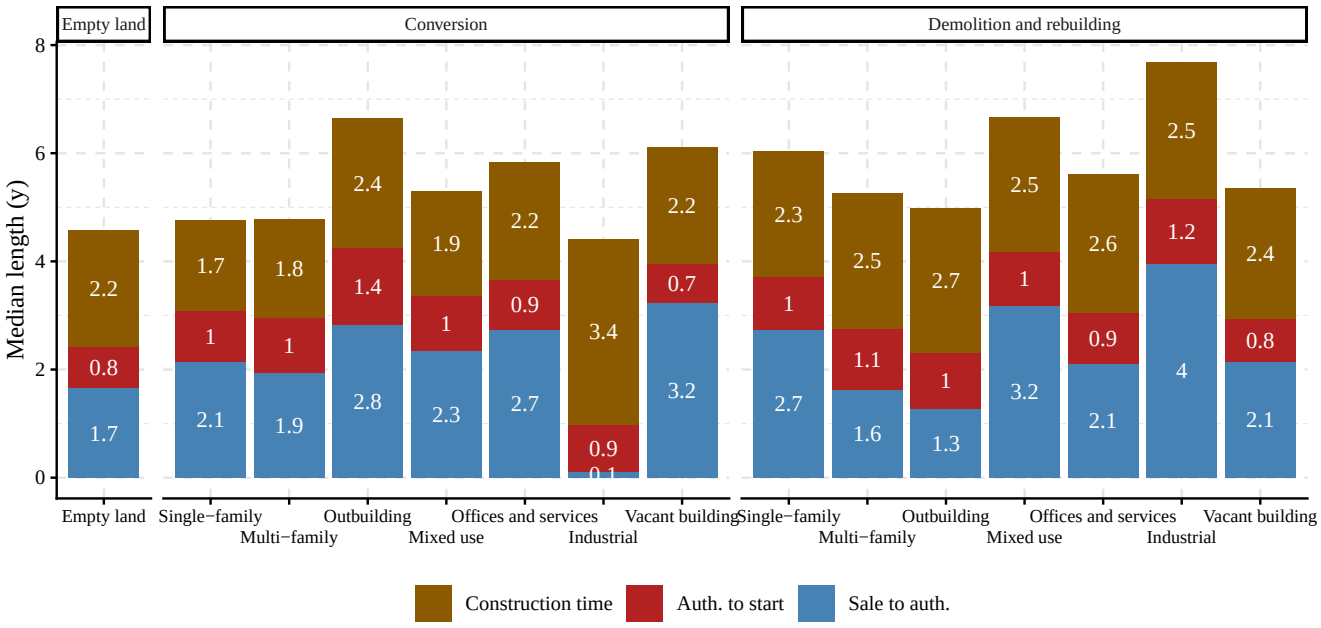


Figure J1: Time from the sale of a parcel to completion, by type of project and initial use

Data: transaction register, permit register. Median time from the sale of the parcel to the permit, from the permit to the start of works, and from the start of works to completion, by type of project and initial use, for permits creating at least one housing unit.

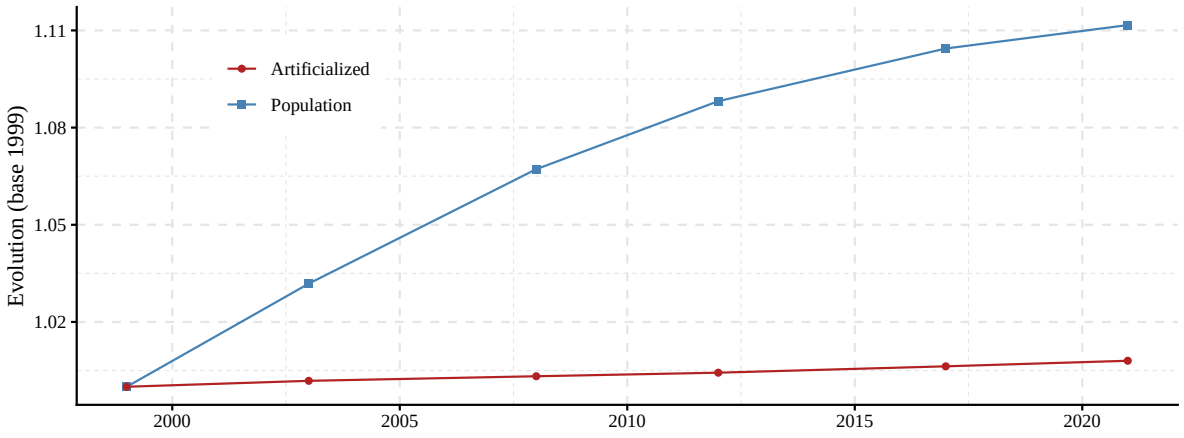


Figure J2: Population and built-up land in the Paris area, 1999–2021

Data: population censuses. Built-up land from the regional land-use inventory. Indices, 1999 = 1.

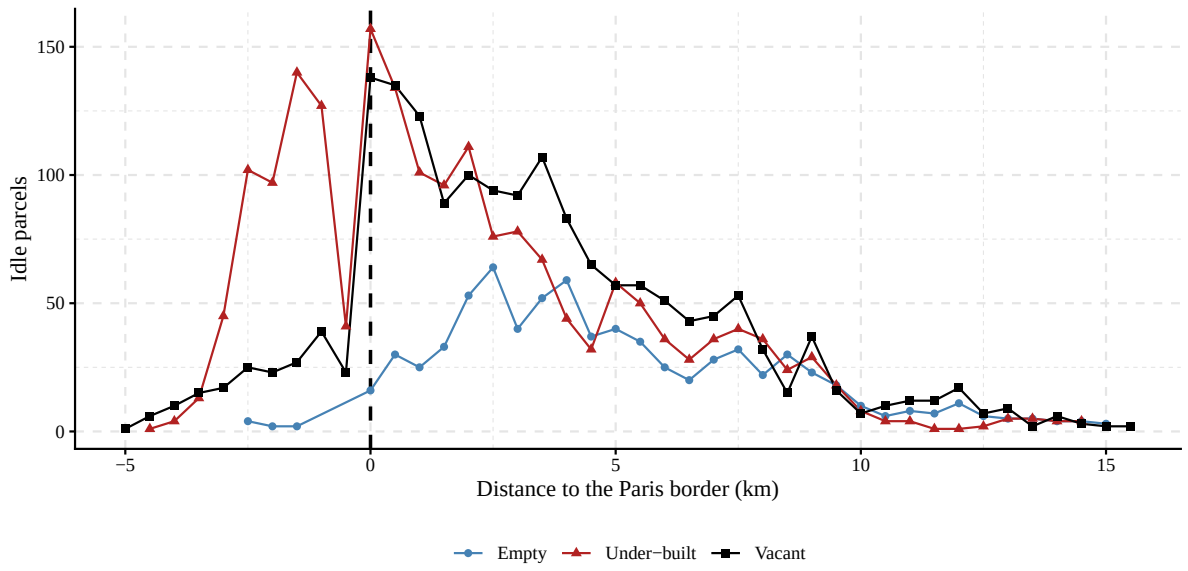


Figure J3: Idle parcels by distance to Paris

Data: land tax records, building database, permit register. Empty parcels carry no building. Vacant parcels carry a building with no taxed unit and a documented past economic activity. Under-built parcels carry a building less than a quarter of the mean height of the buildings that touch it. Data as of 2024.

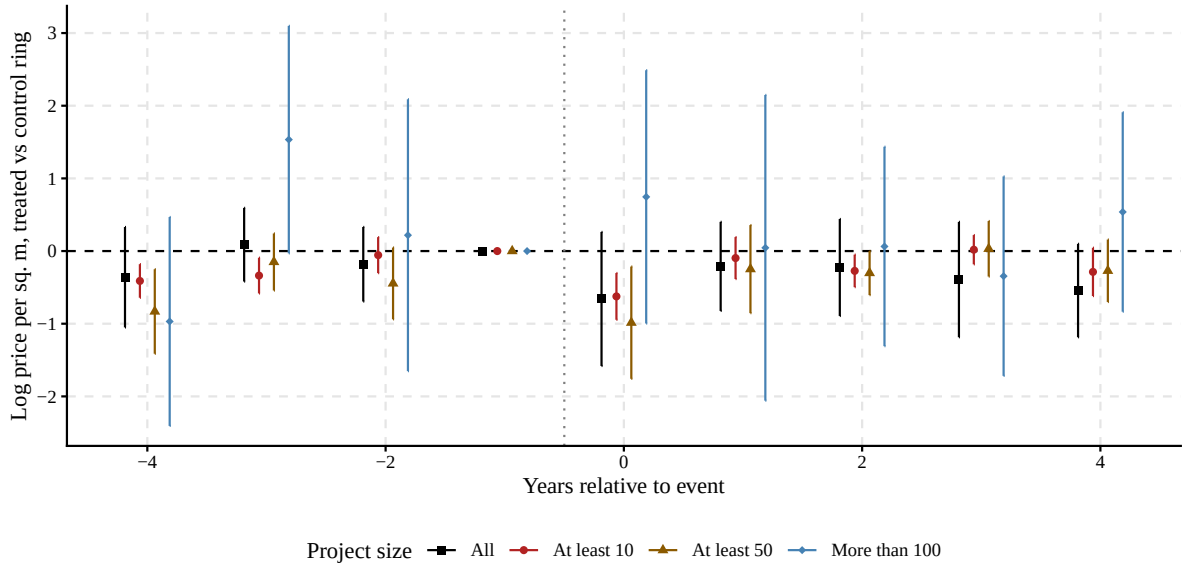


Figure J4: Event study: change in price after the permit

Data: transaction register, permit register, building database. Inner ring 20–100 m, buffer 100–150 m (excluded), outer ring 150–500 m from the project centroid. Event-time coefficients ξ_k of (A2), relative to the year before the permit ($k = -1$), with 95% confidence intervals. Each colour and shape is a different sample of projects by size.

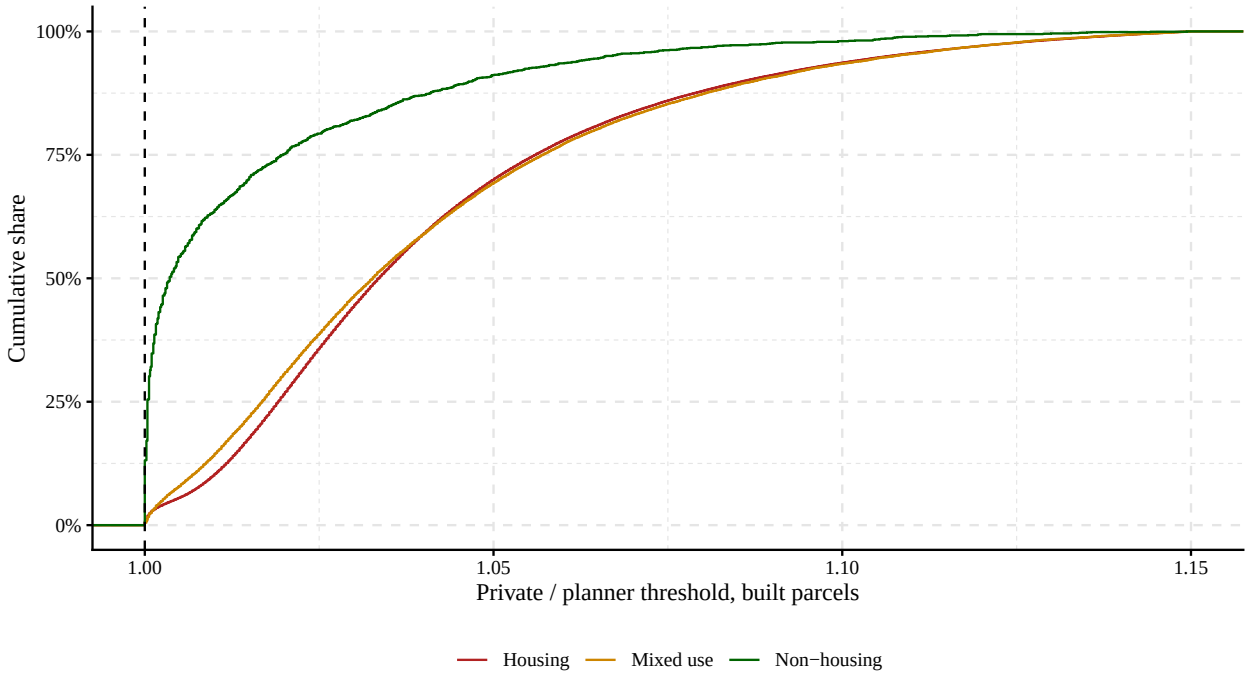


Figure J5: Redevelopment threshold relative to the constrained-efficient threshold

Data: model estimation. Cumulative distribution, by initial use, of $R_i^\star / R_i^{\text{CE}}$ over the parcels of the estimation sample with a standing building. Horizontal axis truncated at 1.15.

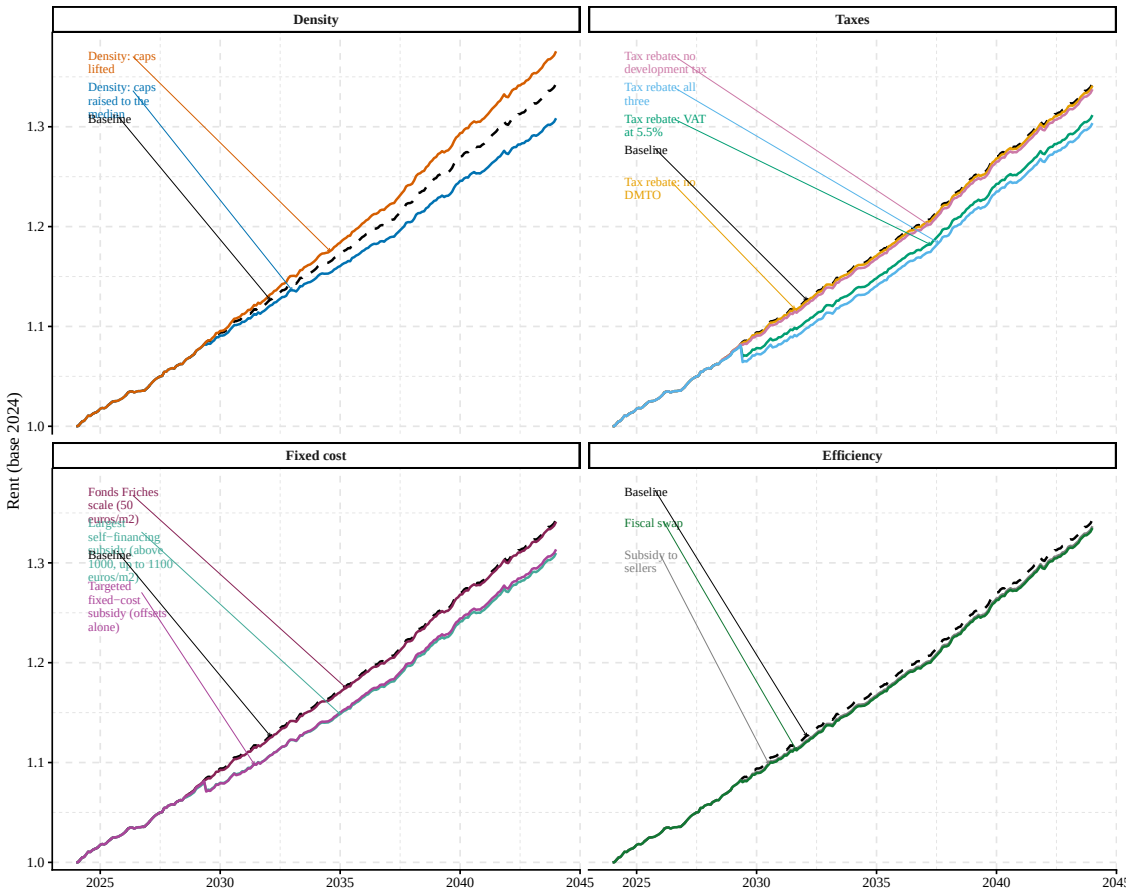


Figure J6: Rents under each policy

Data: model estimation. Mean regional market rent across simulated demand paths, by policy, relative to its 2024 level, over the twenty years of the horizon. Dashed: private equilibrium, repeated in every panel. The same demand paths are used in every scenario.

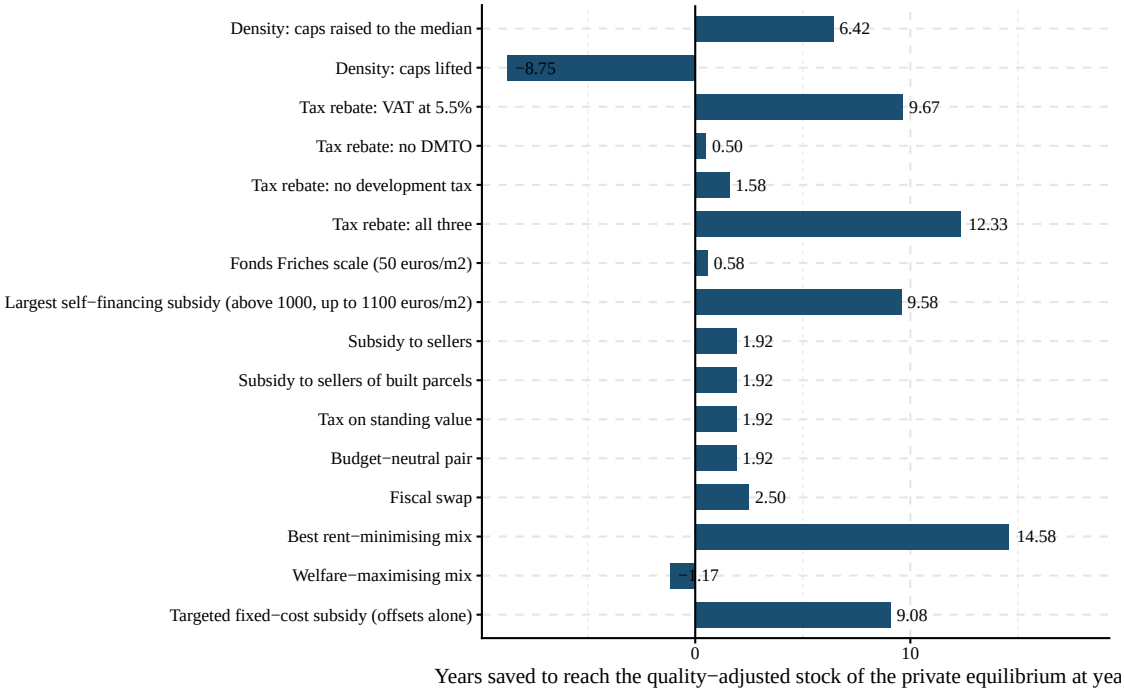


Figure J7: Time saved by each policy

Data: model estimation. Years saved to reach the quality-adjusted stock that the baseline reaches in 20 years, equation (25).